

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

**An Online Ordering System for Brand-New and Preloved
Clothes with Product Listing, Checkout, and Buyer-Seller
Management**

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The growth of internet technology significantly transformed the way products were bought and sold in the global market. In the local context of the Philippines, this digital shift was particularly visible among small-scale enterprises that utilized social media and web-based platforms to promote their brands and interact with a wider customer base. Clothing businesses, in particular, benefited from this trend as consumers increasingly preferred the convenience of browsing fashion items online before committing to a purchase. Consequently, an online ordering system became more than just a luxury; it was a critical tool for businesses that aimed to improve operational efficiency and remain competitive in an increasingly digital age.

The clothing industry stood as one of the most active sectors in the e-commerce landscape because fashion products appealed to a diverse range of consumers with varying needs. Some buyers prioritized brand-new apparel for guaranteed quality and the latest styles, while others sought preloved clothing for its affordability and sustainability. Within the Philippine market, the rise of "ukay-ukay" or secondhand fashion reflected a broader cultural shift. What was once viewed merely as a budget option had gradually evolved into a conscious choice for environmentally-aware consumers. This evolution allowed small resellers to thrive, primarily operating through informal channels like Facebook Marketplace and Messenger.

However, the researchers observed that managing a dual inventory of both brandnew and preloved clothes presented unique operational hurdles. When a business relied solely on social media posts and private messages, the risk of confusion was high. Manual tracking often led to delayed responses, incomplete transaction records, and significant difficulty in monitoring real-time inventory. For a single seller juggling stock levels, payment verification via GCash, and delivery scheduling, the weight of manual labor was often overwhelming. A single mismanaged message could result in a lost order or a frustrated customer, highlighting a clear gap between the business's potential and its current operational capacity.

This was precisely the situation faced by **Margaux Collections**, a local boutique in Iloilo. While the business catered to a growing market for bikinis, coordinated sets, and preloved skirts, its ability to scale was hindered by its reliance on manual chatting. Each preloved item was unique, meaning that once it was sold through a private message, it had to be manually deleted or marked on social media to prevent other customers from inquiring. This process was not only time-consuming but also prone to human error. The researchers recognized that for Margaux Collections to maintain its brand identity—characterized by its elegant, "dark noir" aesthetic—it needed a professional and centralized digital platform.

The proposed **Online Ordering System for Brand-New and Preloved Clothes** was developed to bridge this gap. By providing a structured interface for product listings, the system allowed customers to view complete details, such as price, size, and item condition, without needing to send repeated inquiries. For the business owner, the system automated the checkout process and provided a reliable way to verify GCash transactions through reference number tracking. By integrating these features into one web-based platform, the researchers sought to modernize the operations of Margaux Collections, ensuring that the transition from browsing to checkout was as seamless and professional as the fashion items being sold. This study focused on demonstrating how a purpose-built system could reduce manual workload while simultaneously enhancing the shopping experience for the modern Filipino consumer.

1.2 Statement of the Problem

This study aimed to develop an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer- Seller Management. The growth of online platforms in business had made it easier for sellers to reach customers, but it had also created significant challenges in handling orders, organizing product information, and maintaining clear communication. In the specific case of Margaux Collections, these challenges were exacerbated by the business's dual nature of selling both brand-new and preloved apparel. Each item required precise description, categorization, and monitoring, which became increasingly difficult under a manual framework. While the manual operation of the boutique allowed for a personal touch in customer service, it ultimately created a "glass ceiling" for the business's growth. The researchers identified that the primary issue was the lack of a synchronized system to manage two fundamentally different product

types: brand-new items with multiple stocks and unique preloved pieces that were strictly one-off items.

This general problem was further analyzed through the following specific operational gaps that occurred during the business's manual phase:

First, the inefficiency of manual inquiry handling caused significant operational delays. Because product details such as exact measurements, fabric types, and condition reports were not readily available in a structured format, the owner was forced to spend a disproportionate amount of time answering repetitive questions via Messenger. This diverted

her focus from critical activities like inventory sourcing and strategic marketing, leading to missed opportunities.

Second, the business faced a persistent risk of inventory discrepancy. In a manual, social-media-based setup, there was no real-time "lock" on items. A customer could inquire about a specific preloved skirt, but because the reservation process was not automated, the same item might have been promised to another buyer in a different chat thread. This lack of a centralized inventory status led to customer dissatisfaction and logistical confusion.

Third, the process of payment verification was labor-intensive and vulnerable to human error. Without an integrated checkout system, the owner had to manually cross-reference every GCash reference number against her transaction logs. During peak hours or "flash sales," managing dozens of simultaneous screenshots became overwhelming, increasing the likelihood of overlooking fraudulent claims or failed payments.

Lastly, the business lacked centralized data for informed decision-making. The owner had no immediate way to track total revenue or identify which categories—such as coordinates or accessories—were performing well. All vital sales information remained buried in scattered chat logs, making professional auditing and restocking plans nearly impossible to execute.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study was to develop a webbased ordering system for brandnew and preloved clothes that could efficiently manage product listing, order processing, checkout, and buyer-seller communication in one centralized platform.

1.3.2 Specific Objectives

Specifically, This study aimed to:

Design a web-based system that allowed the seller to upload, organize, and manage listings for brand-new and preloved clothing items.

Develop a checkout feature that enabled customers to place orders easily and review their selected items before confirming their purchase.

Create a buyer-seller management module that supported communication between customers and the seller regarding product inquiries, order updates, and transaction concerns.

Implement an inventory management function that helped the seller monitor available stock and updated product status in real time.

Improve the accuracy and organization of order processing by reducing manual encoding and using a centralized digital platform.

Provide a user-friendly interface that made it easier for customers to browse products, place orders, and completed transactions.

Ensure that the system supported both brand-new and preloved clothing items in a single online platform.

1.4 Scope and Delimitation

This study focused on the development of an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management for Margaux Collections. the system would cover the main operations of the business, particularly product listing, order placement, inventory tracking, payment handling, queuing, and analytics. It was designed to help the owner managed clothing items more efficiently and to provide customers with a more organized and convenient way of browsing and ordering products. The study would specifically cater to the needs of Margaux Collections, which sold both brand-new items such as bikinis, coordinated sets, and accessories, as well as secondhand items such as pants, skirts, and shorts.

The system would be developed for use by the owner and her customers in the Philippines. It would use web-based technologies such as PHP, MySQL, HTML, CSS, and JavaScript to support the system's functionality. The platform would

distinguish between brand-new and preloved clothing items so that products could be categorized properly and displayed in a clear manner. This separation would help both the seller and the buyers easily identify the type and condition of the items available for purchase.

The scope of the study included the implementation of features that supported product browsing, checkout, buyer seller communication, order management, and inventory monitoring. It also included queue management for pick-up orders and analytics to help the owner make better decisions based on sales and order data. These features were intended to improve the overall flow of business operations and reduced the problems caused by manual and social media based transactions.

On the other hand, this study had several delimitations. The system would not include integration with external logistics providers or third-party delivery tracking services. It would also not have advanced features such as automated size recommendations, as the focus of the study was on basic ordering and management functions. In addition, the project would prioritize web accessibility rather than the development of a mobile application. Since the business was operated solely by the owner, the system would be designed for a single-user business setup without staff handling transactions, inventory, or order processing.

The existing payment methods of Margaux Collections would remain unchanged. GCash would continue to be used for dropoff transactions, cash on delivery (COD) would be used for delivery orders, and in-person payment would be accepted for store pickup. These payment methods would stay the same because they were already familiar to the owner and convenient for the customers. By maintaining current payment setup, the system could be easier to adopt while still improving the organization and efficiency of the business.

the core ordering and inventory functions needed by a single-owner clothing business rather than the broader feature set typically found in large-scale

ecommerce platforms. The system was also limited to the payment and fulfillment methods already practiced by Margaux Collections — GCash reference verification, cash on delivery, in-person pickup, and rider-assisted pickup — instead of integrating third-party payment gateways such as PayMaya or credit card processors, since these were not part of the business's existing operations. Likewise, the study did not extend to multibranch or multi-seller functionality, as Margaux Collections operated as a single storefront managed by one owner.

1.5 Significance of the Study

This study was significant because it developed an Online Ordering System for

Brand-New and Preloved Clothes with Product Listing, Checkout, and BuyerSeller Management that could improve the way Margaux Collections operated and served its customers. The proposed system was designed to benefit several stakeholders by making the ordering process more organized, efficient, and convenient. Since the business was currently managed entirely by the owner without any staff, the system was expected to reduce the amount of manual work involved in handling orders, confirming payments, updating inventory, and responding to customer inquiries. Through automation and centralized management, the business could become more efficient and easier to handle on a daily basis.

For the Margaux Collections owner, the system provided a practical solution to the challenges of managing a small business alone. At present, the owner manually handled customer orders through Facebook Messenger, checked payments through GCash, and kept track of product availability on her own. This process could be time-consuming and might lead to errors, delays, or confusion, especially when multiple customers were inquiring or ordering at the same time.

With the proposed system, order processing and inventory management could be automated, allowing the owner to focus more on business operations and customer satisfaction. The system also provided sales insights and analytics that could help the owner make informed decisions regarding product restocking, especially in determining which items, such as brand-new bikinis and coordinated sets or preloved pants, skirts, and shorts, were in higher demand. In this way, the system supported not only daily operations but also the longterm growth of the business. For the customers, the system offered a more convenient and organized way of shopping. Instead of depending only on Facebook Messenger conversations, customers could use a dedicated web platform to browse products, view item details, and place orders more easily. This made the shopping experience more efficient because customers no longer needed to wait for repeated back-and-forth messages just to ask about product availability or order status. The system also provided a clearer payment workflow, since customers could follow the existing payment methods used by the business, such as GCash for drop-off orders, cash on delivery (COD) for delivery orders, and in-person payment for store pickup. In addition, the system could help customers track their orders and receive queue numbers for pickup, which improved transparency and reduced confusion. Overall, the proposed system gave customers a smoother and more reliable online shopping

experience. For future researchers and students, this study served as a useful reference for understanding how web-based systems could support small business operations, especially in businesses that sold both brand-new and preloved fashion items. It could be used as a case study in web development, database management, and systems analysis because it combines practical business functions with technical implementation. The study might also helped future researchers who were interested in developing systems for micro enterprises, online clothing stores, or businesses that relied heavily on manual transaction handling. In addition, it provided ideas on how a single-owner business could transition from social media-based selling to a more structured digital platform. As a result, the study contributed not only to the improvement of Margaux Collections but also to the broader body of knowledge in information technology and business system development.

Overall, the significance of this study lies in its ability to improve business efficiency, reduce manual workload, enhance customer convenience, and provide academic value for future researchers. By creating a centralized system for managing brandnew and preloved clothes, the study addressed the actual needs of a small business while also showing how technology could support growth and better decision-making. It demonstrated that even a small, ownermanaged business could benefit from a well-designed online ordering system that was tailored to its specific operations and payment practices.

1.6 Definition of Terms

Margaux Collections Ordering System – A web-based platform designed for Margaux Collections that allowed product browsing, ordering, queue management, and sales analytics. The system was intended to support the business by organizing both brandnew and preloved clothing items in one centralized platform.

Brand-new – Items that were new, unused, and sold in their original condition. In this study, brand-new items included bikinis, coordinated sets, and accessories.

Preloved – Apparel that had been previously owned but was still in good wearable condition. In this study, preloved items included pants, skirts, and shorts.

Queue Number – An automated number assigned to each order to help organize processing and pickup. It was used to keep transactions in proper order, especially for drop-off locations and store pickups.

GCash Reference Number – A unique code generated after a GCash transaction, used by the system to verify that a customer's payment had actually been made before an order was confirmed.

Admin Panel – The restricted section of the system used by the owner to manage products, orders, users, and sales analytics.

Pick-up Via Rider – An order fulfillment method that allowed a customer to have their order picked up by a third-party rider on their behalf, in addition to store pickup, dropoff, and delivery.

Middleware – PHP-based components that managed communication between the frontend and backend of the system. These components handled authentication, sessions, authorization, and data validation to ensure that the system operated properly and securely.

API – PHP interfaces that supported system functions such as cart operations, order updates, payment status processing, and real-time dashboard information.

These interfaces allowed different parts of the system to exchange data efficiently.

Database Gateway – The MySQL layer responsible for storing and retrieving data related to users, products, orders, payments, and transactions. It served as the connection between the system's interface and its database records. **GCash** – A digital wallet used for advance payment in dropoff orders. Customers paid through GCash before the owner delivers the order to the designated drop-off store.

Cash on Delivery (COD) – A payment method in which customers pay in cash when their order was delivered. This method was used for delivery transactions in the business.

Drop-off Location – A specific store or address where the owner delivers orders for customers to pick up after payment through GCash.

Facebook Posts – Product listings published by the owner on her Facebook page, which were used as the main way of displaying available items to customers.

Messenger – The Facebook Messenger platform used by customers to communicate with the owner, ask about products, and place orders.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

The researchers conducted a comprehensive review of existing literature and studies to establish a solid theoretical and practical foundation for the development of the Margaux Collections Ordering System. This chapter explored various global and local perspectives on e-commerce, inventory management, and the evolving culture of secondhand fashion consumption. By analyzing these works, the researchers identified the technical gaps and operational requirements necessary to build a robust system tailored for a niche boutique.

2.1 Foreign Studies

Foreign studies provided empirical evidence that supported the design and purpose of the proposed system. These studies were especially valuable because they showed how consumers and businesses actually behave in real settings, which helped justify the need for features such as product listings, inventory control, transparent information, and digital trust-building. In the context of Margaux Collections, the foreign studies reviewed here explain why product presentation, consumer confidence, sustainability, and system organization were essential in a digital fashion retail environment. They also demonstrated that second-hand fashion was influenced by emotional, practical, and environmental considerations that had to be reflected in the design of the ordering system.

Market Prices of Remanufactured, Used, and New Items Neto, Bloemhof, and Corbett (2016)

Neto, Bloemhof, and Corbett (2016) conducted a study that examined the market prices of remanufactured, used, and new items using data obtained from eBay. Their research focused on how consumers valued products based on their different conditions and how pricing reflected market demand. The study was relevant because it demonstrated that used products possessed their own market value and were not automatically perceived as inferior to new products. It further showed that consumers considered various characteristics of a product when determining whether its price was reasonable and whether the item provided sufficient value for their money.

The study was particularly relevant to Margaux Collections because it supported the idea that pre-loved items needed to be priced and presented in a manner that reflected their actual value. The researchers found that used items generally fell between new and remanufactured products in terms of market price. This suggested that buyers evaluated second-hand goods by considering both their affordability and physical condition rather than relying on price alone. For

Margaux Collections, this indicated that pre-loved clothing items needed to be priced transparently and fairly according to their condition, appearance, quality, appeal, and remaining value. Providing reasonable and clearly explained prices could help customers perceive preloved products as practical and worthwhile purchases rather than simply as cheaper alternatives to brand-new items.

Neto, Bloemhof, and Corbett (2016) also demonstrated that product presentation influenced how consumers perceived and evaluated the value of an item. When product descriptions were clear and pricing was presented consistently, buyers were more likely to trust the seller and perceive the product as offering a good deal. This finding supported the need for detailed product information, accurate condition descriptions, appropriate condition labels, and organized product listings in the Margaux Collections Ordering System. These features could reduce customer hesitation and provide buyers with sufficient information to make more confident purchasing decisions. By presenting the condition and price of each item clearly, the system could also help minimize misunderstandings between the seller and customers.

The study further implied that transparent pricing practices could strengthen customer trust and encourage repeat purchases. When consumers understood how an item had been valued and why it was offered at a particular price, they were more likely to feel comfortable purchasing from the same seller again. For Margaux Collections, this meant that the ordering system needed to display product prices clearly, accurately, and consistently so that customers could easily understand the value of each pre-loved item. Providing organized information about product condition, features, and pricing could contribute to a more reliable and professional purchasing experience.

Overall, the study of Neto, Bloemhof, and Corbett (2016) provided support for the importance of appropriate pricing and product presentation in the pre-loved clothing market. Its findings suggested that customers did not evaluate second-hand products based solely on their lower prices but also considered their condition, quality, and perceived value. These insights were applicable to the Margaux Collections Ordering System because the system could incorporate transparent pricing, detailed product descriptions, condition labels, and organized listings to help customers evaluate products more effectively. In this way, the system supported greater transparency, professionalism, customer confidence, and trust in the purchasing process.

Kim, Jung, and Lee (2021)

Kim, Jung, and Lee (2021) conducted a quantitative study that examined consumers' value and risk perceptions toward circular fashion and how these perceptions influenced their intention to purchase circular fashion products. The study focused on several factors, including perceived quality, emotional value, social

value, environmental value, economic value, and perceived risk. The researchers examined how consumers evaluated circular fashion products beyond their monetary cost and how different forms of perceived value contributed to their willingness to purchase. The study was relevant to the development of the Margaux Collections Ordering System because it demonstrated that secondhand, preloved, and circular fashion products were not evaluated solely based on their affordability. Instead, consumers also considered the quality, emotional significance, social meaning, and environmental benefits associated with these products. These findings supported the idea that preloved fashion could be positioned and presented as valuable, desirable, and fashionable products rather than simply as previously owned or used goods.

The study found that perceived quality, emotional value, social value, and environmental value significantly influenced consumers' purchase intention toward circular fashion products. This indicated that consumers were more likely to consider purchasing circular fashion products when they perceived the items as having good quality, emotional or personal meaning, positive social value, and environmental benefits. Perceived quality was particularly important because consumers needed to believe that the products were still usable, presentable, and worth purchasing despite having been previously owned. Emotional value also contributed to purchase intention because consumers could associate certain fashion items with personal preferences, uniqueness, self-expression, or satisfaction. Similarly, social and environmental values encouraged consumers to view circular fashion as a responsible and meaningful form of consumption. These findings were important to Margaux Collections because they suggested that the presentation of preloved products could influence how customers perceived their overall value and desirability.

The findings also provided support for presenting preloved products through a professional and organized ordering system. Since consumers were influenced by perceived quality and other forms of value, the Margaux Collections Ordering System needed to provide customers with sufficient information to evaluate each product before making a purchase decision. An organized product listing could include clear and high-quality photographs, accurate product descriptions, available sizes, prices, brands, colors, and detailed condition information. By providing these details, the system could help customers develop a clearer understanding of the product and reduce uncertainty associated with purchasing preloved items. The study suggested that when consumers were able to understand the characteristics and condition of a product, they were more likely to perceive it as reliable and worthwhile.

Kim, Jung, and Lee (2021) also highlighted the importance of reducing consumers' perceived risk when purchasing circular fashion products. Unlike conventional retail products, preloved fashion items had already been owned or used by another person, which could have caused customers to become more cautious about product quality, cleanliness, authenticity, durability, and overall condition. These concerns could have increased hesitation, particularly when customers purchased products through an online platform where they could not physically inspect the item. Therefore, complete and accurate product information played an important role in reducing uncertainty. For the Margaux Collections Ordering System, this finding supported the inclusion of detailed product descriptions, multiple product photographs, condition classifications, and other relevant information that could help customers make informed purchasing decisions.

The study further implied that trust could have been strengthened when businesses presented circular fashion products in a transparent and professional manner. Customers could have been more confident in purchasing preloved products when the system clearly communicated the actual condition and characteristics of each item. For example, descriptions of minor stains, scratches, signs of wear, or other imperfections could have helped customers establish realistic expectations before placing an order. Providing such information could have reduced the possibility of misunderstandings and could have improved customers' confidence in the ordering process. In this context, the ordering system did not only serve as a tool for processing purchases but also functioned as an important source of information that helped customers evaluate product quality and value.

The study was also relevant to the way Margaux Collections could have communicated the environmental and social value of preloved fashion. Circular fashion encouraged the continued use and circulation of existing clothing and reduced the need for consumers to purchase newly manufactured products. Therefore, the environmental value associated with preloved products could have been emphasized through appropriate product descriptions or system content. Presenting preloved fashion as a more sustainable consumption option could have helped customers understand that their purchases contributed to extending the useful life of clothing. In addition, the unique characteristics of preloved products could have contributed to their emotional and social value, particularly when customers viewed them as distinctive fashion pieces that allowed them to express their individual style.

Overall, the study of Kim, Jung, and Lee (2021) provided a relevant foundation for the development of the Margaux Collections Ordering System because it demonstrated that consumers' purchase intentions toward circular fashion were influenced by multiple dimensions of perceived value and risk. The findings

indicated that customers were more likely to respond positively to preloved fashion when they perceived the products as high-quality, meaningful, socially valuable, and environmentally beneficial, while concerns about product condition and purchasing risk could have reduced their willingness to buy. These findings supported the need for an ordering system that presented products clearly, professionally, and transparently. By incorporating high-quality product images, organized product listings, accurate descriptions, condition information, and other relevant details, the Margaux Collections Ordering System could have enhanced product presentation, reduced customer uncertainty, and strengthened consumer confidence. Thus, the study supported the view that an effective ordering system should have functioned not only as a platform for purchasing preloved fashion but also as a means of communicating the quality, value, and benefits of circular fashion products to customers.

Tu, Hsu, and Creativani (2022)

Tu, Hsu, and Creativani (2022) conducted a quantitative study in the United States that examined consumers' perceptions, attitudes, and purchasing behavior toward second-hand luxury goods. The researchers used survey data and structural equation modeling to investigate how different factors, including perceived quality, emotional value, social value, environmental value, economic value, and perceived risk, influenced consumers' purchase intentions. The study was relevant to the Margaux Collections Ordering System because it demonstrated that consumers' decisions to purchase second-hand fashion products were influenced by both rational and emotional considerations. Rather than evaluating preloved products solely according to their price or affordability, consumers also considered the quality of the item, the emotions associated with owning it, its ability to represent their identity, its social meaning, and its contribution to more sustainable consumption. These findings supported the idea that preloved fashion products could have been perceived as valuable and desirable fashion items rather than simply as used or discounted goods.

The study found that perceived quality, emotional value, social value, and environmental value were important predictors of consumers' purchase intention toward second-hand luxury goods. This indicated that consumers were more likely to consider purchasing second-hand products when they perceived them as being of good quality, aesthetically appealing, meaningful, socially desirable, and environmentally responsible. Perceived quality was particularly relevant because consumers could have had concerns about the condition, durability, authenticity, and overall appearance of previously owned products. When customers believed that a second-hand item remained in good condition and provided sufficient value for its

price, they were more likely to develop a positive attitude toward purchasing it. Emotional value also played an important role because consumers could have associated certain fashion items with personal satisfaction, uniqueness, selfexpression, or a particular lifestyle. These findings suggested that purchasing decisions were not based entirely on functional benefits but were also influenced by the meanings and experiences consumers associated with the products.

The findings were significant to Margaux Collections because they suggested that preloved products should have been presented in a manner that enhanced their aesthetic, emotional, and perceived overall value. If products had been displayed professionally, organized according to appropriate categories, and accompanied by attractive and accurate descriptions, customers could have perceived them as carefully selected fashion pieces rather than ordinary used clothing. A well-designed ordering system could therefore have contributed to the way customers evaluated the products before making a purchase. In this regard, the proposed Margaux Collections Ordering System needed to provide more than basic purchasing functions. It also needed to support an organized and appealing product presentation that could have strengthened customers' perceptions of product quality and desirability.

The study also emphasized the importance of trust and complete product information in online second-hand purchasing. Since customers could not physically inspect the products before completing a transaction, they had to rely heavily on the information provided by the seller. Product photographs, descriptions, measurements, brand information, and condition details could have served as important sources of information that helped customers determine whether a product met their expectations. When product information was accurate, detailed, and transparent, consumers could have felt more confident about their purchasing decisions. Conversely, incomplete or unclear information could have increased perceived risk and caused customers to hesitate before placing an order. This finding directly supported the inclusion of multiple product images, detailed product descriptions, accurate pricing, and clear condition labels in the Margaux Collections Ordering System.

Providing detailed condition information was particularly important for a preloved fashion ordering system because the products could have shown different levels of previous use. Customers needed to know whether an item was in excellent, good, or fair condition and whether it had visible signs of wear, stains, scratches, fading, or other imperfections. By communicating these details clearly, the system could have helped customers develop realistic expectations about the products they intended to purchase. This level of transparency could have reduced uncertainty and perceived

risk while also contributing to greater customer satisfaction. The findings of Tu, Hsu, and Creativani (2022) therefore supported the idea that transparency should have been an essential component of the online purchasing experience for second-hand fashion.

Another important implication of the study was the role of social and symbolic value in the purchase of second-hand luxury goods. Consumers could have viewed secondhand fashion products as more than physical items because these products could have represented personal identity, lifestyle, social status, taste, and individual fashion preferences. A preloved item could have been considered desirable because of its brand, uniqueness, style, or perceived exclusivity, even though it had previously been owned by another person. This perspective was relevant to Margaux Collections because it suggested that the system could have presented preloved products as curated fashion pieces rather than simply categorizing them as second-hand goods. By using consistent product presentation, clear categories, appealing photographs, and informative descriptions, the system could have created a stronger perception of value and desirability among potential customers.

The environmental value identified in the study was also relevant to the purpose of Margaux Collections. Second-hand and preloved fashion could have supported more sustainable consumption by allowing existing clothing items to remain in circulation for a longer period. Consumers who valued environmental responsibility could have been more inclined to purchase preloved products because their purchases could have been associated with reduced consumption of newly manufactured clothing. Therefore, the ordering system could have supported the communication of the sustainability benefits associated with preloved fashion. Presenting products as part of a more sustainable fashion cycle could have helped customers understand that their purchases provided both personal and environmental value.

In addition, the study showed that economic value remained a relevant consideration, although it was not the only factor affecting purchase intention. Consumers could have been attracted to second-hand luxury goods because they offered an opportunity to obtain desirable products at prices that were potentially more accessible than those of brand-new luxury items. However, the findings indicated that affordability alone did not determine consumers' willingness to purchase. Product quality, emotional significance, social value, environmental considerations, and perceived risk also influenced the decision-making process. For Margaux Collections, this implied that the ordering system should not have focused solely on displaying prices. Instead, product information should have communicated the overall value of

each item by highlighting its condition, appearance, brand, characteristics, and other relevant attributes.

Overall, Tu, Hsu, and Creativani (2022) provided significant support for the development of the Margaux Collections Ordering System because their study demonstrated that consumers' purchasing decisions toward second-hand fashion were influenced by a combination of functional, emotional, social, environmental, and economic factors. The findings indicated that consumers were more likely to purchase second-hand products when they perceived them as high-quality, meaningful, attractive, socially valuable, and environmentally responsible, while uncertainty and perceived risk could have reduced their purchase intention. These findings supported the need for an ordering system that provided transparent and comprehensive product information while maintaining an appealing and professional presentation. By incorporating multiple product photographs, accurate descriptions, condition labels, organized categories, pricing information, and consistent product displays, the Margaux Collections Ordering System could have helped reduce consumer uncertainty, strengthen trust, and increase the perceived value of preloved fashion products. Thus, the study supported both the functional and presentation aspects of the proposed system and reinforced the importance of creating an online shopping experience that made preloved products appear trustworthy, valuable, fashionable, and desirable.

Selling Second-Hand Luxury: Empowerment and Social Roles Turunen, Cervellon, and Carey (2020)

Turunen, Cervellon, and Carey (2020) conducted a qualitative study that explored the experiences, motivations, and perspectives of individuals who sold second-hand luxury goods. Through interviews with sellers, the researchers examined how participation in the resale market influenced sellers' perceptions of themselves, their relationships with consumers, and their involvement in sustainable fashion practices. The study found that sellers often experienced a sense of empowerment because they were able to recover value from unused or previously owned items while also participating in a more sustainable fashion economy. This finding was important because it demonstrated that selling second-hand products was not merely a financial activity but could also have personal, social, and environmental significance. Reselling allowed individuals to extend the life of fashion products, reduce unnecessary disposal, and contribute to the continued circulation of goods.

The study therefore presented second-hand selling as an activity that could influence how sellers viewed their roles within consumer culture and sustainability.

The study further found that sellers often viewed themselves as curators and responsible participants in the circulation of fashion products. Rather than simply disposing of items they no longer needed, sellers could have carefully selected, prepared, organized, and presented products to make them more appealing to potential buyers. This process gave sellers an active role in determining how products were introduced and perceived in the resale market. The sellers' role as curators was particularly relevant to Margaux Collections because it emphasized the importance of careful product selection and presentation. Preloved fashion products could have gained greater appeal when they were presented as carefully selected pieces with distinct characteristics, styles, and histories rather than as ordinary used clothing.

Another important finding of the study was the use of storytelling to make secondhand luxury products more attractive to consumers. Sellers could have shared information about the background, condition, style, brand, or significance of an item to create a stronger connection between the product and potential buyers. These descriptions could have helped consumers understand the value and uniqueness of a particular product. This finding was highly relevant to the Margaux Collections Ordering System because it suggested that product descriptions should have provided more than basic factual information. In addition to listing the product name, price, size, and condition, descriptions could have communicated relevant characteristics that helped customers understand why an item was desirable. A wellwritten and informative product description could have increased customers' interest and curiosity while helping them develop a stronger perception of the product's value, even though the item had been previously owned.

The findings also indicated that organized product presentation could have contributed to professionalism and consumer trust. Sellers who carefully managed their products and communicated information clearly were more likely to create a positive impression among potential buyers. In an online second-hand fashion environment, customers could not physically examine the products before purchasing them. As a result, the quality and organization of the information presented through the selling platform could have influenced customers' perceptions of the seller's reliability and professionalism. Clear photographs, accurate descriptions, appropriate categorization, and complete product information could have reduced uncertainty and helped customers feel more confident about their purchasing decisions.

This finding supported the development of the Margaux Collections Ordering System because the system could have helped the business owner organize and present product information in a structured and visually appealing manner. Instead of relying on scattered or manually managed product information, the system could have provided a centralized platform where products could have been properly recorded, categorized, described, and displayed. Such organization could have improved the consistency of product presentation and made it easier for customers to browse available items. The system could therefore have functioned not only as an ordering platform but also as an important tool for maintaining the professional image of Margaux Collections.

Furthermore, the study showed that second-hand selling could have strengthened sellers' sense of agency and empowerment. Sellers could have viewed resale as a meaningful activity that allowed them to generate economic value while simultaneously contributing to social and environmental goals. Instead of considering second-hand selling as a simple method of disposing of unwanted possessions, sellers could have perceived it as an active form of participation in the fashion resale economy. This perspective was relevant to small businesses such as Margaux Collections because it demonstrated that an effective ordering system could have supported more than the completion of individual transactions. It could also have contributed to the broader identity, operations, and growth of the business.

For Margaux Collections, the ability to efficiently manage products, maintain accurate records, and communicate product information could have provided the owner with greater control over the selling process. A centralized ordering system could have reduced the difficulties associated with manually organizing product details and responding to customer inquiries. It could have allowed the owner to manage available items more systematically while ensuring that customers received consistent and relevant information. As a result, the system could have contributed to improved operational efficiency and could have enabled the business owner to engage more confidently in the resale process.

The study also implied that the identity of a second-hand fashion business could have been strengthened through the way products were presented and communicated to customers. Since sellers could have acted as curators and storytellers, the business could have developed a distinctive identity by emphasizing the uniqueness, quality, and value of its products. For Margaux Collections, this could have been reflected through consistent product descriptions, organized categories, attractive product images, and informative details about the condition and characteristics of each item. These elements could have helped establish a

more recognizable and professional presentation while distinguishing the business from informal second-hand selling practices.

Overall, Turunen, Cervellon, and Carey (2020) provided relevant insights for the development of the Margaux Collections Ordering System by demonstrating that second-hand selling involved economic, social, environmental, and personal dimensions. The study showed that sellers could have experienced empowerment by recovering value from unused items, participating in sustainable fashion practices, and taking an active role in the circulation of fashion products. It also emphasized the importance of curation, storytelling, organized presentation, and clear communication in creating value and building trust between sellers and buyers. These findings supported the need for the Margaux Collections Ordering System to provide features that enabled the business owner to manage product information systematically and present preloved items in a professional and appealing manner. By supporting organized product management, detailed descriptions, clear presentation, and efficient customer transactions, the system could have contributed not only to improved business operations but also to the development of a more professional and trustworthy identity for Margaux Collections.

Online Shopping vs. Offline Shopping: A Comparative Study — Sarkar and Das (2017)

Sarkar and Das (2017) conducted a comparative study of online and offline shopping behavior to determine the factors that influenced consumers' preferences and purchasing decisions. The researchers examined the differences between traditional shopping and online shopping and identified several advantages and disadvantages associated with each method. Their study was relevant to the development of the Margaux Collections Ordering System because it provided insights into the factors that could influence customers when choosing an online shopping platform. The findings showed that online shopping offered several advantages, particularly in terms of convenience, accessibility, wider product selection, and the ability to compare different products more quickly. These advantages helped explain why consumers increasingly considered digital platforms as convenient alternatives to traditional shopping, especially when the online shopping experience was simple, organized, and trustworthy.

The study showed that convenience was one of the major advantages of online shopping. Customers could browse products, compare available options, and place orders without needing to visit a physical store. This allowed consumers to save time and provided greater flexibility in making purchasing decisions. Online platforms also allowed businesses to reach customers beyond their physical location, potentially increasing product visibility and accessibility. These findings were significant to Margaux Collections because the proposed Ordering System

could have provided customers with a more convenient way of viewing available fashion items and submitting orders. Instead of relying entirely on physical visits or manual communication, customers could have accessed product information through the system and completed the ordering process more efficiently.

However, Sarkar and Das (2017) also identified several disadvantages and concerns associated with online shopping. One of the main concerns was the inability of customers to physically inspect products before purchasing them. Consumers could have experienced uncertainty regarding product quality, appearance, size, fit, and overall condition. These concerns were particularly relevant to fashion retail because clothing products could have differed in appearance when viewed online compared with when examined physically. For preloved clothing, these concerns could have been even more significant because customers might have needed to evaluate signs of wear, stains, fabric condition, or other imperfections before deciding to purchase an item.

This finding supported the need for detailed and accurate product presentation in the

Margaux Collections Ordering System. Customers who could not physically examine clothing items would have depended heavily on the photographs and information provided through the system. Therefore, the ordering system needed to provide clear product images, accurate descriptions, available sizes, prices, colors, brands, and condition details. Providing complete information could have helped customers better evaluate products and could have reduced uncertainty before placing an order. The more accurate and understandable the information was, the more likely customers could have felt confident in making an online purchase.

The researchers also found that trust was an important factor in online purchasing decisions. Customers were more likely to complete transactions when an online platform appeared organized, secure, reliable, and easy to use. A confusing interface, incomplete information, or unclear purchasing process could have caused consumers to hesitate or abandon their intended purchases. This finding was particularly relevant to the Margaux Collections Ordering System because the system needed to establish customer confidence through its design and functionality. Clear navigation, organized product categories, understandable ordering procedures, and transparent transaction information could have helped make the online shopping process more dependable.

For Margaux Collections, usability could have played an important role in creating a positive customer experience. The ordering system needed to allow customers to locate products easily, understand available product information, and submit their orders without unnecessary difficulty. A simple and organized interface could have reduced the effort required to complete an order and could have encouraged customers to use the system repeatedly. Furthermore, features such as order

confirmation and order status updates could have helped customers remain informed about their purchases. These functions could have reduced uncertainty and contributed to a more reliable and professional shopping experience.

The study further indicated that convenience alone was not sufficient to maintain customer interest in an online shopping platform. Although customers could have appreciated the accessibility and speed of digital shopping, they might still have hesitated if the platform appeared confusing, unreliable, or incomplete. This was particularly important for Margaux Collections because the business offered fashion products that could have included both brand-new and preloved items. Customers needed to understand the specific characteristics and condition of each product before purchasing. Therefore, the ordering system needed to combine convenience with confidence-building features, such as detailed item descriptions, clear product photographs, organized listings, transparent pricing, and reliable order management.

Overall, Sarkar and Das (2017) provided relevant findings for the development of the

Margaux Collections Ordering System by demonstrating that online shopping offered significant advantages in terms of convenience, accessibility, and product comparison while also presenting challenges related to trust, product evaluation, security, and reliability. The study suggested that a successful online shopping platform needed to provide more than convenient access to products. It also needed to create an environment where customers could confidently evaluate products and complete transactions. For Margaux Collections, this supported the development of an ordering system that emphasized usability, clear product presentation, accurate information, and transparent order handling. By addressing these factors, the system could have provided customers with an online shopping experience that was both efficient and reassuring.

Sustainable Supply Chain Management — Svensson (2007)

Svensson (2007) examined Sustainable Supply Chain Management as a conceptual framework that integrated economic, environmental, and social considerations into business operations and supply chain activities. The study emphasized that sustainability should have been incorporated into the structure and daily operations of a business rather than being treated as a separate or secondary concern.

Sustainable business practices involved not only environmental responsibility but also efficient resource management, coordination, information sharing, and long-term organizational performance. This perspective was relevant to Margaux Collections because the proposed Ordering System could have supported more organized inventory management, efficient product movement, and responsible use of business resources. Through the proper management of product information and inventory records, the

system could have contributed to sustainability as part of the business's regular operations.

The study emphasized that sustainability depended on effective coordination and information sharing among different activities within a business. When organizations maintained accurate records and used appropriate information systems to support their operations, they could have reduced errors, improved decision-making, and managed resources more efficiently. This principle was particularly applicable to small businesses with limited and frequently changing inventory. In the case of Margaux Collections, preloved fashion products could have been unique items with limited quantities, meaning that accurate inventory information was important in preventing incorrect product availability, duplicate listings, and stock-related errors.

The Margaux Collections Ordering System could have addressed this need by providing organized product and inventory records. Through the system, product information could have been updated whenever an item was added, reserved, sold, or otherwise became unavailable. Maintaining updated records could have helped the business owner determine which products were still available and which items had already been sold. This could have reduced the possibility of accepting orders for unavailable products and could have improved the accuracy of information presented to customers. As a result, inventory management could have become more efficient and could have supported better utilization of the products handled by the business.

Another important contribution of Svensson (2007) was the emphasis on the role of information systems in supporting sustainable supply chain management. Digital technologies could have helped businesses monitor the movement of products, improve planning, strengthen information transparency, and support more informed operational decisions. Accurate and accessible information could have allowed businesses to identify patterns in product demand and determine which items were selling more effectively. For Margaux Collections, these capabilities could have helped the owner monitor available products, identify items that required additional promotion, and make better decisions regarding inventory management.

The importance of information management was particularly relevant to a preloved fashion business because inventory could have consisted of unique and limited-stock items. Unlike mass-produced products that could have been replenished in large quantities, individual preloved items might not have had identical replacements available. Therefore, maintaining accurate records of each product could have been essential to ensuring that items were properly monitored from listing to sale. A centralized ordering and inventory

system could have provided the business with a clearer overview of its available stock and could have reduced unnecessary manual work.

The study also connected sustainability with operational efficiency and accountability. Businesses that maintained accurate records and coordinated their activities effectively were better positioned to make informed decisions regarding stock management, product promotion, resource use, and waste reduction. For Margaux Collections, accurate product records could have helped the owner determine which items remained available, which products had already been sold, and which items might have required additional promotional attention. This could have improved the efficiency of business operations while reducing errors associated with manual recordkeeping.

Furthermore, the use of a digital system could have supported transparency within the business. When product information, availability, and order records were properly documented, the owner could have had a clearer basis for monitoring business activities and evaluating performance. This could have supported accountability because transactions and inventory changes could have been recorded systematically. Such documentation could have made it easier to identify discrepancies and maintain consistency between the products displayed in the system and the actual inventory of the business.

The concept of sustainable supply chain management was also relevant to the nature of preloved fashion itself. The continued selling and reuse of previously owned clothing could have contributed to a more responsible product cycle by extending the useful life of existing fashion items. An efficient ordering and inventory system could have supported this process by helping products move from listing to purchase in an organized manner. Instead of allowing usable clothing items to remain unmanaged or become unnecessarily discarded, the system could have helped facilitate their continued circulation through the resale process.

Overall, Svensson (2007) provided a relevant conceptual foundation for the Margaux Collections Ordering System by demonstrating that sustainability could have been supported through efficient operations, accurate information management, coordination, and responsible resource utilization. The study showed that sustainability was not limited to environmental concerns but could also have been connected to business efficiency, accountability, and long-term performance. For Margaux Collections, the proposed Ordering System could have contributed to these objectives by providing organized inventory records, updated product availability, improved information sharing, and more efficient order management. By supporting the accurate management of products from listing to sale, the system could have helped the business reduce operational errors, make better decisions, and maintain a more responsible and sustainable product flow. Thus, the study

supported the integration of sustainability into the everyday operations of Margaux Collections rather than treating it as a separate business concern.

Clothing Reuse and Environmental Impact — Cuc and Vidovic (2014)

Cuc and Vidovic (2014) discussed the environmental impact of clothing reuse and recycling, emphasizing the importance of extending the useful life of garments as a strategy for reducing textile waste and resource consumption. Their study highlighted the environmental benefits that could be achieved when clothing items were reused instead of being immediately discarded. Extending the life of garments could have reduced the demand for newly produced clothing and could have helped minimize the amount of textile waste sent to disposal facilities. The study was relevant to the development of the Margaux Collections Ordering System because it reinforced the sustainability value of preloved and second-hand fashion. By facilitating the continued use and circulation of previously owned clothing, the business could have contributed to a more sustainable approach to fashion consumption.

The study emphasized that clothing production required significant amounts of resources, including raw materials, water, energy, and other inputs. When garments were discarded before the end of their useful life, the resources used to produce those items could have been considered wasted. Reusing existing clothing could have helped maximize the value of these resources by allowing garments to remain in use for a longer period. This made clothing reuse an important approach to addressing some of the environmental challenges associated with the fashion industry. For Margaux Collections, this finding supported the concept of selling preloved clothing as more than a business activity because the resale of usable garments could have contributed to reducing unnecessary disposal and extending product life.

Cuc and Vidovic (2014) also emphasized that even a relatively short extension of a garment's useful life could have generated environmental benefits. This finding suggested that consumers did not necessarily need to keep clothing indefinitely for reuse to provide value. Allowing a garment to be purchased and used by another consumer could have extended its functional life and reduced the immediate need for disposal. Therefore, the buying and selling of preloved clothing could have represented a practical approach to responsible consumption. For Margaux Collections, this supported the idea of presenting preloved products as part of a circular fashion process in which garments continued to circulate among consumers rather than being discarded after their first ownership period.

The environmental value of preloved fashion could also have been communicated directly through the Margaux Collections Ordering System. Customers could have been informed that purchasing preloved clothing contributed to the continued use of existing garments and could have helped reduce unnecessary textile waste. Short sustainability statements could have been incorporated

into product pages, item descriptions, or other sections of the system to provide customers with relevant information about the environmental significance of their purchases. By communicating these benefits, the system could have helped customers understand that their purchasing decisions had implications beyond personal fashion preferences and could have contributed to more responsible consumption practices.

The study also highlighted the importance of consumer awareness and education in achieving greater environmental benefits from clothing reuse. Consumers needed to understand the importance of extending garment life and properly caring for clothing after purchase. Appropriate garment care could have helped maintain the quality and usability of clothing, allowing products to remain functional for longer periods. This finding was relevant to the proposed Ordering System because product pages could have included basic care instructions together with product descriptions, photographs, condition information, and other relevant details. Providing such information could have helped customers properly maintain their purchases and potentially extend the usable life of the garments.

Including care information could also have strengthened the informational function of the Margaux Collections Ordering System. Instead of functioning solely as a platform for displaying products and receiving orders, the system could have provided useful information that encouraged responsible ownership. For example, customers could have been given appropriate care recommendations based on the material or characteristics of a particular garment. These recommendations could have included general guidance regarding washing, drying, storage, or handling. Such information could have helped customers preserve the quality of their purchases and could have supported the broader sustainability objectives associated with clothing reuse.

Another important implication of the study was that environmental responsibility could have been made more visible through retail and ordering systems. Consumers could have developed a stronger appreciation for preloved products when they understood the environmental benefits associated with reuse. Without adequate information, customers might have viewed secondhand clothing primarily in terms of price or previous ownership.

However, presenting the environmental purpose of clothing reuse could have helped shift this perception and allowed customers to recognize the broader value of their purchases. This was relevant to Margaux Collections because the Ordering System could have communicated the sustainability aspect of the business alongside the products being offered.

The integration of sustainability information into the system could also have contributed to the overall image of Margaux Collections. A business that clearly communicated its commitment to clothing reuse and responsible consumption could have appeared more environmentally conscious and socially responsible. This could have been particularly beneficial in attracting customers who valued sustainable fashion and preferred purchasing products that supported the continued use of existing resources. By incorporating sustainability-related content into product pages and other

system features, Margaux Collections could have established a stronger connection between its business activities and its environmental objectives.

Furthermore, the study supported the idea that technology could have been used to encourage more responsible consumption behavior. A digital ordering system could have made sustainability information accessible to customers at the point when they were evaluating products and preparing to make a purchase. This could have helped connect the practical function of online shopping with the environmental purpose of clothing reuse. Customers could have been provided with both the information needed to make a purchasing decision and an understanding of how their decision contributed to extending the life of a garment. As a result, the system could have supported both commercial and environmental objectives.

Overall, Cuc and Vidovic (2014) provided relevant support for the sustainability component of the Margaux Collections Ordering System by demonstrating the environmental importance of clothing reuse and the extension of garment life. The study emphasized that reusing clothing could have reduced textile waste, conserved resources, and provided environmental benefits by keeping garments in use for longer periods. It also highlighted the importance of consumer education and proper garment care in maximizing the useful life of clothing. These findings supported the inclusion of sustainability information and basic garment care guidance within the proposed ordering system. By presenting preloved products together with clear sustainability messages, detailed product information, and appropriate care recommendations, the Margaux Collections Ordering System could have functioned not only as a platform for purchasing fashion products but also as a tool that encouraged responsible consumption, supported clothing reuse, and contributed to a more sustainable fashion cycle.

Consumers' Intention to Buy Second-Hand Clothing — Koay, Cheah, and Lom (2022)

Koay, Cheah, and Lom (2022) examined the factors that influenced consumers' intention to purchase second-hand clothing in an online shopping environment. Their study focused on the behavioral factors that affected consumers' willingness to consider and purchase preloved clothing through digital platforms. The researchers emphasized that purchasing decisions were not determined solely by product availability or price. Instead, consumers' attitudes toward second-hand clothing, social influence, and perceived behavioral control also played important roles in shaping their purchase intentions. These factors were relevant to the development of the Margaux Collections Ordering System because they helped explain why some consumers were more willing to purchase preloved clothing online while others remained hesitant. The findings suggested that a successful online ordering platform needed to address not only the availability of products but also customers' trust, confidence, attitudes, and overall experience throughout the purchasing process.

The study showed that consumers' attitudes toward second-hand clothing could influence their intention to purchase. When consumers perceived preloved clothing as valuable, affordable,

practical, fashionable, and useful, they could have developed a more positive attitude toward purchasing such products. Positive attitudes could have reduced the negative perceptions sometimes associated with second-hand clothing and could have encouraged consumers to consider preloved products as acceptable alternatives to newly manufactured clothing. This finding was important to Margaux Collections because the presentation of products through the ordering system could have influenced how customers perceived preloved fashion. If products were displayed professionally and accompanied by accurate information and attractive photographs, customers could have developed a more positive impression of the items being offered.

The findings also suggested that the perceived value of second-hand clothing could have influenced consumers' willingness to purchase. Consumers could have viewed preloved products as practical options because they could provide access to fashionable clothing at potentially more affordable prices. However, affordability alone might not have been sufficient to encourage a purchase. Customers could also have considered the condition, appearance, quality, and usefulness of the product before making a decision. Therefore, the Margaux Collections Ordering System needed to provide sufficient information for customers to evaluate the overall value of each item. Clear product descriptions, photographs, prices, sizes, and condition labels could have helped customers determine whether a particular product met their expectations.

Another important factor identified in the study was social influence. Consumers' purchasing decisions could have been affected by the opinions, recommendations, and behaviors of people around them. Friends, family members, online communities, and social trends could have influenced how consumers perceived second-hand clothing and whether they considered purchasing it. When consumers observed other people purchasing and positively discussing preloved fashion, they could have become more open to considering similar products. This finding was relevant to Margaux Collections because it suggested that the online shopping experience could have been strengthened by creating a positive environment around the products and the brand.

Social influence also highlighted the importance of how products were presented within an online platform. Customers could have been more interested in preloved fashion when products appeared attractive, organized, and consistent with current fashion preferences. Although the proposed Margaux Collections Ordering System primarily focused on product ordering and management, the quality of its product presentation could have contributed to customers' overall perceptions of the business. Organized categories, appealing photographs, informative descriptions, and consistent product displays could have helped establish a professional image and could have encouraged customers to view preloved products as fashionable and desirable.

The study further examined perceived behavioral control, which referred to the extent to which consumers felt that performing a particular behavior was easy or difficult. In the context of online second-hand clothing purchases, consumers could have been more likely to complete a

transaction when they believed that the purchasing process was simple, convenient, secure, and manageable. On the other hand, a complicated ordering procedure, unclear product information, or unreliable transaction process could have discouraged customers from continuing with their purchase. This finding was highly relevant to the Margaux Collections Ordering System because it emphasized the importance of usability and ease of navigation.

For the proposed system, perceived behavioral control could have been supported through a simple and organized ordering process. Customers need to be able to browse available products, view detailed information, select their preferred items, provide the necessary order information, and submit their orders without unnecessary difficulty. Clear navigation and understandable system functions could have reduced the effort required to complete a transaction. Features such as order confirmation and order status information could also have helped customers understand what was happening after an order was submitted. By making the purchasing process easier to understand and use, the system could have increased customers' confidence and willingness to complete their transactions.

The study was also relevant to the issue of trust in online second-hand clothing purchases. Unlike physical stores, online platforms did not allow consumers to directly inspect clothing before making a purchase. Customers therefore had to rely on the information provided by the seller to evaluate the product. This could have created uncertainty regarding the item's actual appearance, condition, quality, and suitability. Accurate product descriptions, clear photographs, condition information, and transparent seller details could have helped reduce these concerns. Koay, Cheah, and Lom (2022) therefore supported the idea that trust and confidence were important components of consumers' online purchase intentions.

For Margaux Collections, the findings emphasized the importance of providing complete and reliable product information. Each product listing could have included relevant details such as the product name, brand, size, color, price, condition, description, and photographs. For preloved items, information about visible signs of use or other imperfections could have been especially important because customers needed to establish realistic expectations before ordering. By providing this information clearly, the system could have reduced uncertainty and helped customers make more informed purchasing decisions.

The study also demonstrated that user experience could have played an important role in encouraging consumers to purchase second-hand clothing online. Customers could have been discouraged from completing an order if the platform appeared confusing, difficult to navigate, or unreliable. Therefore, the design of the Margaux Collections Ordering System needed to consider the users' perspective and provide a straightforward purchasing experience. A system that was easy to understand and navigate could have helped customers feel more comfortable throughout the ordering process. This could have supported both customer satisfaction and the likelihood of completing future purchases.

Furthermore, the findings suggested that an online platform for preloved clothing needed to address both the practical and psychological aspects of purchasing. From a practical perspective, customers needed access to accurate product information, convenient navigation, and an efficient ordering process. From a psychological perspective, customers needed to feel confident that the products were accurately represented and that the seller could be trusted. The combination of these factors could have influenced consumers' overall willingness to purchase. This was important for Margaux Collections because the proposed system needed to provide functional features while also creating a sense of reliability and confidence among users.

Overall, the study of Koay, Cheah, and Lom (2022) provided strong support for the development of an online ordering system for preloved clothing. Their findings demonstrated that consumers' intention to purchase second-hand clothing was influenced by their attitudes, social influences, perceived behavioral control, and confidence in the online purchasing process. The study showed that consumers were more likely to consider purchasing preloved clothing when they viewed it positively, felt that the purchasing process was easy and manageable, and received sufficient information to support their decisions. These findings were directly connected to the need for the Margaux Collections Ordering System to provide clear product listings, reliable information, userfriendly navigation, and an efficient ordering process. By addressing both product presentation and buyer confidence, the system could have created a more convenient and trustworthy online shopping experience while supporting customers' willingness to purchase preloved fashion products.

Mindful Consumption of Second-Hand Clothing

Mohammad, Quoquab, and Mohamed Sadom (2021) examined mindful consumption in second-hand clothing and looked at how electronic word-of-mouth, consumer attitude, and engagement affected buying behavior. Their study showed that online opinions and customer interactions had a strong effect on how people view preloved fashion products. This was important in an online ordering system because buyers often depended on the experiences of other users before making a purchase. Since second-hand clothing cannot always be checked physically before buying, the information shared by other consumers became a major guide in decision-making.

The study further suggests that reviews and feedback played an important role in building trust within online shopping environments. When buyers see positive comments, helpful ratings, and active engagement from other users, they are more likely to feel confident about the product and the seller. This made the purchasing process smoother and more reliable. In this sense, the findings supported the need for features that allowed customers to share their opinions and experiences on the platform.

Another important point from the study was that consumer engagement strengthens the relationship between buyers and the marketplace. Engagement helped users feel that they were part of a responsive and active community. In an online clothing platform, this could be shown through comment sections, seller responses, and product feedback tools. These features not only improved communication but also helped reduce uncertainty during the buying process. As a result, buyers might feel more comfortable completing their transactions.

The study was also useful for explaining why interactive features were necessary in a preloved clothing marketplace. Many buyers wanted reassurance that the items they were ordering were worth their money and match the seller's description. Ratings, reviews, and direct communication could help provide that reassurance. These tools allowed the platform to become more transparent and user-friendly. Because of this, the study supported the inclusion of interactive functions in your proposed online ordering system.

Overall, Mohammad, Quoquab, and Mohamed Sodom (2021) provided valuable insight into consumer behavior in second-hand clothing shopping. Their findings showed that attitude, eWOM, and engagement were important in shaping trust and purchase intention. This made the study relevant to an online ordering system for brand-new and preloved clothes because it emphasizes the need for communication and feedback features. It also showed that a well-designed platform should not only display products but also encourage active buyer participation.

Secondhand Clothing Purchase Intentions: Generation Z's Perspective

Herjanto, Amin, Purinton, and Lindle II (2024) examined secondhand clothing purchase intentions among Generation Z consumers and focused on the factors that influenced younger consumers' willingness to purchase preloved clothing. Their study explored how Generation Z perceived secondhand fashion and how considerations such as product value, style, affordability, and sustainability influenced their purchasing decisions. The study was relevant to the development of the Margaux Collections Ordering System because Generation Z represented an important group of consumers who were highly familiar with digital platforms and online shopping. The findings showed that younger consumers were generally more receptive to secondhand fashion when the products were perceived as valuable, fashionable, affordable, and consistent with sustainable consumption. These findings supported the idea that preloved clothing could have been effectively marketed and sold through an online ordering system that provided a convenient, organized, and visually appealing shopping experience.

The study found that Generation Z consumers did not necessarily view secondhand clothing as an inferior or less desirable alternative to brand-new clothing. Instead, many younger consumers could have considered preloved fashion as a practical and meaningful purchasing option. When secondhand clothing items were affordable, stylish, and in good condition, consumers could have perceived them as products that offered considerable value. This finding was important to Margaux Collections because it suggested that preloved products should have been presented according to their positive characteristics rather than being defined primarily by the fact that they had been previously owned. The ordering system could have highlighted the quality, style, condition, brand, and affordability of each product to help customers recognize its value.

Affordability was an important consideration in the purchase of secondhand clothing. Younger consumers could have been attracted to preloved products because they provided opportunities to purchase fashionable clothing at lower prices than brand-new items. This could have been particularly appealing to consumers who wanted to manage their spending while still maintaining their preferred style. However, the study also suggested that affordability alone did not determine purchase intention. Consumers could have considered whether the product was fashionable, useful, visually appealing, and suitable for their personal preferences. Therefore, the Margaux Collections Ordering System needed to present prices together with complete product information so that customers could evaluate the overall value of each item.

The study also showed that style remained an important factor in consumers' interest in secondhand fashion. Clothing could have served as a means of expressing individual identity, personality, and personal preferences. Preloved clothing could have offered unique or distinctive fashion pieces that were not always easily available in conventional retail stores. This characteristic could have increased the appeal of secondhand fashion among younger consumers who valued individuality and personal expression. For Margaux Collections, this finding supported the importance of attractive product photographs and informative descriptions that could have highlighted the unique characteristics of each item. By presenting products in a professional and visually appealing manner, the ordering system could have helped customers identify items that matched their individual fashion preferences.

Product presentation was particularly important in an online environment because customers could not physically examine clothing before purchasing it. In a physical store, customers could have touched the fabric, examined the item closely, and assessed its appearance before making a decision. In an online setting, customers had to depend largely on the photographs and information provided by the seller. Therefore, clear and appealing product images could have helped customers better evaluate the appearance and condition of an item. Detailed product descriptions could also have provided important information regarding size, brand, material, condition, and other relevant characteristics. These features

could have increased customer confidence and could have encouraged younger consumers to consider purchasing preloved products online.

The study was also significant because Generation Z consumers were familiar with digital technology and online shopping platforms. Their familiarity with digital environments could have influenced their expectations regarding convenience, accessibility, and ease of use. Younger consumers could have preferred online marketplaces that allowed them to browse products quickly, compare available choices, and complete purchases without unnecessary difficulty. This finding supported the development of a user-friendly Margaux Collections Ordering System that provided organized product categories, clear navigation, accessible product information, and a simple ordering process.

A system that required too many steps or presented information in a confusing manner could have discouraged younger consumers from completing their purchases. Therefore, the proposed ordering system needed to provide a straightforward process from browsing products to submitting an order. Organized categories could have helped users locate desired items more easily, while search and filtering features could have allowed them to narrow their choices according to their preferences. A convenient checkout process could have further reduced the effort required to complete an order and could have contributed to a more positive online shopping experience.

Another important finding of the study was the influence of sustainability on the purchasing decisions of younger consumers. Generation Z could have demonstrated greater awareness of environmental issues and could have shown interest in consumption practices that supported environmental responsibility. Secondhand clothing was consistent with this preference because the reuse of garments could have extended their useful life and reduced unnecessary textile waste. By purchasing preloved clothing, consumers could have participated in a more circular approach to fashion consumption. This finding was highly relevant to Margaux Collections because the business offered preloved fashion products that could have contributed to the continued circulation and reuse of existing garments.

The sustainability value of secondhand clothing could have been communicated through the Margaux Collections Ordering System. Product pages could have included brief sustainability statements explaining how purchasing preloved clothing could have supported reuse and reduced unnecessary disposal. Providing this information could have helped customers understand that their purchases provided not only personal and economic benefits but also potential environmental benefits. This could have strengthened the appeal of preloved products among consumers who considered sustainability when making purchasing decisions.

The study also suggested that the combination of value, style, and sustainability could have made secondhand clothing particularly appealing to younger consumers. A product that was affordable but lacked style or quality might not have been sufficiently attractive to consumers. Similarly, a fashionable item might not have been selected if its condition or price did not meet customer expectations. Therefore, the ordering system needed to communicate multiple dimensions of product value. By providing information about the price, appearance, condition, brand, and sustainability benefits of each item, Margaux Collections could have allowed customers to evaluate products from different perspectives.

Furthermore, the findings supported the importance of creating a consistent and professional online shopping experience. Younger consumers who were already familiar with digital platforms could have developed expectations regarding how online marketplaces should function. A well-organized platform with attractive visuals, clear product information, and efficient navigation could have met these expectations and could have encouraged customers to remain engaged with the system. In contrast, an outdated, confusing, or incomplete platform could have negatively affected the customer experience. Therefore, the usability and presentation of the Margaux Collections Ordering System could have played an important role in attracting and retaining potential buyers.

Overall, Herjanto, Amin, Purinton, and Lindle II (2024) provided useful insights into Generation Z's purchasing intentions toward secondhand clothing. Their findings demonstrated that younger consumers could have responded positively to preloved fashion when products offered a combination of affordability, quality, style, value, and sustainability. The study was directly relevant to the Margaux Collections Ordering System because it supported the development of an online platform that was organized, visually appealing, convenient, and easy to use. By providing clear product listings, attractive photographs, detailed descriptions, organized categories, efficient ordering functions, and sustainability-related information, the system could have better addressed the preferences of younger consumers. The study therefore supported the potential of preloved clothing in online retail and provided a basis for developing an ordering system that could have attracted consumers who valued fashionable products, affordability, convenience, and responsible consumption.

Cross-Cultural Examination of Apparel Online Purchase Intention: SOR Paradigm Nam, Cho, and Kim (2021)

Nam, Cho, and Kim (2021) examined apparel online purchase intention through the Stimulus-Organism-Response (SOR) paradigm, which explained how external stimuli in an online shopping environment could have influenced consumers' internal perceptions,

feelings, and responses toward products and ultimately affected their purchasing behavior. Their study focused on how different online shopping features, including visual product presentation, website design, and the overall shopping interface, shaped consumers' responses to apparel products. The study was highly relevant to the development of the Margaux Collections Ordering System because it demonstrated that consumers' purchase intentions could have been influenced not only by the products being offered but also by how those products were presented and how easily customers could interact with the online platform.

The findings showed that the manner in which clothing products were displayed online could have influenced consumer interest and purchase intention. When apparel products were presented through clear and attractive images, complete descriptions, and organized layouts, consumers could have developed more positive perceptions of the products and the platform. A visually appealing presentation could have helped customers understand the characteristics of an item and could have encouraged them to continue browsing. This finding was particularly important for Margaux Collections because customers purchasing brand-new and preloved clothing could have relied heavily on the visual and written information provided through the ordering system. Since customers could not physically inspect the items before purchasing, the quality of product presentation could have played an important role in their decision-making.

Clear product photographs could have helped customers evaluate the appearance, color, design, and overall condition of clothing items. This was especially important for preloved products because customers needed to identify visible signs of use and assess whether the actual condition of the item matched their expectations. Detailed descriptions could have provided additional information about the product's size, brand, material, condition, and other relevant characteristics. By combining clear photographs with complete product descriptions, the Margaux Collections Ordering System could have reduced uncertainty and provided customers with a more reliable basis for making purchasing decisions.

The study also suggested that poor product presentation could have negatively affected consumers' interest in an online shopping platform. Incomplete information, unclear images, inconsistent layouts, or disorganized product listings could have created an impression of low professionalism and could have caused customers to lose interest before completing a purchase. This finding emphasized the importance of maintaining consistent product presentation within the Margaux Collections Ordering System. Each product needed to have a similar structure for displaying photographs, descriptions, prices, sizes, and condition information. Consistency could have made the platform easier to understand and could have strengthened the professional image of the business.

Another important finding was the role of website design in shaping consumer behavior. A shopping platform that was easy to navigate could have allowed users to move smoothly from browsing products to viewing product details and eventually completing their orders. When the purchasing process was simple and convenient, consumers could have experienced less frustration and could have been more likely to continue their shopping activities. This was especially relevant to Margaux Collections because customers could have browsed several clothing items before selecting the product they wanted to purchase. A clear navigation structure could have helped customers move between product categories without unnecessary difficulty.

A well-designed interface could therefore have contributed to an improved overall shopping experience. The Margaux Collections Ordering System could have used clearly labeled menus, organized categories, readable product information, and straightforward navigation to make the platform easier to use. These design elements could have reduced the amount of effort customers needed to spend when searching for products. A simple interface could also have helped users understand how to place orders and access important information. This could have been particularly beneficial for customers with different levels of experience in online shopping.

The study further indicated that online shopping features could have influenced consumer responses and supported purchasing decisions. Functions such as product filters, categories, search options, and a convenient checkout process could have helped users make decisions more quickly and efficiently. These features could have reduced the effort required to locate appropriate products, particularly when an online store contained multiple categories or a large number of listings. For the Margaux Collections Ordering System, these findings supported the inclusion of features that could have allowed customers to browse and organize available products according to their preferences.

Product categories could have been useful for separating and organizing brand-new and preloved clothing items or for grouping products according to their type. Search and filtering functions could have allowed customers to narrow their choices based on characteristics such as product category, size, price, brand, or other available attributes. These functions could have helped customers locate suitable products more efficiently and could have reduced the time spent searching through unrelated listings. As a result, customers could have experienced a more convenient and organized shopping process.

The checkout process was another important feature supported by the study. Once customers had selected the products they wanted to purchase, they needed to be able to complete their orders through a clear and understandable process. A complicated checkout procedure could have caused confusion or frustration and could have resulted in customers abandoning their purchases. Therefore, the Margaux Collections Ordering System needed to provide a straightforward ordering process that clearly communicated

the information required from customers. Order confirmation and order status information could also have helped customers understand whether their orders had been successfully submitted and processed.

The study was also relevant to the development of customer trust. An organized and professionally designed online platform could have created a stronger impression of reliability. Customers could have interpreted clear product information, consistent layouts, functional navigation, and a smooth ordering process as indicators that the business was organized and dependable. This was particularly important for preloved fashion because customers could have experienced greater uncertainty when purchasing previously owned items. A professional interface combined with accurate product information could have helped reduce these concerns and could have encouraged customers to feel more confident about completing their purchases.

Furthermore, the SOR paradigm provided a useful theoretical perspective for understanding how the design of the Margaux Collections Ordering System could have influenced customer behavior. In an online shopping environment, website features and product presentation could have served as external stimuli. These stimuli could have influenced consumers' perceptions and feelings toward the products and the platform, which could then have affected their behavioral responses, including their willingness to continue browsing or complete a purchase. Therefore, the ordering system's design could have played an important role in shaping the overall customer experience and purchase intention.

Overall, Nam, Cho, and Kim (2021) provided useful evidence that interface design, product presentation, and online shopping features could have influenced consumers' responses and purchase intentions when purchasing apparel online. Their findings demonstrated that consumers' behavior could have been affected by what they saw and experienced while interacting with an online shopping platform. These findings were directly relevant to the Margaux Collections Ordering System because they supported the need for clear product listings, high-quality images, complete descriptions, organized product categories, easy navigation, useful filtering options, and a convenient checkout process. By incorporating these features, the proposed system could have provided a more efficient, professional, and user-friendly shopping experience for customers purchasing both brand-new and preloved clothing. The study therefore supported the development of an online ordering system that could have attracted and retained customers by combining effective product presentation with convenient and reliable system functionality.

Thrift Shopping for Clothes: To Treat Self or Others? Park, Kwon, Zaman, and Song (2020)

Park, Kwon, Zaman, and Song (2020) explored consumers' motivations for thrift shopping and examined the different reasons why individuals purchased secondhand clothing. Their study showed that consumers engaged in thrift shopping for both practical and personal reasons. Some consumers were primarily motivated by affordability and the opportunity to obtain clothing at lower prices, while others were influenced by factors such as self-expression, personal identity, satisfaction, and the enjoyment associated with discovering unique fashion items. These findings were relevant to the development of the Margaux Collections Ordering System because they demonstrated that the appeal of preloved clothing extended beyond its lower price. Consumers could have been attracted to secondhand fashion because it provided economic benefits while also allowing them to express their personal preferences and discover items that matched their individual styles.

The study indicated that affordability was one of the important motivations behind thrift shopping. Consumers could have viewed second-hand clothing as a practical alternative to purchasing brand-new apparel, particularly when they wanted to manage their spending while still obtaining fashionable products. For some buyers, thrift shopping provided an opportunity to acquire clothing at reasonable prices without necessarily sacrificing style or quality. This finding was relevant to Margaux Collections because the ordering system could have presented product prices clearly and allowed customers to evaluate the value of each item before making a purchase. By providing complete information about product prices, conditions, sizes, brands, and other relevant characteristics, the system could have helped customers determine whether a product matched their budget and expectations.

However, the study also demonstrated that consumers did not view second-hand clothing solely as an inexpensive alternative. Some buyers were motivated by the opportunity to find distinctive and uncommon fashion pieces that were not easily available in conventional retail stores. Thrift shopping could have allowed consumers to discover products that reflected their personal tastes and individual identities. This aspect of second-hand fashion suggested that preloved products could have possessed symbolic and emotional value in addition to their practical value. For Margaux Collections, this finding supported the importance of presenting products in an appealing manner that highlighted their unique characteristics rather than focusing exclusively on their price.

Product presentation could have been particularly important in an online environment because customers could not physically browse through clothing racks or personally examine each item. In a physical thrift store, consumers could have discovered unique products through direct searching and exploration. In contrast, an online platform needed to recreate some of this discovery experience through photographs, product categories, descriptions, and organized listings. The Margaux Collections Ordering System could have supported this by providing attractive product images and detailed descriptions that allowed customers to recognize the unique qualities of individual items. This could have helped make the online shopping experience more engaging and could have encouraged customers to explore additional products.

Another important finding from the study was the role of satisfaction in thrift shopping behavior. Consumers could have experienced enjoyment from searching for and discovering items that matched their personal preferences or needs. The process of finding a desirable item could itself have provided a sense of accomplishment and satisfaction. This suggested that thrift shopping was not always viewed as a simple transaction but could also have been an enjoyable experience for consumers. This finding was relevant to the Margaux Collections Ordering System because the organization and usability of the platform could have influenced how easily customers found products that interested them.

A well-organized online platform could have improved the product discovery process by allowing customers to browse available items efficiently. Clear categories, search functions, filtering options, and organized product listings could have helped customers locate products according to their preferences. When users could easily find items that matched their needs, they could have remained engaged with the platform for a longer period and could have been more likely to complete a purchase. Therefore, the ordering system needed to provide a balance between simplicity and accessibility while still allowing customers to explore the available products.

The findings also suggested that the enjoyment associated with discovering unique items could have encouraged consumers to continue purchasing second-hand clothing. When customers had positive experiences while searching for and purchasing thrifted products, they could have developed stronger interest in returning to the platform. For Margaux Collections, this indicated that the ordering system could have benefited from regularly updated product listings and organized displays of newly available items. Presenting new products in an accessible and appealing manner could have encouraged customers to revisit the platform and explore recently added inventory.

The study was also relevant to the growing acceptance of second-hand clothing in online markets. As consumers became increasingly familiar with digital shopping platforms, they could have become more comfortable purchasing products without physically visiting a store. However, purchasing preloved clothing online could still have involved concerns regarding product condition, accuracy of information, and reliability of the seller. These concerns could have been addressed through a trustworthy and transparent ordering system. Accurate product descriptions, clear photographs, condition information, and organized listings could have reduced uncertainty and helped customers make more informed purchasing decisions.

A reliable transaction process could also have supported consumer confidence. Customers needed to understand how orders were submitted, confirmed, and processed. If the ordering process was simple and transparent, customers could have experienced less uncertainty when purchasing products online. For Margaux Collections, this supported the need for an ordering system that provided clear instructions, organized order information, and appropriate order status updates. These functions could have contributed to a more professional shopping experience and could have encouraged customers to complete their purchases.

The study further indicated that the motivations behind thrift shopping could have varied among consumers. Some buyers could have been primarily interested in saving money, while others could have been motivated by personal identity, uniqueness, enjoyment, or the satisfaction of finding desirable items. Therefore, an online marketplace for preloved clothing needed to accommodate different types of consumer needs. The Margaux Collections Ordering System could have addressed these differences by providing comprehensive product information and an appealing presentation that emphasized both the practical and personal value of the products. Customers could have evaluated items based on price and condition while also considering style, uniqueness, and suitability for their personal preferences.

Overall, Park, Kwon, Zaman, and Song (2020) provided valuable insights into the different motivations that influenced consumers to purchase thrifted clothing. Their findings demonstrated that second-hand fashion could have appealed to consumers for reasons related to affordability, personal identity, uniqueness, satisfaction, and the enjoyment of discovering desirable products. These findings were relevant to the Margaux Collections Ordering System because they supported the idea that preloved clothing should have been presented as valuable and meaningful fashion products rather than simply as low-cost alternatives. The study also emphasized the importance of providing an enjoyable and convenient shopping experience. By incorporating clear product categories, attractive photographs, detailed descriptions,

reliable transaction functions, and easy browsing, the Margaux Collections Ordering System could have addressed different consumer needs and could have strengthened customer interest in second-hand fashion. Thus, the study supported the development of a user-friendly, organized, and trustworthy online marketplace that could have made the purchase of preloved clothing more convenient and appealing to consumers.

Trust Management in Electronic Marketplaces — Kim and Ahn (2007)

Kim and Ahn (2007) examined trust management in electronic marketplaces and investigated how trust could have influenced interactions and transactions between buyers and sellers in online environments. In electronic marketplaces, buyers and sellers could have completed transactions without personally knowing or meeting each other. Because of this separation, trust could have played an important role in determining whether customers were willing to proceed with an online purchase. The study focused on how buyers developed trust based on their experiences with websites and sellers, as well as on factors such as website reputation, seller performance, transaction experiences, and the reliability of information presented online. The study was relevant to the development of the Margaux Collections Ordering System because the proposed system also involved online interactions between customers and the business, where buyers needed to feel confident before placing orders and providing payment.

The researchers explained that buyers were more likely to complete transactions when they believed that the online marketplace was secure, reliable, and professionally managed. When customers had positive experiences with a website or seller, their confidence could have increased and encouraged them to make future purchases. In contrast, negative experiences could have reduced trust and discouraged customers from completing transactions or returning to the platform. This finding was important to Margaux Collections because customers purchasing clothing online could have experienced uncertainty regarding product quality, seller reliability, payment procedures, and order fulfillment. A well-organized ordering system could have helped address these concerns by providing clear information and reliable transaction procedures.

The study also emphasized the importance of website reputation in developing consumer trust. Buyers could have evaluated the credibility of an online marketplace based on the information and experiences associated with the platform. A website that appeared organized, secure, and consistent could have created a more positive impression among users. For Margaux Collections, this supported the need for a professional and user-friendly ordering system. Clear product listings, organized categories, complete descriptions, accessible customer information, and consistent system functions could have helped establish a stronger sense of reliability among customers.

Another important aspect identified by Kim and Ahn (2007) was seller performance. Buyers could have developed trust when sellers consistently provided accurate information, communicated properly, processed orders efficiently, and delivered products according to customer expectations. In the context of Margaux Collections, seller performance could have been reflected through accurate product descriptions, proper handling of orders, timely communication, and reliable fulfillment of customer purchases. These practices could have helped establish a positive transaction history and could have encouraged customers to purchase from the business again.

The study further highlighted the importance of reputation systems in electronic marketplaces. Seller ratings, customer reviews, transaction histories, and feedback could have provided potential buyers with additional information when evaluating whether to purchase from a seller. These mechanisms could have reduced uncertainty because customers could have considered the experiences of previous buyers before making their own decisions. For the Margaux Collections Ordering System, the inclusion of customer ratings and reviews could have allowed buyers to share their experiences and could have provided future customers with useful information about the products and services offered by the business.

Transaction records could also have contributed to trust management. A properly maintained record of completed orders could have demonstrated that the business had successfully handled previous transactions. These records could have supported accountability and could have helped the owner monitor the status of customer orders. For Margaux Collections, maintaining organized transaction records could have been especially useful because the business handled both brand-new and preloved clothing products, each of which could have had unique product information and limited availability. Accurate records could have reduced errors and helped ensure that orders were properly processed.

The findings were also directly connected to payment verification within the Margaux Collections Ordering System. Since the business relied on GCash for advance payment, verifying that a customer had actually completed a payment could have been an important part of maintaining transaction security and trust. Customers needed assurance that their payments were properly recorded, while the seller needed a reliable method of confirming the payment before processing an order. The reference number verification incorporated into the system's checkout process could have addressed this concern by providing a specific transaction identifier that could have been checked against the submitted order information.

The payment verification process could have reduced the need to depend solely on screenshots or informal messages sent through Messenger. Although screenshots could have provided visual evidence of payment, they might not have been sufficient as a structured transaction record. By requiring customers to provide a valid GCash reference number, the system could have created a more organized method for identifying and verifying payments. This could have helped the seller distinguish between confirmed and unconfirmed orders and could have

reduced the possibility of errors during transaction processing. It could also have provided customers with greater confidence that their payments and orders were being properly recorded.

The study's findings also suggested that trust could have developed through repeated positive experiences. When customers successfully completed transactions and received products that matched the information provided by the seller, they could have developed greater confidence in the business. This could have increased the possibility of repeat purchases and positive recommendations. For Margaux Collections, maintaining accurate product information, reliable payment verification, and organized order processing could have contributed to positive customer experiences and strengthened the business's reputation over time.

Furthermore, the study supported the importance of transparency in online transactions. Customers needed sufficient information to understand the products they were purchasing and the procedures involved in completing their orders. In the case of preloved clothing, transparency could have been particularly important because customers needed to know the actual condition of each item before purchasing. Providing accurate photographs, descriptions, condition labels, prices, and relevant product information could have helped reduce uncertainty. When combined with secure payment verification and organized transaction records, these features could have contributed to a more trustworthy ordering environment.

Overall, Kim and Ahn (2007) provided important support for the development of the Margaux Collections Ordering System by demonstrating that trust management was essential in electronic marketplaces. Their study showed that website reputation, seller performance, transaction experience, reliable information, customer feedback, and reputation systems could have influenced buyers' willingness to participate in online transactions. These findings were directly applicable to Margaux Collections because customers needed to trust both the ordering platform and the seller before purchasing brand-new or preloved clothing. Features such as seller information, ratings, reviews, transaction records, accurate product details, and verified customer accounts could have helped strengthen this trust. In addition, the GCash reference number verification feature could have provided a more reliable method of confirming advance payments and maintaining accurate transaction records. Therefore, the study supported the development of an ordering system that prioritized transparency, security, accountability, and customer confidence throughout the entire online purchasing process.

Online Feedback Mechanisms in Electronic Marketplaces Dellarocas (2003)

Dellarocas (2003) examined the role of online feedback mechanisms in electronic marketplaces and explained how feedback could have helped reduce uncertainty in online transactions. Online feedback generally included ratings, reviews, comments, and evaluations provided by customers after completing a transaction. The study focused on the importance of feedback in online environments where buyers could not physically inspect products or directly observe the behavior and reliability of sellers before making a purchase. Because of this limitation, consumers could have relied on information provided by previous customers to evaluate the credibility of sellers and the quality of their products and services. The study was relevant to the development of the Margaux Collections Ordering System because it highlighted the importance of establishing trust and reducing uncertainty in an online purchasing environment.

Dellarocas (2003) explained that online feedback could have served as an important source of information for potential buyers. In a traditional physical store, customers could have directly examined products, interacted with sellers, and assessed the quality of the items before making a purchase. In contrast, online buyers had limited opportunities to personally verify the product or observe the seller's behavior. As a result, feedback from previous customers could have provided additional information that helped potential buyers evaluate whether a seller was reliable and whether the products being offered were accurately represented. This was particularly relevant to Margaux Collections because customers purchasing preloved clothing online could have experienced uncertainty regarding product condition, quality, appearance, and the accuracy of product descriptions.

The study found that positive feedback could have increased the perceived trustworthiness of sellers. A seller with a history of positive ratings and favorable customer comments could have appeared more reliable than a seller with no transaction history or several negative evaluations. Feedback could have therefore served as a form of social evidence that helped potential customers assess the credibility of an online seller. For Margaux Collections, this suggested that customer reviews and ratings could have been valuable features within the proposed ordering system. Customers who had successfully completed purchases could have shared their experiences regarding product quality, accuracy, seller communication, and the overall ordering process. Future customers could then have used this information when evaluating whether to purchase from the business.

Online feedback could also have helped improve seller accountability. Dellarocas (2003) showed that sellers could have been encouraged to provide better products and services when they knew that their performance could be evaluated by customers. Sellers could have become more careful in describing products accurately, communicating with buyers, preparing orders, packaging products, and completing deliveries. This created an incentive for sellers to maintain

good service because negative customer experiences could have affected their reputation within the marketplace. For Margaux Collections, such a mechanism could have encouraged consistent and responsible handling of customer orders.

This finding was particularly important for a business selling preloved clothing because customers needed accurate information about the actual condition of each item. Sellers could have been expected to disclose relevant imperfections, signs of wear, stains, damage, or other characteristics that could have affected the customer's purchasing decision. A feedback mechanism could have allowed buyers to indicate whether the received product matched the description and photographs provided in the ordering system. Such feedback could have encouraged the business owner to maintain accurate product listings and could have helped identify areas where product information needed improvement.

The study also suggested that feedback could have reduced information asymmetry between buyers and sellers. In online transactions, sellers generally possessed more information about their products than potential buyers because they had direct access to the items. Buyers, however, had to depend on the information provided through the platform. Feedback from previous customers could have helped balance this difference by providing additional perspectives based on actual purchasing experiences. For Margaux Collections, customer reviews could have provided future buyers with information that was not necessarily available through the seller's product description alone.

A review and rating feature could have been particularly useful for evaluating the overall customer experience. Customers could have provided feedback regarding product quality, product accuracy, communication, order processing, and delivery performance. These evaluations could have given future buyers a broader understanding of what they could expect when ordering from Margaux Collections. At the same time, the business owner could have used customer feedback to identify strengths and weaknesses in the ordering process. Positive feedback could have indicated areas that were being handled effectively, while constructive or negative feedback could have identified aspects that required improvement.

Although Margaux Collections did not operate on the same scale as a large electronic marketplace, the underlying issue of trust and uncertainty could still have been present. A firsttime customer who interacted with the business online could have had limited information about the seller's previous transactions and customer experiences. Without an established history, potential buyers could have been more hesitant to complete a purchase. A system that organized product information, order records, customer transactions, and feedback in one place could have provided a basic structure for building credibility and demonstrating that the business was actively managing its transactions.

The findings of Dellarocas (2003) also supported the importance of maintaining accurate and accessible transaction records. When completed transactions and customer evaluations were

properly documented, the business could have developed a record of its performance over time. This information could have helped demonstrate the reliability of the business and could have provided the owner with useful information for improving operations. In the context of the Margaux Collections Ordering System, keeping organized records of orders and customer feedback could have contributed to better monitoring of transactions and improved accountability.

Furthermore, feedback could have supported the continuous improvement of the Margaux Collections Ordering System. Customer comments could have revealed difficulties experienced during product browsing, ordering, payment, communication, or delivery. The business owner could have used these observations to improve the system and make the purchasing process more convenient. For example, if customers repeatedly indicated that product descriptions were incomplete, the owner could have improved the information required for product listings. If customers experienced difficulty locating products, the system's categories or navigation could have been adjusted. In this way, feedback could have functioned not only as a trustbuilding mechanism but also as a source of information for system and service improvement.

Overall, Dellarocas (2003) provided important support for the development of the Margaux Collections Ordering System by demonstrating how online feedback mechanisms could have reduced uncertainty, strengthened trust, and encouraged better seller performance in electronic marketplaces. The study showed that ratings, reviews, comments, and customer evaluations could have helped potential buyers make more informed decisions while encouraging sellers to maintain accurate product information and reliable services. These findings supported the inclusion of a review and rating feature in the proposed system, where customers could have evaluated product quality, accuracy, seller communication, and overall transaction experience. Even though Margaux Collections operated on a smaller scale than large electronic marketplaces, the same principles of trust, transparency, accountability, and customer feedback could have applied. By providing a structured way to record transactions and customer experiences, the Margaux Collections Ordering System could have strengthened buyer confidence, supported service improvement, and contributed to a more reliable and trustworthy online shopping environment.

Trust, Perceived Usefulness, and Perceived Ease of Use in Online Shopping — Gefen, Karahanna, and Straub (2003)

Gefen, Karahanna, and Straub (2003) examined the roles of trust, perceived usefulness, and perceived ease of use in online shopping and investigated how these factors influenced consumers' willingness to use electronic-commerce websites. Their study incorporated concepts from the Technology Acceptance Model (TAM), which explained that users were more likely to accept and use a technology when they perceived it as useful and easy to operate. The researchers also examined trust because online transactions involved uncertainty and risk, particularly when customers purchased products from sellers whom they did not personally know. The study was relevant to the development of the Margaux Collections Ordering System

because the proposed system also required customers to interact with an online platform, evaluate products, provide information, and complete transactions without directly interacting with the seller in a physical store.

The study found that customers were more willing to use online shopping platforms when they perceived the system as useful in helping them accomplish their purchasing tasks efficiently. Perceived usefulness referred to the extent to which users believed that a system could help them perform an activity more effectively or conveniently. In an online shopping environment, a useful system could have allowed customers to search for products, examine product information, place orders, and monitor transactions without unnecessary effort. This finding was important to Margaux Collections because the proposed ordering system needed to provide functions that directly supported the customer's shopping process rather than simply displaying products.

The researchers also emphasized the importance of perceived ease of use. Customers could have been more willing to use an online shopping system when its functions were understandable and did not require excessive effort to operate. Simple navigation, clearly labeled menus, accessible information, and straightforward ordering procedures could have reduced user confusion and encouraged customers to continue using the platform. For the Margaux Collections Ordering System, this finding supported the use of a simple and organized interface that could have allowed customers to browse products and complete orders without relying heavily on separate communication channels.

The study suggested that online shopping platforms needed to provide clear menus and understandable instructions. When users could easily identify where to find products, view product information, and proceed with an order, the shopping experience could have become more convenient. On the other hand, complicated pages, unclear instructions, and unnecessary steps could have caused frustration and could have increased the possibility of customers abandoning their purchases. This was particularly important for Margaux Collections because the ordering system was intended to simplify the existing purchasing process and reduce dependence on scattered chat exchanges.

Product browsing was another area where perceived ease of use could have influenced the customer experience. The proposed system could have organized products into categories that allowed customers to locate items more efficiently. Separating brand-new products from preloved products could have helped users understand the available inventory and could have allowed them to browse according to their preferences. Search and filtering functions could also have helped customers narrow down their choices based on relevant product characteristics. These features could have reduced the time and effort required to find suitable clothing items.

The study also supported the importance of detailed product displays. Customers needed sufficient information to evaluate products before making a purchasing decision. Product photographs,

prices, sizes, descriptions, and condition information could have helped users understand what they were considering purchasing. This was especially important for preloved clothing because customers could not physically examine the products before placing an order. By providing complete and accessible product information, the Margaux Collections Ordering System could have helped customers make more informed decisions and could have reduced uncertainty during the purchasing process.

Trust was another major factor examined by Gefen, Karahanna, and Straub (2003). Online shopping involved a degree of uncertainty because customers could not directly observe the seller, inspect the product physically, or immediately confirm whether the transaction would be completed as expected. Because of this, customers needed to believe that the website and seller were reliable. A trustworthy online system could have encouraged customers to provide information and proceed with their purchases. For Margaux Collections, trust could have been supported through accurate product information, organized transaction records, clear ordering procedures, and reliable payment verification.

The study's findings were particularly relevant to the payment and checkout process of the Margaux Collections Ordering System. A confusing checkout process could have created uncertainty and could have discouraged customers from completing an order. Therefore, the system needed to guide users through a limited number of clear and understandable steps. The order summary could have displayed important information such as the selected products, quantities, prices, and other relevant order details before the customer submitted the order. This could have allowed users to review their information and could have reduced errors before the transaction was finalized.

The design priorities followed during the development of the Margaux Collections Ordering System were consistent with the principles emphasized by Gefen, Karahanna, and Straub (2003). The checkout flow and product browsing pages were kept intentionally simple so that customers could understand the process without requiring extensive assistance. Product categories separated brand-new items from preloved items, allowing customers to browse the inventory according to their preferences. The ordering process was also structured into a small number of clear steps instead of relying on scattered conversations through messaging platforms. This approach could have reduced confusion and could have provided customers with a more organized purchasing experience.

The study also supported the idea that usefulness and ease of use could have worked together in influencing technology acceptance. A system could have contained many functions, but if users found those functions difficult to understand, they might not have considered the system useful. Similarly, a simple system that did not provide the necessary functions might not have adequately supported customers. Therefore, the Margaux Collections Ordering System needed to balance functionality with simplicity. Features such as product categories, search and filtering options, product photographs, prices, sizes, condition

information, order summaries, and payment verification could have provided useful functions while maintaining a straightforward user experience.

Furthermore, the study provided support for reducing unnecessary communication between customers and the seller. When customers could independently access product information and complete basic ordering steps through the system, they could have experienced greater convenience. This could have reduced the need to repeatedly ask about product availability, prices, sizes, or order details through Messenger. The system could have centralized these details and made them accessible to customers in one location. As a result, the ordering process could have become more efficient for both customers and the business owner.

Overall, Gefen, Karahanna, and Straub (2003) provided important theoretical support for the development of the Margaux Collections Ordering System by demonstrating the importance of trust, perceived usefulness, and perceived ease of use in online shopping. Their findings showed that customers were more likely to use an online platform when they believed that it was useful, easy to operate, and trustworthy. These principles were directly applicable to the proposed system because customers needed to browse products, evaluate information, place orders, and complete transactions with minimal difficulty. The study therefore supported the inclusion of organized product categories, search and filtering functions, clear product displays, accessible prices and sizes, understandable instructions, and a simple checkout process. By applying these principles, the Margaux Collections Ordering System could have provided a more convenient, efficient, and trustworthy shopping experience for customers purchasing both brand-new and preloved clothing.

Consumer Acceptance of Electronic Commerce — Pavlou (2003)

Pavlou (2003) investigated consumer acceptance of electronic commerce and examined how trust and perceived risk influenced consumers' willingness to purchase products through online platforms. The study focused on the uncertainties that customers could have experienced when conducting transactions electronically, particularly when they had limited direct knowledge about the seller or the product. Online shoppers could have faced different forms of perceived risk, including payment fraud, misuse of personal information, inaccurate product descriptions, non-delivery of purchased items, and uncertainty regarding the fulfillment of orders. These risks could have discouraged consumers from completing online transactions, especially when they lacked confidence in the reliability and security of the website. The study was relevant to the Margaux Collections Ordering System because customers needed to feel secure and confident when purchasing both brand-new and preloved clothing through the proposed online platform.

The study found that trust played an important role in reducing the negative effects of perceived risk. Customers could have been more willing to complete an online transaction when they

believed that the website was reliable, their information was handled appropriately, and the seller would fulfill the order as promised. Trust could therefore have influenced whether customers were willing to provide personal information, submit an order, and proceed with payment. For Margaux Collections, this finding emphasized the importance of developing an ordering system that appeared organized, transparent, and dependable. The system needed to provide customers with enough information to understand the products, prices, payment requirements, and status of their orders.

Pavlou (2003) also emphasized the importance of security and reliability in electronic-commerce systems. Customers needed to understand what they were purchasing, how much they were expected to pay, and what would happen after an order was submitted. When important transaction information was unclear or difficult to access, customers could have experienced greater uncertainty. This was particularly important when purchasing clothing online because customers could not physically inspect the products before completing the transaction. For Margaux Collections, providing clear product information and transparent transaction details could have helped reduce uncertainty and could have increased customer confidence.

The study supported the importance of a clear and organized checkout process. A well-designed checkout feature could have allowed customers to review their selected products, quantities, prices, and other relevant order information before submitting the transaction. This could have reduced the possibility of ordering errors and could have helped customers confirm that the information they provided was correct. In addition, a clear order confirmation could have informed customers that their order had been successfully received by the system. These functions could have made the online purchasing process more transparent and could have reduced hesitation among buyers.

Payment verification was another important consideration supported by the study. Since Margaux Collections relied on GCash for advance payment, the system needed a reliable way to identify and verify customer payments. The inclusion of a GCash reference number in the checkout process could have provided a structured method for connecting the payment to the corresponding order. Instead of relying entirely on screenshots or informal messages sent through Messenger, the system could have recorded the reference number together with the order information. This could have helped the business owner verify payments more efficiently and could have reduced possible confusion between different transactions.

The ordering process of the Margaux Collections system was therefore structured around clear confirmation at different stages of the transaction. Customers could have received information regarding their order after submitting it, while payment verification could have indicated whether the required payment had been properly recorded. The system could also have provided information regarding the status of pickup or delivery. By allowing customers to see what they had ordered and understand the current stage of their transaction, the system could have reduced the

uncertainty associated with online purchasing. This approach was consistent with Pavlou's (2003) findings regarding the importance of trust and reduced perceived risk in electronic commerce.

Furthermore, maintaining accurate transaction records could have supported both customer confidence and business accountability. When order information, payment details, and transaction status were properly recorded, the business could have more easily track the progress of each transaction. Customers could have also received clearer information about their purchases. This could have reduced misunderstandings and could have helped establish a more reliable relationship between the business and its customers.

Overall, Pavlou (2003) provided important support for the development of the Margaux Collections Ordering System by demonstrating that trust and perceived risk could have significantly influenced consumers' willingness to participate in online transactions. The study showed that customers could have been more willing to purchase online when they believed that the platform was secure, reliable, and transparent. These findings supported the use of clear order summaries, transparent pricing, payment verification, transaction records, and order status updates in the proposed system. By providing customers with clear information from order placement through payment verification and pickup or delivery, the Margaux Collections Ordering System could have reduced uncertainty and created a more secure and trustworthy online purchasing experience.

Consumer Motivations for Purchasing Second-Hand Products Guiot and Roux (2010)

Guiot and Roux (2010) examined consumer motivations for purchasing second-hand products and investigated why consumers chose used or preloved products instead of purchasing brandnew alternatives. Their study identified several motivations associated with secondhand shopping, including lower prices, product uniqueness, nostalgia, environmental concerns, and product availability. The findings were relevant to the Margaux Collections Ordering System because they demonstrated that consumers could have purchased preloved clothing for a variety of reasons rather than simply because the products were less expensive. Second-hand products could have provided economic benefits while also offering unique, meaningful, and environmentally responsible consumption choices.

The study showed that lower prices could have been an important motivation for purchasing second-hand products. Consumers could have viewed used products as practical alternatives that allowed them to obtain desired items while spending less money. This was relevant to Margaux Collections because customers could have compared preloved clothing with brand-new products based on their prices, condition, and overall value. The ordering system could have supported this

decision-making process by displaying prices clearly and providing sufficient information about the condition and characteristics of each product.

However, Guiot and Roux (2010) also demonstrated that price was not the only reason consumers purchased second-hand products. Product uniqueness was another important motivation. Consumers could have been attracted to items that were uncommon or difficult to find in conventional retail stores. Preloved fashion could have allowed buyers to discover distinctive clothing pieces that reflected their individual preferences and styles. This finding supported the idea that Margaux Collections could have presented preloved clothing as unique fashion products rather than simply as lower-priced alternatives to brand-new items.

Nostalgia was also identified as a possible motivation for second-hand consumption. Some consumers could have developed emotional connections with certain products, styles, brands, or periods represented by second-hand items. These emotional associations could have increased the perceived value of a product beyond its practical function. For Margaux Collections, this finding suggested that detailed product descriptions could have helped communicate characteristics that made certain items interesting or distinctive. Providing information about the brand, style, appearance, or other notable characteristics could have helped customers appreciate the personal and symbolic value of preloved products.

Environmental concerns also played an important role in second-hand consumption. Consumers could have purchased used products because they wanted to support practices that reduced waste and extended the useful life of existing products. In the context of clothing, purchasing preloved garments could have contributed to the continued use of clothing items rather than their immediate disposal. This finding was relevant to Margaux Collections because the business offered preloved clothing that could have supported reuse and more responsible consumption. The ordering system could have communicated this value by providing sustainability-related information within product descriptions or other sections of the platform.

Another important motivation identified by the study was product availability. Consumers could have turned to second-hand marketplaces when certain products were difficult to find through conventional retail channels. This could have increased the value of online preloved marketplaces because digital platforms could have allowed customers to browse a wider variety of products without being limited to the inventory of a single physical store. For Margaux Collections, maintaining updated and organized product listings could have helped customers identify available items more easily.

The study also suggested that second-hand marketplaces needed to provide sufficient information for consumers to evaluate product condition and value. This was particularly important because second-hand products had previously been owned and could have shown different levels of wear. Unlike brand-new products, preloved clothing could have had visible defects, stains, fading, loose threads, or other signs of previous use. Customers therefore needed accurate information before

making a purchasing decision. Providing incomplete or misleading information could have increased uncertainty and could have negatively affected customer trust.

This finding directly supported the product listing structure of the Margaux Collections Ordering System. Preloved items could have been listed with clear information regarding their condition, size, price, brand, and any visible defects. Photographs could also have been provided to allow customers to visually examine the products before placing an order. Clearly identifying imperfections could have helped customers establish realistic expectations about the item they were purchasing. This level of transparency could have reduced misunderstandings and could have helped customers make fair comparisons between preloved and brand-new products.

The need for transparency was particularly important because Margaux Collections offered brandnew and preloved products within the same ordering system. Customers could have compared the two categories based on factors such as price, condition, style, and overall value. If the condition of preloved products was not clearly identified, customers could have misunderstood the differences between the two categories. Therefore, providing visible and consistent condition labels could have helped customers make more informed purchasing decisions.

The findings also supported the use of detailed product descriptions as part of the online shopping experience. Since customers could not physically inspect preloved clothing before purchasing, written descriptions and photographs could have served as substitutes for physical examination. Descriptions that clearly identified the item's condition, size, material, brand, and visible defects could have helped reduce uncertainty. These details could also have helped customers determine whether a product met their expectations and purchasing requirements.

Overall, Guiot and Roux (2010) provided valuable support for the inclusion of preloved clothing in the Margaux Collections Ordering System. Their study demonstrated that consumers could have purchased second-hand products because of various motivations, including affordability, uniqueness, nostalgia, environmental responsibility, and product availability. The findings showed that preloved products could have possessed economic, personal, and environmental value beyond their lower prices. The study also emphasized the importance of providing sufficient and accurate product information so that customers could properly evaluate secondhand items. These findings supported the way preloved products were expected to be presented in the Margaux Collections Ordering System, where condition, size, price, and visible defects needed to be clearly indicated for each item. By providing transparent product information and distinguishing brand-new products from preloved items, the system could have helped customers make informed decisions and could have created a more trustworthy and organized online shopping experience.

Resnick and Zeckhauser (2002) Resnick and Zeckhauser (2002) examined trust among strangers in online transactions. Their study focused on reputation systems and how customers used seller records before purchasing products online. In physical stores, buyers were able to communicate directly with sellers and inspect products. In online marketplaces, however, buyers depended on available information, seller reputation, and the experiences of previous customers. The study explained that a seller's transaction history served as evidence of reliability. Positive records showed that sellers communicated effectively with buyers, provided accurate products, and completed orders properly. The researchers also showed that reputation systems regulated behavior in online marketplaces. Sellers maintained good ratings because negative feedback reduced their ability to attract future buyers. The study supported the buyer–seller management component of the proposed system. Seller profiles, completed-order records, customer feedback, and ratings encouraged responsible transactions. Although Margaux Collections operated as a single-seller platform rather than an open marketplace with multiple vendors, the same principle applied at a smaller scale — a completed order history and consistent transaction record gave the owner a form of credibility that repeat customers could recognize, even without a formal public rating system.

Ladhari, Gonthier, and Lajante (2019) Ladhari, Gonthier, and Lajante (2019) examined the purchase intentions of young consumers toward second-hand clothing. Their study focused on why consumers considered purchasing used fashion products online. The researchers considered attitude, perceived value, fashion interest, affordability, uniqueness, and environmental awareness. These factors explained why consumers were willing to purchase preloved clothes through digital platforms. The study indicated that consumers viewed second-hand clothing positively when the products were affordable, fashionable, unique, and attractively presented. Product presentation influenced how customers perceived preloved clothing. The findings suggested that online clothing platforms should provide highquality images, complete product information, and organized browsing features. These features reduced uncertainty and made used clothing more appealing to consumers. The study was related to the proposed system because it focused on online second-hand fashion shopping. The system supported both brand-new and preloved products through categories, photos, descriptions, sizes, prices, and condition labels.

This was particularly relevant to Margaux Collections because its customer base leaned toward younger buyers who were already familiar with browsing fashion items on social media before deciding to purchase. Presenting preloved pieces with clear photos and complete details, rather than relying only on a caption and a Messenger reply, matched the expectations this younger market already had from other online shopping experiences.

Perceived Value and Trust in Online Resale Fashion, Zhang, Xu, and Wang

(2023) explored how perceived value and seller trust influenced purchasing decisions in the online second-hand fashion market. Their study measured how buyers weighed quality, price, and emotional appeal when they considered pre-owned clothing. The researchers found that customers were far more likely to complete a purchase when they believed the item offered fair value for its condition and when the seller appeared consistent and reliable. Transparent pricing and honest descriptions reduced hesitation significantly, even when buyers compared preloved items directly against new alternatives. The study also revealed that trust acted as a critical bridge between interest and action; without confidence in the seller, attractive pricing alone rarely convinced shoppers to proceed. These findings reinforced the importance of clarity and openness in the Margaux Collections system. Every listed item was presented with consistent pricing, clear condition notes, and honest descriptions so that buyers understood exactly what they received and trusted the value they paid.

Visual Presentation and Buyer Confidence in Online Clothing Shopping, Saha and Das

(2019) investigated how visual presentation and product imagery shaped consumer confidence in online clothing purchases. Their research compared buying decisions across platforms that used few photographs versus those that displayed multiple images from different angles. The study showed that buyers felt substantially more secure when they viewed detailed, varied photographs that revealed an item's true appearance, fabric, and condition. Because online shoppers could not touch or try on clothing personally, they depended almost entirely on visuals to judge quality and fit. Poor or limited imagery raised doubts and caused many potential buyers to abandon their carts. The researchers concluded that comprehensive visual content functioned as a substitute for physical inspection. For Margaux Collections, this meant that the ordering system included space for multiple clear photographs per listing so that customers examined each piece thoroughly before deciding. Better visuals reduced uncertainty and helped buyers feel confident about their choice.

Listing Organization and Purchase Decision-Making, Chen, Liu, and Wu (2021)

examined how inventory visibility and product organization affected the shopping experience in multi-category online fashion stores. Their study tracked how quickly shoppers made decisions and how often they returned when listings were organized clearly versus when items appeared mixed together without structure. The researchers discovered that buyers completed purchases faster and returned more often when they could easily identify whether an item was new or pre-owned, what size it was, and whether it remained available. Confusing listings or uncertain availability caused frustration and delayed decisions. The study also noted that separated categories and clear status labels helped shoppers compare options fairly without guessing quality or origin. This result guided the organization of the Margaux Collections system. The platform separated brand-new items from preloved pieces and displayed availability status clearly so that buyers browsed

efficiently, understood what they viewed, and made decisions without unnecessary delays.

Sustainability Messaging and Appeal of Preloved Fashion, Mair and Gazzola (2020) Mair and Gazzola (2020) analyzed how sustainability-related messaging influenced appeal and purchase intention in the resale and circular fashion sector. Their study surveyed shoppers to learn whether environmental information alongside product descriptions changed how people valued secondhand items. The researchers found that buyers viewed preloved clothing more positively when sellers explained that reuse extended garment lifespans, reduced waste, and conserved resources. Shoppers considered such items not merely cheaper alternatives but meaningful choices that aligned with responsible consumption. Sustainability messaging did not distract from style or quality; instead, it added value that many buyers appreciated. The study suggested that mentioning environmental benefits helped second-hand fashion stand out as desirable and purposeful. For Margaux Collections, this meant that brief, clear references to sustainability could accompany listings. Customers understood that buying preloved supported responsible fashion, and that awareness strengthened their appreciation of each item's worth beyond its price.

Payment Clarity and Transaction Trust in Online Shopping, Chou, Chen, and Lin (2016) Chou, Chen, and Lin (2016) studied how payment clarity and verification procedures influenced trust in online peer-to-peer commerce. Their research compared systems where payment details and confirmation processes appeared openly against those where financial steps remained unclear or unconfirmed. The study showed that buyers hesitated significantly when they did not understand how to pay, when funds would be collected, or how they would receive confirmation. Uncertainty about payment became one of the strongest reasons shoppers abandoned transactions. In contrast, when steps were outlined plainly and confirmation was provided reliably, buyers felt protected and completed orders with greater confidence. The researchers concluded that trust depended not only on the product but also on a predictable, verifiable transaction process. This finding directly supported the payment verification approach used in Margaux Collections. The ordering system guided buyers through clear steps and required confirmation of payment so that both parties confirmed transactions reliably, and customers always understood exactly where they stood.

Trust and Mobile Wallet Adoption in Micro-Retail, Smith and Harrison (2022)

Smith and Harrison (2022) examined the factors that influenced the adoption of mobile wallet payments among small-scale retail consumers. Their study focused on the 'Perceived Security' and 'Transaction Confirmation' stages of the buying process. The researchers discovered that consumers were 45% more likely to complete a transaction when the system provided an immediate field for payment reference input, as it created a psychological sense of accountability. The study concluded that for small businesses, integrating a structured payment verification step was more critical than having an automated bank API because it allowed for human-in-the-loop verification which customers found more trustworthy.

This study was directly instrumental in the researchers' decision to design the GCash Reference Number field in the Margaux Collections checkout module. Instead of a purely automated system which could be prone to downtime, the researchers followed Smith and Harrison's findings by implementing a manual-input verification system. This ensured that every order placed in Margaux Collections was backed by a verifiable transaction ID, thereby bridging the trust gap between the home-based boutique owner and her digital customers.

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2.2 Local Studies

Local studies provided the empirical basis for the proposed system because they showed how Filipino businesses actually used digital tools in practice. These studies were especially valuable because they reveal the operational benefited, limitations, and adoption patterns of systems designed for small enterprises in the Philippines. In the case of Margaux Collections, the local studies reviewed here supported the need for a more structured ordering platform that could improve efficiency, accuracy, and customer trust. They also demonstrated that even smallscale businesses could benefit significantly from digital solutions when these were designed according to their actual workflow, resources, and operational challenges.

Developing a Point-of-Sale Inventory Management System for

Lords Grace General Merchandising

Antonio, Reyes, and Torres, 2017

The development of web-based point-of-sale and inventory management systems has become an important area of applied computing research in the Philippines, particularly for small businesses that need affordable, practical, and reliable digital tools. Small enterprises often operate with limited capital, personnel, equipment, and technical support. Despite these limitations, they still need to manage product records, monitor stock levels, record sales, prepare reports, and respond to customer orders efficiently. Web-based systems provide a possible solution because they can centralize business information and make important records easier to organize, update, and access.

Antonio, Reyes, and Torres (2017) developed a PHP- and MySQL-based pointofsale and inventory management system for Lord's Grace General Merchandising, a small retail business in the Philippines. The project was completed as an undergraduate capstone at the Polytechnic University of the Philippines and demonstrated how information systems could be applied to address practical business problems. The system served as a local example of how a small enterprise could use accessible web technologies to improve inventory control, sales recording, reporting, and general operational efficiency. The project also

showed that the successful adoption of technology did not necessarily require a business to invest in an expensive or highly complex commercial platform.

The study was important because it demonstrated that digital systems could be designed according to the actual needs and limitations of a local business. Rather than creating a system with unnecessary features, the developers focused on functions that directly supported the daily activities of Lord's Grace General Merchandising. These included managing products, monitoring available stock, recording sales, generating receipts, and preparing basic reports. By addressing the specific workflow of the business, the system provided a more practical solution than a generalized application that might not match the organization's operations.

The system developed by Antonio et al. (2017) included several modules that supported routine retail activities. The product management module allowed authorized users to add new products, edit product details, update information, and remove records when necessary. This function helped the business maintain an organized catalog of the items it sold. Instead of relying on separate notebooks, paper lists, or scattered files, the business could store product information in a centralized database. This made it easier to search for records, correct errors, and maintain updated product information.

The inventory tracking module monitored stock levels and helped users determine which products were available, sold, or approaching a low-stock level. Low-stock alerts were particularly useful because they helped the business identify products that needed replenishment before supplies were completely exhausted. Without this type of monitoring, owners might discover shortages only after customers had already requested the product. The inventory module therefore supported more timely purchasing decisions and reduced the likelihood of stockouts.

The sales recording module provided a structured way to document customer transactions. It logged the products purchased, recorded the quantities sold, calculated the total amount, and generated receipts. Recording transactions digitally helped reduce dependence on handwritten sales notes and made it easier to review previous purchases. It also connected sales activities with inventory monitoring

because the quantity of a product could be updated after each transaction. This connection reduced the possibility that the recorded stock would remain unchanged after an item had been sold.

The reporting module generated summaries of sales performance over specific periods. These reports could help the business owner review daily, weekly, or monthly sales and identify products that performed well. The system could also identify top-selling items, allowing the owner to understand customer preferences and make better decisions about future purchasing or stock replenishment. By converting transaction records into organized reports, the system supported decision-making based on actual business data rather than memory or informal observation.

Together, these modules created a more organized workflow than manual recordkeeping. Under a manual system, owners or employees might have needed to check different notebooks, receipts, and paper lists to determine whether a product was available or how many units had been sold. This process could be slow and vulnerable to mistakes. In contrast, a centralized digital system allowed authorized users to access relevant information in one location. The system could make routine tasks more efficient and reduce the amount of time spent searching for or comparing separate records.

The study also showed that adopting a digital system could reduce common operational problems. Manual methods often resulted in misplaced records, delayed updates, incorrect stock counts, duplicated entries, and inconsistencies between actual inventory and recorded inventory. These issues became more serious when several employees handled sales or when the owner had to manage multiple responsibilities simultaneously. A computer-based system helped address these problems by providing a consistent structure for recording and updating information.

Inventory accuracy was important not only for internal business management but also for customer satisfaction. When records were inaccurate, the business might advertise or promise products that were no longer available. A customer could place an order based on outdated information and later be informed that the product had

already been sold. Such situations could cause delays, cancellations, and dissatisfaction. By keeping product and sales information updated, the system helped the business provide more accurate information to customers and respond to orders more reliably.

Digital recordkeeping also supported better monitoring of product demand. When sales data were stored systematically, the owner could identify which products were frequently purchased and which items remained in stock for longer periods. This information helped the business adjust purchasing, pricing, promotions, and product selection. For example, products with consistently strong sales could be replenished more frequently, while slow-moving items could be promoted, discounted, or reviewed to determine why they were not selling. This type of analysis was more difficult when sales records were incomplete or scattered across different sources.

Another important contribution of the study was its demonstration of the practicality of using PHP and MySQL for local business applications. PHP and MySQL were widely used in academic projects and small-scale system development because they were accessible, flexible, and suitable for database-driven websites. PHP could be used to develop the system's web interface and business functions, while MySQL could store product, inventory, sales, and user information in a structured database. This combination allowed developers to create a functional system without requiring the business to purchase expensive proprietary software.

The use of PHP and MySQL was especially relevant to small businesses in the Philippines because budget limitations often prevented them from adopting costly enterprise platforms. A small business might not need a complex system with advanced accounting, multi-branch management, or extensive supply-chain functions. Instead, it might need a focused application that handled the most important activities at a reasonable development and maintenance cost. The system developed by Antonio et al. (2017) showed that a small enterprise could benefit from a practical digital platform built with commonly available technologies.

The study also demonstrated that a business did not need an excessively complex system to improve its operations. What it needed was a platform that was functional, understandable, reliable, and aligned with its workflow. A system with too many

unnecessary features could make daily operations more difficult, especially for users with limited technical experience. For small businesses, usability was therefore as important as technical capability. The system had to make work easier rather than add another complicated process that employees needed to learn.

These findings were highly relevant to Margaux Collections because the business also needed a reliable method for managing products, sales, customer orders, and inventory. Like the business examined by Antonio et al. (2017), Margaux Collections may experience difficulties when product information and transactions are managed informally. Manual records, chat messages, memory, or separate files can make it difficult to determine which products are available, which items have already been sold, and which orders still need to be processed.

The proposed Margaux Collections Ordering System was designed to address these concerns by providing a structured platform for product management, sales recording, inventory monitoring, and order processing. The system could help the business owner store product details in one centralized database and update product status as transactions occurred. This would make it easier to identify available, reserved, sold, and unavailable items. It could also reduce confusion when the business received multiple inquiries or orders from different customers.

The relevance of the study was especially strong because Margaux Collections handled both brand-new and pre-loved fashion items. Brand-new items might be available in several quantities, sizes, or colors, while pre-loved clothing pieces were often unique or limited to one item. This difference required careful inventory control. If a unique pre-loved garment was recorded incorrectly or was not removed from the available product list after being sold, another customer might attempt to purchase the same item. A digital system could reduce this risk by assigning each product an organized record and updating its availability after every transaction.

The Ordering System could also include product information such as item name, product code, category, brand, size, color, condition, price, quantity, and product image. For pre-loved items, the system could include additional details about measurements, visible flaws, fabric condition, or other characteristics that customers might need to know before making a purchase. Complete product

records would support accurate listings and help the owner respond more efficiently to customer questions.

Sales recording would be another important function for Margaux Collections. Each completed order could be recorded with information about the customer, selected products, quantities, prices, payment status, and delivery details. This would help the owner monitor the progress of each transaction and reduce dependence on separate messages or handwritten notes. The system could also update inventory after a sale, ensuring that the product's availability reflected its actual status.

The system could further assist the owner in tracking customer orders from the time they were placed until they were completed. Orders might be categorized as pending, confirmed, paid, packed, shipped, completed, or cancelled. This type of order monitoring would help the business identify which transactions still required attention. It would also make it easier to provide customers with accurate updates about payment confirmation, preparation, and delivery.

The system's reporting functions could provide Margaux Collections with useful information about sales performance. Reports might summarize total sales, bestselling products, order frequency, remaining stock, and product categories with the highest demand. Such information could help the owner determine which types of clothing should be sourced more frequently and which items might require additional promotion. Reports could also support future planning by showing changes in sales patterns over particular periods.

The study by Antonio et al. (2017) also supported the importance of centralizing product records and monitoring stock movements. Centralized information could help prevent duplication, inconsistent entries, and confusion between old and updated product details. For example, if the owner changed the price or availability of an item, the update could be reflected in the central product record instead of needing to be changed across several separate documents. This could make product management more efficient and reduce the likelihood of outdated information being shown to customers.

The relevance of the study extended beyond technical similarity. It also supported the broader idea that an ordering system could serve as a practical management tool rather than simply a platform for receiving customer orders. A properly designed system could improve the entire business process by connecting product management, inventory monitoring, sales recording, and order tracking. This integration could help the owner understand the relationship between product availability, customer demand, and sales performance.

The study also emphasized that digital systems were most effective when designed around the actual workflow of the business. This was an important consideration for Margaux Collections because the proposed Ordering System was not intended to be a generic e-commerce platform. Instead, it was being developed to address the specific requirements of a small fashion business. These requirements included managing clothing products, distinguishing brand-new items from pre-loved items, recording unique product details, monitoring limited inventory, and tracking customer orders.

For Margaux Collections, tailoring the system to the business workflow could improve both usability and effectiveness. The owner should be able to perform common activities without unnecessary steps, such as adding products, editing item information, updating stock, recording sales, and checking order status. The system should also use terms and categories that are familiar to the business rather than requiring the owner to adapt to unfamiliar or irrelevant features. This approach would increase the likelihood that the system would be used consistently.

A user-friendly interface would be particularly important because small business owners often perform multiple roles. The owner might manage inventory, communicate with customers, prepare products, arrange deliveries, record sales, and make purchasing decisions. The Ordering System should therefore make routine tasks straightforward and efficient. Clear navigation, organized forms, readable product listings, and visible stock-status indicators could help reduce the time required to complete each activity.

Reliability would also be essential. The system needed to save product and transaction records accurately and make them available when the owner needed them. If information was lost, delayed, or incorrectly updated, the system could create additional operational problems. Regular backups, controlled user access, and careful testing could help maintain the integrity of the records. Reliability would also contribute to customer trust because accurate product availability and order information would allow the business to provide more dependable service.

The use of PHP and MySQL could also be appropriate for the Margaux Collections Ordering System because it could provide a practical foundation for a small business application. A PHP-based web interface could allow the owner to access the system through a browser, while a MySQL database could store information about products, customers, orders, sales, and inventory changes. The system could later be expanded if the business required additional functions, such as user accounts, automated notifications, customer records, or more detailed sales reports.

Overall, Antonio, Reyes, and Torres (2017) provided a strong local precedent for the development of the Margaux Collections Ordering System. Their project demonstrated that a PHP- and MySQL-based point-of-sale and inventory solution could be applied effectively in a small Philippine retail business. It showed that digital tools could improve product organization, inventory control, sales recording, reporting, and operational efficiency without requiring an expensive or overly complicated platform.

More importantly, the study established that the value of an information system depended on how well it responded to the actual needs of the business. A system could be technically functional but still ineffective if it was difficult to use, disconnected from the business workflow, or filled with unnecessary features. For Margaux Collections, the system needed to remain focused on practical functions that directly supported the management of fashion products and customer orders.

The findings therefore supported the development of an organized, user-friendly, and dependable Ordering System for Margaux Collections. By centralizing product information, monitoring inventory movements, recording transactions, tracking orders, and generating useful reports, the system could help the owner reduce manual errors and improve customer service. It could also provide a stronger

foundation for future business growth by making the enterprise more organized, data-informed, and capable of responding efficiently to changes in customer demand.

Analysis of Inventory Management Systems of Small-Sized

Restaurants in Quezon Province

Santos, Dela Cruz, and Lopez, 2021

Santos, Dela Cruz, and Lopez (2021), in their study published in the Philippine Journal of Hospitality and Service Management, examined the inventory management practices of small-sized restaurants in Quezon Province, Philippines. The research sought to identify the methods used by restaurant owners to monitor supplies, determine common inventory-related problems, and develop recommendations for improving stock control and operational efficiency. The study considered practical issues such as purchasing, recording inventory, monitoring stock levels, preventing spoilage, and maintaining accurate records. Its findings were particularly relevant to small enterprises because these businesses often operated with limited staff, limited financial resources, and limited time, yet they still needed dependable records to support daily operations and business decisions. [mseuf.edu](https://mseuf.edu.ph/research/read/819)

The researchers used a mixed-methods approach that combined inventory audits, interviews with business owners, and comparisons between manual and digital inventory management practices. Through these methods, the study examined how restaurant owners recorded incoming supplies, monitored available stock, identified items that needed replenishment, and handled problems such as overstocking, shortages, and spoilage. The research also considered how inventory management affected other areas of the business, including purchasing decisions, cash flow, food preparation, pricing, customer service, and overall operational control.

Although the study focused on the restaurant industry, its findings extended beyond food service. Restaurants and retail businesses may sell different types of products, but both need to maintain accurate information about available stock, recorded sales, product movement, and replenishment requirements. In both settings, inaccurate records can result in unavailable products, unnecessary purchases, excess stock, financial losses, and customer dissatisfaction. The study therefore

offered valuable insights for other small businesses that needed to track inventory carefully while working within limited operational capacity.

One of the central findings of the research was that many small restaurants continued to rely on manual inventory methods. These methods included handwritten logs, paper-based stock lists, notebooks, spreadsheets, and records based partly on memory. Although manual systems could be inexpensive and easy to begin using, they often became unreliable when inventory changed frequently. Restaurant supplies were regularly purchased, consumed, sold, or discarded, and records were not always updated immediately after each activity. As a result, the inventory information available to the owner did not always reflect the actual stock on hand.

Manual inventory systems frequently resulted in delays, recording errors, incomplete information, and inconsistencies between written records and physical inventory. Business owners or employees might forget to record a purchase, fail to update the quantity of an item that had been used, or enter incorrect figures into a stock list. In some cases, inventory was counted only when a problem became noticeable, such as when an ingredient ran out unexpectedly or when the owner discovered that too many supplies had been purchased. This reactive approach made it difficult to identify problems early and limited the owner's ability to make timely decisions.

The study also found that some restaurants experienced difficulty maintaining consistent inventory monitoring because of the volume and frequency of their purchases. Restaurant supplies were not limited to a small number of standardized products. They could include food ingredients, beverages, packaging materials, cleaning supplies, and other operational items. Managing all of these categories manually required time and attention, especially when different employees were responsible for receiving, storing, preparing, and using the supplies. When responsibilities were unclear, records could become incomplete or duplicated.

One specific problem identified in the study involved purchasing decisions being based on the chef's estimates or immediate judgment. While the chef might have had practical knowledge of daily consumption, relying primarily on personal judgment could result in overstocking or understocking. Overordering could create unnecessary costs, occupy storage space, and increase the possibility of spoilage. Underordering, on the other hand, could interrupt food preparation and prevent the restaurant from serving certain menu items. These problems demonstrated that purchasing decisions needed to be supported by accurate and updated inventory information rather than by memory or informal estimates alone.

The research also identified problems related to large purchasing volumes, inaccurate food and beverage inventory counts, and the spoilage of perishable items such as vegetables, meat,

fish, and poultry. When inventory was not monitored consistently, restaurants could purchase more supplies than they could use before expiration. Spoilage not only represented a direct financial loss but could also affect the restaurant's profitability and cash flow. In addition, inaccurate records could distort the business's understanding of its actual financial position because the value of available inventory did not match the value recorded in its documents.

To address the problem of spoilage, the study recognized the importance of inventory rotation methods such as First In, First Out (FIFO). Under FIFO, items received earlier are used before newer deliveries. This method helps prevent older supplies from remaining unused while newer supplies are consumed first. The researchers also identified First Expiry, First Out (FEFO) as another useful strategy for products with different expiration dates. Under FEFO, items with the earliest expiration dates are prioritized regardless of when they were received. These practices are especially important for businesses handling perishable products, although the broader principle of systematic stock rotation can also be applied to fashion retail.

The study emphasized that inventory management was not merely an administrative activity. It was a critical operational function that directly affected business continuity, cost control, service quality, and customer satisfaction. When a restaurant knew what products were available, what supplies had been used, and what items needed to be purchased, it could plan its operations more effectively. Accurate inventory records helped reduce unexpected shortages and limited the possibility of spending money on supplies that were not immediately needed. In contrast, poor inventory control could lead to disrupted operations, financial waste, and reduced customer trust.

In comparison with manual methods, digital inventory systems provided stronger control over the movement of stock. Digital systems allowed owners to record purchases, sales, and adjustments in a more organized manner. Information could be updated more quickly, making it easier to determine the current quantity of available products. Digital records also reduced the likelihood of calculation errors, misplaced documents, and incomplete entries. When properly maintained, a digital system provided a clearer view of inventory changes and helped business owners make decisions based on more accurate information.

Digital inventory systems also made it easier to identify patterns in product movement. Owners could determine which products were used or sold quickly, which items remained in stock for extended periods, and which supplies required frequent replenishment. This information supported better purchasing and planning decisions. Fast-moving products could be monitored closely so that they would not run out, while slow-moving items could be purchased in smaller quantities or evaluated for possible changes in pricing, promotion, or product strategy.

Another advantage of digital inventory management was the possibility of improving communication among individuals involved in business operations. When inventory information was recorded in one organized system, owners and employees could refer to the same data rather than maintaining separate or conflicting lists. This reduced confusion regarding available stock, recent purchases, and items that needed replenishment. It also created a clearer record of who entered or changed information, which could improve accountability and reduce errors.

The study identified three important factors that influenced the successful adoption of digital inventory management systems among small Philippine businesses: simplicity of the user interface, system reliability, and perceived usefulness in comparison with the previous manual method. These factors were especially important for microbusinesses and small enterprises because owners often performed several responsibilities at the same time. A business owner might be responsible for purchasing, sales, customer service, staff supervision, financial management, and inventory monitoring. Such owners could not afford to spend excessive time learning complicated software.

For this reason, an inventory system needed to be simple and easy to understand. Users should be able to add products, update quantities, record sales, and check stock levels without going through unnecessary steps. An interface that was confusing or difficult to navigate could discourage consistent use. Even a system with advanced technical functions would not be effective if the intended users avoided it because they found it too complicated. Simplicity therefore represented an important condition for ensuring that digitalization improved rather than complicated daily operations.

System reliability was another essential factor. Business owners needed to trust that the information recorded in the system would be saved accurately and remain accessible when needed. A system that frequently crashed, lost information, produced incorrect calculations, or became unavailable could create problems similar to those associated with manual recordkeeping.

Reliability also involved the system's ability to respond consistently when users recorded new stock, updated product quantities, or entered completed sales. If the system performed dependably, owners would be more likely to use it as the primary source of inventory information.

Perceived usefulness also influenced whether business owners accepted digital inventory systems. Small businesses were more likely to adopt technology when they could immediately recognize how it improved their work. If the system saved time, reduced recording errors, improved stock visibility, and made purchasing decisions easier, owners could see a direct reason to use it. In contrast, if the system provided features that were unrelated to the business's daily needs, it might be viewed as an unnecessary expense or additional responsibility.

This finding suggested that technology adoption among small businesses was strongly shaped by practical benefits rather than by innovation alone. Business owners generally needed to see how a new system solved existing problems. A digital system was more likely to be embraced when it simplified recordkeeping, reduced stress, improved organization, and produced information that could be used immediately. The system had to demonstrate its value through actual improvements in daily operations rather than through abstract claims about modernization.

The findings of Santos et al. (2021) were highly relevant to Margaux Collections, even though the original study focused on restaurants rather than fashion retail. Both types of businesses needed to monitor product availability, record sales, identify fast-moving and slow-moving items, and determine when stocks required replenishment. Both also faced the risk that recorded information might not correspond to actual inventory. If a product was listed as available when it had already been sold, the business could accept an order that it could not fulfill. If a product was not recorded properly, the business might overlook a potential sale or make inaccurate purchasing decisions.

For Margaux Collections, accurate inventory management was particularly important because the business handled both brand-new and pre-loved fashion items. Each product needed to be recorded carefully according to its name, category, size, color, condition, price, and available quantity. Brand-new items might be available in multiple quantities or sizes, while pre-loved products were often unique or limited to a single piece. This difference increased the possibility of confusion if stock information was recorded manually or updated inconsistently.

Pre-loved fashion items required especially careful monitoring because a single garment might have a unique design, size, condition, or brand. Once that item was sold, it might not be possible to replace it with an identical product. An inaccurate inventory record could therefore result in duplicate orders, incorrect product availability, or customer disappointment. A digital system could reduce these risks by assigning each product an organized record and updating its status when an item was reserved, sold, or removed from inventory.

The proposed Margaux Collections Ordering System should therefore allow the business owner to monitor available items in an organized and timely manner. When a customer places an order, the system should record the transaction and update the product's availability. If an item is reserved, the system could also distinguish between available, reserved, sold, and unavailable statuses. These functions would help prevent the same item from being offered to multiple customers and would give the owner a clearer understanding of which products remained in stock.

The system should also support the recording of product details. Important information could include the product name, item code, category, size, color, brand, condition, price, quantity, and product image. For pre-loved clothing, the system could also include notes regarding visible flaws, measurements, fabric type, or other details that customers might need before purchasing. Organized product records would improve the accuracy of online listings and make it easier for the business owner to locate and update specific items.

In addition, the system should support sales tracking and order monitoring. Sales records could help the owner identify which products had already been purchased, which customers had placed orders, and which orders were still being processed. Order monitoring could also provide information about payment status, delivery status, and order completion. These functions would reduce dependence on handwritten notes, chat histories, or memory and would make it easier to review past transactions.

A useful system should also help the business owner identify fast-moving and slow-moving products. Fast-moving items could indicate strong customer demand, popular styles, preferred sizes, or effective pricing. Slow-moving items might require additional promotion, price adjustments, improved product presentation, or reconsideration of the purchasing strategy. This type of information would support better decision-making because the owner could use actual sales records rather than relying only on assumptions about customer preferences.

The study also supported the idea that the Margaux Collections Ordering System should be designed around the actual needs of the business. The system should not contain unnecessary features that make it difficult to use or that do not contribute to daily operations. Instead, it should prioritize practical functions such as product listing, inventory updating, sales recording, order monitoring, and stock-status management. These functions would directly support the main activities of the business and help ensure that the system remained relevant to the owner and future users.

The system should also be flexible enough to accommodate the differences between brandnew and pre-loved products. Brand-new products may require quantity tracking across several units, sizes, or colors, while pre-loved items may require individual records because each item may be unique. A practical system could allow the owner to identify whether an item was new or preowned and record the appropriate information for each type. This would improve inventory organization while maintaining a straightforward user experience.

Ease of use would be particularly important for Margaux Collections because the business owner might have limited time and may not have specialized technical training. The system should allow the owner to complete routine tasks without extensive instruction. Clear labels, simple navigation, understandable buttons, organized forms, and visible status indicators could make the system easier to operate. The design should focus on reducing the amount of time needed to add products, update stock, and process orders.

Reliability would also be essential to the success of the proposed system. The owner must be able to depend on the system when checking whether an item was available, recording a sale, or reviewing an order. If information was not saved correctly or stock updates were delayed, the system could create the same problems it was intended to solve. Regular data backups, appropriate access controls, and consistent updating procedures could help protect the accuracy and availability of the business records.

Overall, Santos, Dela Cruz, and Lopez (2021) provided a strong local basis for the development of the Margaux Collections Ordering System. Their study demonstrated that small businesses benefited from organized inventory systems because accurate records improved stock monitoring, reduced errors, supported purchasing decisions, and increased operational efficiency. The research also showed that inventory management problems could affect broader

business outcomes, including cash flow, product availability, service quality, and customer satisfaction.

More importantly, the study emphasized that successful digital system adoption depended on simplicity, reliability, and usefulness. These qualities were essential for Margaux Collections because the proposed system needed to support the practical requirements of a small fashion business without creating unnecessary complexity. By providing organized product records, updated stock information, sales tracking, and order monitoring, the system could reduce the burden of manual recordkeeping and improve the accuracy of daily operations.

The findings ultimately supported the need for an ordering and inventory system that was userfriendly, dependable, and aligned with the actual activities of Margaux Collections. A welldesigned system could help the business manage brand-new and pre-loved items more effectively, prevent inaccurate product availability, respond to customer orders more efficiently, and make better decisions about its stock. In this way, the study of inventory management in small restaurants remained relevant to the development of a digital ordering solution for a small fashion enterprise.

Impact of Inventory Management on Financial Performance of MSMEs

Gaoat et al. (2023) examined the relationship between inventory management, operational efficiency, and financial performance among micro, small, and medium enterprises (MSMEs) in Santiago City, Philippines. The study used a quantitative, correlational research design and gathered data from MSME owners and managers to determine how inventory management practices affected business performance. The researchers considered financial indicators such as sales, operating margin, profit, return on assets, liquidity, and inventory turnover. Their findings showed a strong positive relationship between effective inventory management and operational efficiency, as well as a moderate positive relationship between inventory management and financial performance. This demonstrated that inventory control was not merely an administrative responsibility but a business function with direct consequences for profitability, liquidity, productivity, and long-term sustainability.

The study was highly relevant to small enterprises because MSMEs generally operated with limited capital, personnel, storage space, and technical resources. Unlike larger corporations, small businesses often had less flexibility to absorb losses caused by unsold products, damaged

goods, inaccurate records, or excessive purchases. A small amount of misplaced or slowmoving stock could affect the business's cash flow and limit its ability to purchase products that customers actually wanted. For this reason, the quality of a business's inventory management could influence not only its daily operations but also its ability to remain financially stable over time.

The research found that MSMEs with more effective inventory management practices tended to perform better operationally and financially. Proper inventory management allowed business owners to monitor the movement of goods, determine available stock, reduce unnecessary purchases, and identify products that required replenishment. It also helped businesses avoid overstocking, minimize losses, and respond more accurately to customer demand. These practices contributed to more efficient use of business resources because owners could direct their limited capital toward products with stronger demand instead of tying up money in items that remained unsold.

Overstocking represented one of the major problems that effective inventory management could help prevent. When a business purchased more products than it could sell within a reasonable period, its capital became tied up in inventory. The business might also need to use additional storage space and spend more time managing products that were not moving quickly. In the case of fashion products, excessive inventory could become outdated, less desirable, or difficult to sell at the original price. A reliable inventory system could help owners identify slow-moving items early and respond through promotions, discounts, bundle offers, improved product presentation, or changes in purchasing strategy.

Understocking created a different set of problems. When a business failed to maintain adequate supplies of popular products, customers might be unable to purchase the items they wanted. Repeated stockouts could lead to missed sales and customer dissatisfaction. Customers might also shift to competing businesses that offered more consistent product availability. Accurate inventory records helped owners recognize which products required timely replenishment and reduced the possibility of discovering shortages only after customers had already placed orders.

The study emphasized that inventory information should be used to guide business decisionmaking rather than treated as a simple recordkeeping requirement. Inventory records should show more than the quantity of items available. When properly organized, they could also help owners understand sales trends, purchasing patterns, product performance, and customer preferences. For example, a business could use its inventory and sales data to determine which products sold quickly, which items remained in stock for extended periods, and which categories generated the greatest revenue.

This type of information was especially important for MSMEs because they often operated with limited working capital. Small businesses could not afford to repeatedly purchase products

that did not sell or miss opportunities to restock items that customers regularly requested. Accurate inventory data allowed owners to allocate their resources more carefully. Instead of relying primarily on memory or assumptions, they could base decisions on recorded sales and stock movements. This improved the quality of purchasing, pricing, promotional, and productselection decisions.

The study also suggested that effective inventory management strengthened a business's ability to respond to customer demand. When owners knew which products were popular and which ones were slow-moving, they could adjust their sales and replenishment strategies accordingly. Fast-moving items could be monitored more closely and reordered when necessary, while slowmoving products could be reviewed to determine whether they required discounts, improved marketing, or changes in product presentation. This responsiveness was essential for small businesses because customer preferences could change quickly and financial margins were often limited.

The relationship between inventory control and customer satisfaction was also important. Customers generally expected businesses to provide accurate information about product availability, price, condition, and delivery. If a product was shown as available but had already been sold, the business might be unable to fulfill the order. This could result in cancellations, delays, or customer complaints. By maintaining accurate and updated inventory records, a business could provide more dependable service and reduce the risk of making promises that it could not fulfill.

The financial benefits of effective inventory management extended beyond the prevention of stockouts and overstocking. Proper control could also reduce losses resulting from damaged products, misplaced items, inaccurate purchasing, and unnecessary storage costs. It could improve liquidity by preventing excessive amounts of cash from being tied up in unsold inventory. In addition, accurate stock information could support better calculation of sales performance and profit. When owners knew the quantity and cost of their products, they could make more informed decisions about prices, discounts, and purchasing.

The findings were particularly relevant to fashion businesses because clothing products could differ according to brand, design, color, size, condition, and style. These characteristics made it important to maintain detailed and accurate product records. A business could have several items that appeared similar but differed in size, condition, or price. If these differences were not recorded properly, the owner could experience confusion when processing orders or updating product availability. Inventory accuracy was therefore essential for ensuring that customers received the correct item and that the business maintained reliable sales records.

For Margaux Collections, the study reinforced the importance of developing an Ordering System with accurate stock tracking, sales summaries, and reporting capabilities. The business handled both brand-new and pre-loved fashion items, which required careful monitoring of product condition, category, size, price, and availability. Brand-new items might be available in several

units, while pre-loved items were often unique or limited to one piece. This difference increased the risk of duplication or incorrect availability if the inventory was managed manually.

A digital system could help ensure that the inventory database was updated whenever products were added, reserved, sold, cancelled, or removed. It could distinguish among different inventory statuses, such as available, reserved, sold, and unavailable. This would help prevent the business from accepting orders for items that were no longer in stock. It could also reduce the possibility of offering the same unique pre-loved item to multiple customers. Accurate stock updates would improve order fulfillment and help protect customer trust.

The Ordering System could also help the owner identify fast-moving and slow-moving products. Fast-moving items might indicate strong customer demand, effective pricing, or successful product presentation. Slow-moving items might need additional promotion, lower pricing, better photographs, or more detailed product descriptions. By reviewing sales summaries and inventory reports, the owner could make informed decisions about which products to promote and which types of items to purchase or source in the future.

Sales reports could further help Margaux Collections assess its financial performance. The system could summarize sales by date, product category, item condition, price range, or product type. These reports could help the owner determine which products generated the greatest number of sales and which items contributed most to revenue. Although an ordering system would not automatically guarantee higher profit, it could provide the information necessary for making more effective pricing, restocking, and promotional decisions.

The study was also relevant because it highlighted the connection between operational efficiency and financial stability. When inventory was managed properly, businesses could reduce avoidable costs and improve the use of available resources. For Margaux Collections, every product represented a financial investment. Money spent on acquiring or preparing products could remain unavailable for other business needs until the item was sold. If products remained in stock for too long, the business could experience slower cash flow and reduced capacity to acquire new merchandise.

A digital inventory system could help the owner obtain a clearer overview of product movement and financial activity. By reviewing which items sold frequently and which products remained unsold, the owner could determine where business resources were being used effectively. The owner could also identify products that required additional attention and avoid repeating purchasing decisions that resulted in excessive slow-moving stock. This would support more careful planning and allow the business to allocate time, money, and promotional effort more efficiently.

Inventory management could also support customer service by improving the speed and accuracy of order processing. When product information was organized in one system, the owner could

respond to customer inquiries more quickly. The owner would not need to search through multiple notebooks, messages, or files to determine whether an item was available.

Accurate information could be provided about product details, price, condition, and order status. This could reduce delays and create a more professional experience for customers.

Another important implication of the study was that inventory management should be viewed as a continuing business process rather than a one-time activity. Inventory information needed to be updated regularly as products entered the business, were listed for sale, reserved, sold, returned, or removed. A system that was accurate only at the beginning of operations would eventually become unreliable if updates were not performed consistently. The value of the Ordering System would therefore depend on the establishment of regular procedures for entering products, recording transactions, and reviewing stock information.

For Margaux Collections, the system also needed to be flexible enough to accommodate changes in product lines and business operations. The business might add new types of fashion items, include accessories, expand its selection of pre-loved products, or change its order and delivery procedures. The system should allow the owner to add new product categories and update existing records without creating confusion. This flexibility would help the system remain useful as the business developed.

The study by Gaoat et al. (2023) also suggested that digital inventory management could support long-term business sustainability. Financial stability depended partly on the ability of a business to control costs, maintain sufficient liquidity, respond to demand, and use resources efficiently. A system that improved visibility over stock and sales could help the owner make decisions that supported these objectives. It could also provide a stronger foundation for future growth by creating organized business records that could be used for planning, evaluation, and performance monitoring.

When considered together with the other local studies, Gaoat et al. (2023) strengthened the justification for the Margaux Collections Ordering System. Antonio et al. (2017) demonstrated that PHP- and MySQL-based systems could improve product management, sales recording, inventory monitoring, and reporting in a small Philippine retail business. Santos et al. (2021) showed that digital inventory systems could provide more organized and reliable stock control than manual methods. Gaoat et al. (2023) extended these findings by showing that effective inventory management was associated not only with better operational efficiency but also with stronger financial performance among MSMEs.

Together, these studies showed that a digital ordering and inventory system could support both the operational and financial needs of a small business. It could help Margaux Collections maintain accurate stock records, reduce errors, monitor sales, identify product demand, control purchasing, and improve customer service. These improvements could contribute to better financial decisions by giving the owner clearer information about product performance and business resources.

Overall, Gaoat et al. (2023) provided a strong research basis for treating inventory management as a strategic function rather than a routine administrative task. For Margaux Collections, the proposed Ordering System should not only record orders but also support inventory analysis and financial decision-making. By maintaining updated records, generating sales summaries, identifying fast- and slow-moving products, and monitoring product availability, the system could help the business remain organized, responsive, and financially stable. Its development was therefore justified not only as a technological improvement but also as a practical strategy for strengthening the long-term performance and sustainability of a small fashion enterprise.

Digital Ukaynomics: Analysis on the Market for Secondhand Clothing Sold through Online Channels (Tomoling, 2024)

Tomoling's (2024) undergraduate thesis, *Digital Ukaynomics: Analysis on the Market for Secondhand Clothing Sold through Online Channels*, examines the growing online market for secondhand clothing in the Philippines. The study focuses on consumers who purchase preloved garments through digital platforms, particularly Facebook groups, which have become an important part of the country's online ukay-ukay ecosystem. By analyzing the sociodemographic characteristics of consumers and the factors that influence their purchasing decisions, the research provides a localized understanding of how e-commerce has reshaped traditional ukay-ukay consumption patterns.

The study was significant because the growth of digital platforms enabled ukay-ukay sellers to reach buyers beyond the geographic boundaries of physical stores and public markets.

Traditional ukay-ukay shopping generally required consumers to visit a physical location, search through clothing racks or piles, and inspect garments personally before purchasing them. Online selling changed this process by allowing buyers to browse products through photographs, descriptions, videos, and posts. Consumers could communicate with sellers, compare items,

negotiate prices, and arrange delivery without visiting a physical store. As a result, ukay-ukay became more accessible, convenient, and integrated into everyday online shopping behavior.

Tomoling employed a quantitative research design using a Google Forms questionnaire distributed through multiple Facebook groups dedicated to the buying and selling of secondhand clothing. The study gathered responses from 248 consumers who purchased secondhand clothes through online channels. Of the total respondents, 200 were female and 48 were male, while the majority were either employed or currently studying. The composition of the sample indicated a strong representation of female consumers in the online secondhand clothing market, although the gender distribution should be interpreted as a characteristic of the study's respondents rather than as a complete representation of all online ukay-ukay consumers in the Philippines. The concentration on Facebook groups was particularly relevant because these communities functioned as informal but highly organized marketplaces where sellers and buyers could interact directly.

Facebook groups provided several advantages for online ukay-ukay transactions. Sellers could post new items, announce restocks, organize live-selling sessions, and communicate directly with prospective buyers. Buyers, on the other hand, could ask questions, comment on posted garments, request measurements, check product availability, and negotiate or clarify delivery arrangements. These groups also created communities where members could share recommendations, provide feedback about sellers, and exchange information about prices and product quality. Although these platforms were informal compared with conventional ecommerce websites, they developed their own systems of trust, communication, reservation, payment, and delivery.

The respondents primarily used Facebook as their online platform for purchasing secondhand clothing. One of the main reasons they bought secondhand garments was the relatively low price. Ukay-ukay allowed consumers to access clothing at costs that were often lower than those of brand-new garments. This affordability was especially attractive to students, employed consumers with limited disposable income, and buyers who wanted to purchase several clothing items without spending a large amount of money. Some consumers were also able to find branded, fashionable, or good-quality pieces at lower prices than they would have paid in conventional retail stores.

However, the study demonstrated that online ukay-ukay consumption was not driven by affordability alone. Although cheap prices remained an important motivation, consumers' demand

was also affected by their demographic characteristics, shopping preferences, online purchasing experience, and logistical considerations. The availability of an item at a low price did not automatically guarantee a purchase. Buyers also had to consider whether the product was worth the cost of delivery, whether they trusted the seller, whether the item matched the description, and whether the purchasing process was convenient and reliable.

Using an ordinary least squares regression model, the study identified several factors that significantly affected consumer demand for secondhand clothing sold online. These factors included age, sex, income, willingness to pay for brand-new clothing, willingness to pay for secondhand clothing, delivery fees, previous buying experience, and browsing time. The inclusion of these variables demonstrated that online ukay-ukay consumption was influenced by a combination of economic, demographic, behavioral, and logistical factors. Demand could therefore not be explained simply by the price of secondhand clothing.

Age and sex were among the demographic factors associated with consumer demand. Different age groups and genders may have had different levels of interest in thrift fashion, different online shopping habits, and different preferences concerning clothing styles and brands. Income also played an important role because consumers' purchasing capacity influenced the number and type of garments they could afford. While lower-income consumers might have been attracted to ukay-ukay because of its affordability, consumers with higher incomes could also participate in the market because of their interest in unique, branded, vintage, or environmentally responsible fashion.

The respondents' willingness to pay for both brand-new and secondhand clothing also affected demand. This factor provided insight into how consumers evaluated secondhand garments in relation to newly manufactured clothing. Consumers who were willing to pay more for brandnew clothing might have had different quality expectations or brand preferences from those who preferred secondhand items. At the same time, willingness to pay for preloved clothing suggested that some consumers were prepared to spend more for garments that offered desirable characteristics such as brand reputation, uniqueness, good condition, durability, or style. Therefore, the market included both highly price-sensitive consumers and buyers who were willing to pay a higher amount for selected secondhand items.

Delivery fees were another important factor influencing online demand. Even when a garment was inexpensive, a high delivery charge could increase the total cost and discourage consumers from

completing the transaction. This was particularly relevant for low-priced ukayukay items, where shipping fees could represent a substantial portion of the final amount paid by the buyer. Consumers had to evaluate the combined cost of the clothing and delivery, rather than considering the listed item price alone. Sellers who offered reasonable shipping rates, combined orders, pickup options, or free-delivery promotions could make their products more attractive to online buyers.

Previous buying experience also influenced consumer demand. Consumers who had purchased secondhand clothing online before were likely to have developed greater familiarity with the process. They may have become more knowledgeable about reliable sellers, payment procedures, common sizing issues, delivery arrangements, and the risks of purchasing garments without physically inspecting them. This experience could reduce uncertainty and make buyers more confident in completing online transactions. First-time buyers, in contrast, might have been more cautious because they had limited knowledge of the seller's reliability or the actual quality of the products.

Browsing time represents another relevant behavioral factor. Consumers who spent more time browsing Facebook groups and online listings had greater opportunities to discover products, compare prices, observe seller activity, and evaluate available choices. Longer browsing periods could also increase the likelihood of finding items that matched a consumer's preferred style, size, brand, or budget. At the same time, extended exposure to product listings might encourage additional purchases, particularly when sellers posted frequent updates, limited offers, or visually appealing collections. Browsing was therefore not merely a passive activity but part of the decision-making process that shaped online ukay-ukay demand.

The study also used logarithmic regression to examine the elasticity of demand for secondhand clothing sold online. The analysis indicated that online ukay-ukay products were normal goods and had inelastic demand. In economics, a normal good is a product for which demand generally increases as consumer income increases. This finding suggested that when consumers' income increased, their demand for online secondhand clothing also tended to increase, although the increase was proportionally smaller than the rise in income. In other words, consumers did not necessarily abandon ukay-ukay when their financial capacity improved. Some continued purchasing secondhand clothing because of its affordability, uniqueness, familiarity, or environmental appeal.

The finding that demand was inelastic suggested that consumer demand did not change dramatically in response to minor price fluctuations. When demand is inelastic, a small change in price generally produces a relatively smaller change in the quantity demanded. This did not mean that consumers were completely unaffected by price increases. Rather, it indicated that other factors such as product uniqueness, seller trust, shopping habits, convenience, and previous experience could help maintain demand even when prices changed slightly. Once consumers became familiar with the online ukay-ukay market, they might have continued purchasing because they valued the overall experience and the benefits associated with secondhand clothing.

This finding had important implications for online ukay-ukay sellers. Sellers could not rely solely on extremely low prices to attract and retain customers. Instead, they could strengthen demand by improving product presentation, accurately describing garment condition, providing clear photographs and measurements, maintaining reasonable delivery fees, and establishing reliable communication practices. Sellers could also develop customer loyalty by offering consistent service, responding promptly to inquiries, creating organized product listings, and building trust through reviews and repeat transactions. These strategies could help businesses maintain consumer interest even when operating costs or prices changed.

The results also suggested that online ukay-ukay sellers needed to pay close attention to the logistical aspects of e-commerce. Product availability, payment procedures, shipping arrangements, delivery fees, packaging, and communication all affected the customer's purchasing decision. In a physical ukay-ukay store, buyers could inspect clothing personally and take the item home immediately. In an online setting, however, buyers had to depend on the seller's information and wait for delivery. Any difficulty involving product descriptions, payment confirmation, shipping schedules, or delivery costs could reduce trust and discourage future purchases.

Tomoling's study also addressed the legal context surrounding the Philippine secondhand clothing market. The research recognized the tension between Republic Act No. 4653, which prohibits the commercial importation of used clothing and rags, and the widespread and socially accepted practice of buying and selling secondhand clothing through physical and digital marketplaces. The presence of online ukay-ukay demonstrated how digital platforms allowed the trade to continue and expand despite legal restrictions on commercial imports. Facebook groups and other online channels made it possible for sellers to reach consumers, advertise products, and organize transactions in ways that were not limited to traditional store locations.

This tension presented an important challenge for policymakers. Regulation of the secondhand clothing sector needed to consider the actual behavior of consumers, sellers, and digital marketplaces. A policy approach focused only on prohibition might fail to account for the economic importance of ukay-ukay, the livelihoods it supported, the demand for affordable clothing, and the growing role of online commerce. At the same time, regulation needed to address concerns related to consumer protection, product quality, sanitation, taxation, business registration, and responsible trade. The study therefore suggested that policy discussions should be informed by the realities of digital marketplace dynamics rather than focusing exclusively on traditional forms of clothing retail.

The research made a significant contribution to the growing body of local literature on secondhand clothing consumption in the Philippines. It provided empirical evidence that digital platforms had transformed ukay-ukay from a primarily physical and necessity-oriented market into a more complex online consumer behavior pattern. Consumers were no longer required to visit a particular market or store to participate in ukay-ukay. Through Facebook groups, they could browse products, communicate with sellers, compare prices, and complete purchases from different locations. This increased accessibility contributed to the expansion and normalization of secondhand clothing consumption.

The study also demonstrated the value of using economic analysis to understand online ukayukay. By examining consumer demand, price sensitivity, income, delivery fees, browsing behavior, and purchasing experience, the research provided insights that complemented cultural and sustainability-focused studies. It showed that ukay-ukay was not only a fashion or environmental practice but also a market shaped by consumer demand and economic behavior. Understanding these factors was important for sellers who wanted to improve their operations and for policymakers who needed to create more realistic and effective approaches to the sector.

Overall, Tomoling's (2024) study revealed that digital ukay-ukay had become an established part of the Philippine secondhand clothing market. Consumers were attracted by affordable prices, but their participation was also shaped by income, demographic characteristics, willingness to pay, delivery expenses, online experience, and browsing behavior. The finding that online secondhand clothing was a normal good with inelastic demand suggested that consumers could remain active participants in the market even as their income or prices changed. The research ultimately showed how e-commerce had expanded the reach of ukay-ukay, created new opportunities for sellers, and transformed secondhand clothing into a significant form of online consumption in the Philippines.

Pre-Loved Clothing and Local Fashion Industry in Kalibo

(Itubay, 2025)

Itubay's (2025) qualitative study, *Pre-Loved Clothing and Local Fashion Industry: Its Marketability, Sustainability and Challenges*, examines the coexistence of pre-loved clothing businesses and local fashion enterprises in Kalibo, Aklan. The research addressed a critical gap in local literature by exploring how these two distinct sectors navigate competition, sustainability concerns, and market dynamics within a single geographic context. Using focused group discussions with 15 business owners (seven from preloved clothing and eight from local fashion), the study provided rich, contextspecific insights into the Philippine fashion economy.

The study identified several key marketing strategies employed by both pre-loved and local fashion businesses. Social media platforms, particularly Facebook, Instagram, and TikTok, emerged as primary tools for visibility and customer engagement. Preloved clothing sellers leveraged live selling, multiple Facebook pages, and integration with e-commerce platforms like Shopee to attract tech-savvy, priceconscious consumers. Meanwhile, local fashion businesses relied more heavily on word-of-ofmouth referrals and personal relationships, though they also increasingly adopted digital channels. Both sectors emphasized quality controlled, flexible pricing, and personalized customer service as critical success factors. Sustainability strategies were another central theme, with both industries demonstrating adaptability to changing fashion trends and consumer preferences. Pre-loved clothing businesses catered to younger, trend-conscious buyers by continuously updating their inventory based on social media influence and celebrity culture. Local fashion businesses maintained consistency in operations through regular posting schedules and customer engagement activities. Both sectors also implemented sustainable pricing strategies, loyalty programs, and promotions such as Buy 1 Take 1 offered to retain customers and move inventory effectively.

Despite their differences, both industries faced significant shared challenges. Sourcing reliable suppliers proved difficult for pre-loved clothing sellers due to the irregular nature of bulk secondhand clothing supply. Quality inconsistencies in preloved items required meticulous inspection and often repairs before sale. Local fashion businesses struggled with workforce management issues, including inconsistent staff performance, motivation, and recruitment difficulties. Both sectors also faced pressure to meet customer expectations regarding product quality, pricing transparency, and delivery timelines.

The study's findings underscore the resilience and adaptability of small fashion businesses in Kalibo. Coping mechanisms included forming strategic partnerships with suppliers, seeking financial support through family loans, providing personalized services to strengthen customer relationships, and leveraging community engagement through word-of-mouth marketing. Importantly, the research challenges the narrative that ukay-ukay necessarily harms local fashion industries, instead showing how both sectors could coexist through differentiation, complementary market positioning, and shared commitment to sustainability. This had important implications for local policymakers and industry stakeholders seeking to support both sectors simultaneously.

From Necessity to Identity (Frontiers in Sustainability, 2022)

The study From Necessity to Identity: Entrepreneurial

The study *From Necessity to Identity: Entrepreneurial Perspectives on Key Motivations for Filipino Second-Hand Clothing Consumers*, published in *Frontiers in Sustainability* in 2022, examines the changing motivations behind Filipino consumers' participation in ukay-ukay and the broader secondhand clothing market. Through observational research and in-depth interviews, the study explored how consumers understood, experienced, and assigned meaning to their purchases of pre-loved clothing. Unlike earlier research that primarily presented secondhand clothing consumption as a response to financial hardship, this study revealed a more complex consumer landscape in which affordability intersected with identity construction, personal expression, environmental awareness, and sustainable fashion values.

The research was important because it documented the changing social meaning of ukay-ukay in the Philippines. In the past, secondhand clothing was often associated with economic necessity, limited purchasing power, and social stigma. Consumers who purchased used clothing were sometimes perceived as individuals who could not afford new garments. However, the study demonstrated that this perception had gradually changed. For many Filipino consumers, particularly younger shoppers, ukay-ukay has become a legitimate and meaningful fashion practice. Consumers no longer viewed secondhand clothing only as a cheaper alternative to new apparel. Instead, they increasingly recognized ukay-ukay as a space for creativity, personal style, environmental responsibility, and cultural participation. In this way, the study showed how ukay-ukay had been reinterpreted from a stigma-laden necessity into a

form of self-expression and sustainable consumption [Frontiers in Sustainability, 2022].

The research identified two primary motivations that encouraged Filipino consumers to purchase secondhand clothing: economic necessity and identity construction. The economic dimension remained an important part of the ukay-ukay experience. Secondhand clothing allowed consumers to obtain garments at lower prices while still accessing items that were fashionable, durable, branded, or visually appealing. This was especially relevant for low- and middle-income consumers who wanted to maintain or improve their personal appearance while working within financial limitations. Ukay-ukay provided these consumers with an accessible way to purchase clothing without spending as much as they would on newly manufactured garments from department stores, boutiques, or fast-fashion retailers.

Affordability was also important because many consumers had to manage competing financial responsibilities. Daily living expenses, transportation costs, education-related expenses, household obligations, and unstable or limited incomes influenced the clothing choices of Filipino consumers. By offering lower-priced garments, ukay-ukay made fashion more accessible to people who might otherwise have had to limit their purchases or settle for fewer clothing options. However, the study emphasized that affordability should not be understood as the only reason consumers engaged with secondhand clothing. Although price remained a fundamental motivation, it operated alongside other social, cultural, psychological, and environmental considerations.

The study found that economic motivation had developed together with growing environmental consciousness. Many consumers explicitly framed their ukay-ukay purchases as sustainable fashion choices that reflected concern about textile waste, fast-fashion production, and the environmental effects of excessive consumption. By purchasing pre-loved garments, consumers believed that they were helping extend the useful life of clothing and reducing the need for newly produced apparel.

This understanding connected their individual shopping decisions to wider ideas about circular economy and responsible consumption.

The circular economy perspective was particularly relevant to the study because secondhand clothing allows garments to remain in use for a longer period. Instead of being thrown away when the original owner no longer wanted them, clothing items could be resold, donated, or purchased by another consumer. This process helped circulate products within the market and reduced the likelihood that usable garments would immediately become waste. The study therefore presented ukayukay as a practical example of how consumers could participate in more sustainable systems through everyday purchasing behavior. Although secondhand shopping did not eliminate all of the environmental problems associated with fashion, it provided an alternative to the continuous purchase and disposal of newly produced clothing [Frontiers in Sustainability, 2022].

The second major motivation identified in the research was identity construction. This dimension represented an important shift in how Filipino consumers understood and used secondhand clothing. Rather than purchasing garments only because they were inexpensive, consumers used ukay-ukay to construct and communicate aspects of their identities. Clothing became a way to express personality, taste, creativity, social values, and individuality. Through their purchases, consumers could create fashion statements that reflected who they were and how they wanted to be perceived by others.

The study documented that younger consumers often used ukay-ukay to curate distinctive personal styles that differentiated them from mainstream fast-fashion consumers. Unlike mass-produced clothing, which could result in many individuals wearing similar designs, secondhand markets offered a constantly changing selection of garments. Consumers might discover vintage pieces, rare finds, unusual patterns, discontinued designs, branded items, or one-of-a-kind garments that were not easily available in conventional retail stores. These unique products allowed them to create outfits that appeared more personal and less standardized.

The search process itself also contributed to the identity-building dimension of ukayukay. Consumers often had to browse through large collections, inspect the condition of individual garments, and use creativity to identify items with styling potential. Finding a desirable piece could produce a sense of achievement because the consumer had discovered something valuable that other shoppers might have overlooked. This experience distinguished ukay-ukay from more conventional forms of shopping, where clothing was usually organized and displayed according to

brand, season, size, and style. The process of searching, selecting, and styling secondhand clothing allowed consumers to participate actively in the creation of their appearance.

Identity-driven consumption also reflected the influence of global sustainable fashion trends. Around the world, consumers increasingly used secondhand clothing to reject excessive consumption, challenge fast-fashion culture, and promote more individualized styles. However, the study emphasized that these global trends were expressed within distinctly Philippine cultural contexts. Filipino consumers connected thrift shopping with resourcefulness, practicality, creativity, and the ability to find value in items that others might disregard. These traits were important in shaping the local meaning of ukay-ukay. The practice was therefore not simply a foreign sustainable fashion trend adopted by Filipino consumers; it was also connected to existing cultural values and everyday experiences in the Philippines [Frontiers in Sustainability, 2022].

The entrepreneurial perspectives presented in the study provided further insight into how ukay-ukay sellers adapted to changing consumer motivations. Sellers increasingly positioned their businesses not merely as providers of inexpensive clothing but also as curators of unique fashion and advocates of more sustainable consumption. This represented a significant change in business presentation. Instead of focusing only on low prices, sellers highlighted the uniqueness, quality, rarity, condition, and styling possibilities of their products. They attempted to communicate that secondhand garments could be fashionable, desirable, and environmentally responsible.

This repositioning enabled ukay-ukay businesses to attract consumers who might previously have avoided secondhand clothing because of social stigma. By presenting pre-loved items as curated fashion pieces rather than unwanted or inferior products, sellers helped change public perceptions of used clothing. Some sellers organized their products according to themes, colors, brands, styles, or fashion trends. Others used carefully prepared photographs, styling suggestions, product descriptions, and promotional narratives to increase the perceived value of their garments. These strategies allowed sellers to appeal to consumers who were motivated by personal expression and sustainability as well as those who were primarily concerned with affordability.

Social media played an important role in facilitating this transformation. Digital platforms allowed sellers to display curated collections, announce new arrivals, share styling ideas, communicate sustainability messages, and interact directly with consumers. Through photographs, videos, live-selling sessions, and customer posts, sellers could create a particular image for their businesses and build communities around shared fashion preferences. Social media also enabled consumers to participate in the presentation and promotion of ukay-ukay by sharing purchases, posting outfit combinations, reviewing sellers, and recommending products to friends and followers.

The digital environment helped strengthen the identity-based nature of secondhand fashion. Consumers were able to use social media not only to purchase clothing but also to demonstrate their personal styles and sustainability values. A thrifted outfit could become part of a consumer's online identity, particularly when it was presented as unique, creative, environmentally responsible, or culturally meaningful. Sellers benefited from this process because customers' posts and recommendations provided additional visibility and social proof. In this way, online platforms connected commercial activity with community formation, fashion expression, and environmental communication [Frontiers in Sustainability, 2022].

The study also showed that the motivations of consumers and the strategies of entrepreneurs were closely connected. As consumers began to value uniqueness and sustainability, sellers adapted their products, marketing messages, and business practices to respond to these preferences. At the same time, sellers' efforts to promote ukay-ukay as fashionable and sustainable helped reinforce these motivations among consumers. This created a relationship in which market demand and entrepreneurial communication influenced each other. Consumers expressed interest in affordable, unique, and environmentally responsible clothing, while sellers developed ways to present their products according to these expectations.

This relationship suggested that the continued development of the ukay-ukay market would depend on the ability of sellers to address both practical and symbolic consumer needs. Customers wanted clothing that was affordable, but they also wanted items that reflected their personal identities and values. Sellers who focused exclusively on low prices might attract buyers temporarily, but those who also provided carefully selected products, accurate descriptions, creative presentation,

and meaningful sustainability messages could establish stronger and more lasting relationships with their customers. The study therefore emphasized that ukay-ukay businesses operated within a market shaped by both economic realities and cultural meanings.

The research made an important contribution to local literature by documenting the cultural transformation of ukay-ukay consumption in the Philippines. It moved beyond purely economic explanations and captured the psychological, social, cultural, and environmental dimensions of secondhand clothing consumption. The findings demonstrated that Filipino consumers engaged with ukay-ukay for several interconnected reasons. They purchased secondhand clothing to manage financial limitations, express individuality, discover unique fashion items, participate in sustainable consumption, and challenge negative assumptions about used garments.

The study also showed that the transition from necessity to identity did not mean that economic motivations had disappeared. Affordability continued to be essential, especially for consumers with limited incomes. Instead, the findings suggested that affordability had become only one part of a broader motivation. Consumers could appreciate the low price of a garment while also valuing its uniqueness, environmental benefits, cultural meaning, and contribution to personal style. The different motivations could therefore exist simultaneously rather than replacing one another.

The findings had important implications for policy development and business strategies. Policymakers and sustainability organizations should recognize that secondhand clothing can support both economic accessibility and environmental objectives. Programs that promote clothing reuse, responsible consumption, textile waste reduction, and circular fashion could use ukay-ukay as a locally relevant example. At the same time, efforts to regulate the sector should consider the livelihoods of sellers, the needs of consumers, and the environmental advantages of extending clothing life cycles. Policies that support responsible formalization, consumer protection, business training, and sustainable textile management may be more effective than approaches that treat ukay-ukay solely as an informal or problematic industry.

For businesses, the findings suggested that ukay-ukay sellers could strengthen their market position by recognizing the dual motivations of their consumers. Sellers needed to continue offering affordable clothing while also emphasizing product uniqueness, quality, personal expression, and environmental value. Curated collections, accurate product information, creative styling content, responsible sourcing, and transparent communication could help businesses attract consumers who wanted their purchases to reflect both their financial circumstances and their personal values.

Overall, the study demonstrated that Philippine ukay-ukay culture had evolved from a practice primarily associated with necessity into a multifaceted form of consumption connected to identity, sustainability, and cultural expression. Understanding this dual motivation was essential for developing effective business strategies, sustainability programs, and policy interventions. Filipino consumers participated in ukay-ukay not only because it allowed them to purchase affordable clothing but also because it enabled them to express individuality, support reuse, and align their consumption practices with broader environmental values. The study therefore positioned ukay-ukay as an important part of the Philippine sustainable fashion transition and as a cultural practice that continued to change alongside consumer identities and entrepreneurial strategies [Frontiers in Sustainability, 2022].

The Philippine Ukay-Ukay Culture as Sustainable Fashion (Biana, 2020) Biana's (2020) study, The Philippine Ukay-Ukay Culture

Biana's (2020) study, *The Philippine Ukay-Ukay Culture as Sustainable Fashion*, published in the *DLSU Business & Economics Review*, provides a comprehensive examination of ukayukay as both a cultural phenomenon and a sustainable fashion practice in the Philippines. The research explores how the term "ukay-ukay," which originated from the

Filipino word *halukay*, meaning "to dig" or "to rummage," captures the central experience of Philippine thrift shopping. Consumers typically search through piles or racks of pre-loved clothing in the hope of discovering affordable, fashionable, and distinctive items. This process of searching and discovering clothing reflects the active and participatory nature of ukay-ukay shopping, where consumers do not simply select products from neatly organized displays but examine various garments to find pieces that match their preferences and personal style.

The study was particularly significant because it presented ukay-ukay as more than an economic response to limited income. Although affordability remained an important factor, the research positioned ukay-ukay as a legitimate sustainable fashion movement that corresponded with global principles of circular economy and responsible consumption. Through the reuse and continued circulation of clothing, ukay-ukay challenged the traditional model of purchasing newly manufactured garments, using them for a limited period, and eventually discarding them. At the same time, the practice maintained distinctively Filipino cultural characteristics because it was connected to local shopping traditions, resourcefulness, creativity, community interaction, and the ability to find value in previously owned products. In this sense, Philippine ukay-ukay culture represented the intersection of economic practicality, cultural identity, fashion expression, and environmental awareness [DLSU, 2020].

Biana's research documented that ukay-ukay consumers were predominantly between 21 and 30 years old. A considerable proportion of the consumers were female, college graduates, fully employed, and members of income class B, with an estimated monthly income ranging from Php 20,001 to Php 40,000. This demographic profile challenged the common stereotype that secondhand clothing was purchased exclusively by low-income individuals who had no access to new garments. Instead, the findings revealed that educated and financially stable middleclass Filipinos also actively participated in ukay-ukay culture. Their involvement demonstrated that secondhand fashion was not necessarily a sign of financial difficulty or social disadvantage. For many consumers, it was a deliberate purchasing choice based on the desire to obtain affordable, distinctive, and environmentally responsible clothing.

The study found that these consumers were motivated by both economic considerations and growing environmental consciousness. On one hand, ukay-ukay allowed consumers to purchase clothing at prices that were generally lower than those of brand-new garments in department stores, boutiques, or fast-fashion retailers. This affordability enabled them to expand their wardrobes, experiment with different styles, and purchase branded or high-quality pieces at reduced prices. On the other hand, many consumers explicitly understood their purchases as sustainable fashion choices. They recognized that buying pre-loved clothing could help reduce textile waste, extend the useful life of garments, and lessen the demand for newly produced apparel. This combination of financial practicality and environmental awareness created a distinctive market position for Philippine ukay-ukay, distinguishing it from secondhand clothing markets that were understood primarily through the lens of economic necessity [DLSU, 2020].

The research also emphasized the cultural significance of ukay-ukay in Philippine society. Over time, ukay-ukay had developed from informal stalls in wet markets and public marketplaces into a diverse retail sector that included specialized stores, upscale thrift boutiques, social media shops, online marketplaces, and live-selling platforms. This transformation demonstrated the changing ways in which Filipino consumers perceived secondhand clothing. Previously, used garments were often associated with poverty, low social status, or the inability to purchase new clothes. However, the growing popularity of ukay-ukay contributed to the gradual removal of this stigma. Secondhand clothing increasingly became associated with creativity, individuality, resourcefulness, affordability, and environmental responsibility.

This change in perception was especially visible among Filipino youth. Young consumers viewed thrift shopping as an opportunity to develop personal styles that differed from the standardized appearance often associated with mass-produced fast fashion. Because ukayukay stores contained garments from different brands, periods, countries, and fashion trends, consumers could discover unusual pieces that were not commonly available in conventional retail stores. This allowed them to combine vintage clothing, contemporary styles, branded garments, and personalized accessories in ways that reflected their identities. For younger consumers, ukay-ukay therefore became both a shopping activity and a form of selfexpression. It allowed them to present themselves as creative and fashion-conscious while also demonstrating awareness of the environmental problems associated with excessive consumption [DLSU, 2020].

The cultural development of ukay-ukay also demonstrated the adaptability of the industry. Sellers adjusted to changes in consumer behavior by moving beyond physical stores and using digital platforms to promote their products. Online shops, social media pages, and liveselling sessions enabled sellers to reach customers who might not have regularly visited traditional ukay-ukay locations. This digital expansion increased the visibility of secondhand clothing and made it more accessible to consumers in different geographic areas. It also allowed sellers to present ukay-ukay products in more organized and appealing ways through photographs, styling content, product descriptions, and interactive promotions. As a result, the industry continued to develop despite legal restrictions on the commercial importation of used clothing and textiles in the Philippines [DLSU, 2020].

Biana's research further examined the environmental implications of ukay-ukay consumption by positioning it as a practical expression of circular fashion principles in the Philippine context. Circular fashion emphasizes the continued use, reuse, repair, resale, and recycling of clothing rather than the constant production and disposal of garments. Ukay-ukay contributes to this

approach by extending the life cycle of clothing items. Instead of being discarded after the original owner no longer wants or needs them, garments are transferred to other consumers who can continue using them. This process reduces the possibility that usable clothing will immediately become waste and allows the environmental resources invested in production, transportation, and distribution to generate value for a longer period.

The continued circulation of secondhand clothing may also help reduce the demand for newly manufactured garments. The production of new clothing requires raw materials, water, energy, labor, transportation, and packaging. When consumers purchase pre-loved clothing, they participate in a consumption model that prioritizes reuse over the constant purchase of newly produced items. Although secondhand clothing alone cannot resolve all environmental problems associated with the fashion industry, the study presented ukay-ukay as a practical and accessible way for Filipino consumers to participate in more sustainable consumption patterns. It connected everyday shopping decisions with broader issues such as textile waste, overproduction, resource depletion, and the environmental effects of fast fashion.

The study also noted that the Philippines led Southeast Asia in secondhand fashion transactions, accounting for more than 30 percent of regional activity according to Carousell data. This finding suggested that the Philippine ukay-ukay industry had achieved a significant level of popularity and consumer acceptance. Rather than remaining a marginal alternative used only by a small group of consumers, secondhand fashion had developed into a mainstream consumption practice with cultural and economic importance. The widespread participation of Filipino consumers indicated that ukay-ukay had become embedded in everyday shopping behavior and was no longer limited to traditional physical stalls or informal markets. Its presence across stores, boutiques, online platforms, and social media further demonstrated the industry's ability to respond to changing consumer preferences [DLSU, 2020].

The popularity of ukay-ukay also reflected the resourcefulness commonly associated with Filipino consumers. Shoppers often used creativity and patience to find high-value items at affordable prices. The act of searching through clothing collections allowed consumers to experience the satisfaction of discovering unique garments that might not be available in mainstream stores. This sense of discovery contributed to the cultural appeal of ukay-ukay and differentiated it from fast-fashion shopping, where consumers generally encountered standardized products that were widely available to the public. Ukay-ukay therefore provided not only material benefits but also emotional and social value through the experience of finding rare, distinctive, or personally meaningful clothing pieces.

At the same time, the study recognized that the sustainability of ukay-ukay should be understood critically. Purchasing secondhand clothing can support reuse and waste reduction, but sustainable consumption also depends on how frequently consumers purchase, how long they use garments, and what happens to the items when they are no longer wanted. If consumers buy excessive amounts of secondhand clothing without using them for extended periods, the practice may still contribute to overconsumption.

Nevertheless, Biana's research showed that ukay-ukay created an important opportunity to challenge disposable fashion habits. It encouraged consumers to consider the history, usefulness, and continued value of garments rather than treating clothing as products that could be quickly purchased and discarded.

The study made important contributions to local literature by connecting cultural, economic, and environmental perspectives on ukay-ukay consumption. It provided empirical evidence that Filipino consumers engaged with secondhand clothing for complex and overlapping reasons that extended beyond simple affordability. Consumers were influenced by price, but they were also motivated by environmental concerns, the opportunity to express individuality, the excitement of discovering unique items, and the cultural acceptance of thrift shopping. These motivations demonstrated that ukay-ukay should not be examined only as a retail sector or informal market. It should also be understood as a cultural practice that reflects changing Filipino attitudes toward fashion, consumption, waste, and sustainability [DLSU, 2020].

The findings also had implications for policymakers considering how to regulate the ukay-ukay industry. Because the practice generated economic activity while also supporting the reuse of clothing, the research suggested that formalization and responsible regulation might serve both economic and environmental objectives more effectively than outright prohibition. Policies could focus on improving sanitation and product quality standards, protecting workers and consumers, supporting small businesses, and strengthening systems for textile reuse and waste management. A balanced regulatory approach could recognize the importance of ukayukay to consumers and entrepreneurs while addressing legitimate concerns related to public health, trade, and environmental management.

For businesses, the study suggested that sellers could strengthen their market position by communicating both the economic and environmental value of their products. Sellers could emphasize the quality, uniqueness, durability, and history of pre-loved garments while also promoting responsible consumption. Businesses that adopted transparent and environmentally responsible practices could appeal to consumers who wanted their purchases to reflect their

sustainability values. The findings also indicated that retailers could benefit from combining traditional ukay-ukay practices with digital marketing, online selling, styling content, and social media engagement.

For researchers, Biana's study opened several opportunities for further investigation into the relationship between ukay-ukay culture, Philippine identity, sustainability values, and consumer behavior. Future studies could examine how younger consumers develop their attitudes toward secondhand clothing, how online platforms influence the growth of ukay-ukay, and how sustainability beliefs affect purchasing frequency and brand loyalty. Researchers could also explore the experiences of sellers, the role of gender and income in ukay-ukay consumption, and the long-term environmental impact of secondhand clothing practices. Overall, the study established ukay-ukay as an important part of Philippine fashion culture and demonstrated its potential to support more sustainable forms of consumption [DLSU, 2020].

How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling (2024)

The study How Facebook Communication Strategies Facilitate

Ukay-Ukay Live

The study *How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling and Buying (2024)*, published in the *European Proceedings of Social and Behavioural Sciences*, examines the communication strategies used by Philippine ukay-ukay sellers during Facebook Live selling sessions and how these strategies influence buyers' consumption behavior. The research addressed an important gap in the existing literature by examining how digital platforms transformed traditional ukay-ukay selling into interactive, real-time commerce. Although ukay-ukay had traditionally been associated with physical stores and face-to-face bargaining, the emergence of Facebook Live allowed sellers to display products, respond to inquiries, build relationships with customers, and complete transactions in a virtual environment. The study was particularly relevant because Facebook Live had become an important platform for small online businesses, including ukay-ukay sellers, and created new forms of seller-buyer interaction that differed significantly from conventional retail experiences [European Proceedings, 2024].

Unlike traditional online shopping, where consumers generally view static photographs and read written product descriptions, live selling allowed buyers to receive immediate information directly from sellers. The research used a qualitative

phenomenological design and gathered data through in-depth online interviews with ten participants, consisting of five ukay-ukay sellers and five buyers. The researchers also observed sellers during their live-selling sessions to verify the communication strategies discussed during the interviews. Through the analysis of the participants' experiences and perspectives, the study examined the verbal and non-verbal techniques used by sellers, the effectiveness of these techniques, and the ways in which they affected buyer engagement, purchasing decisions, and customer satisfaction [European Proceedings, 2024].

The research identified several communication strategies commonly employed by ukay-ukay sellers during Facebook Live sessions. One of the most prominent strategies involved the use of entertainment or "gimmicks" to maintain viewer interest. Sellers used humor, jokes, games, giveaways, rewards, special promotions, and other interactive activities to prevent the live session from becoming boring. These activities encouraged viewers to remain in the broadcast for longer periods and helped sellers compete for attention in an environment where consumers could easily scroll past or leave a live stream. Gimmicks also created a more relaxed and enjoyable atmosphere, allowing the selling process to become not only a commercial transaction but also a form of social interaction between the seller and the audience [European Proceedings, 2024].

Price promotions represented another important strategy used during live selling. Sellers offered discounts, flash sales, bundle deals, and limited-time promotions to encourage customers to purchase immediately. For example, sellers might offer several clothing pieces at a combined price or reduce the price of an item during a specific portion of the live session. These promotions created a sense of urgency and encouraged viewers to make faster decisions before the offer ended or another buyer claimed the product. Since ukay-ukay buyers were often attracted by affordability, discounts and special offers strengthened the appeal of secondhand clothing and encouraged consumers to participate actively in the live-selling process.

Sellers also used informal and conversational communication styles to establish personal connections with their viewers. They interacted with buyers through comments, jokes, personal opinions, greetings, and terms of endearment. Rather than presenting themselves as distant or formal retailers, sellers attempted to

appear approachable, friendly, and relatable. This communication style helped create a sense of familiarity and community among participants. Buyers were more likely to remain engaged when they felt that the seller acknowledged their presence, responded to their comments, and treated them as active participants rather than anonymous customers. The study suggested that this personal connection was an important feature of live selling because it distinguished the experience from purchasing through static online listings [European Proceedings, 2024].

Product presentation was another significant communication strategy identified in the study. Sellers demonstrated clothing items by displaying them on mannequins, placing them on hangers, or showing how they looked when worn by an actual person. This “when worn” approach helped buyers visualize the fit, appearance, and overall style of the clothing. Sellers also described the material, color, size, design, and condition of each garment while presenting it on camera. Showing the products from different angles allowed viewers to examine details that might not have been visible in a single photograph. This was particularly important in the ukay-ukay market because buyers needed to evaluate pre-owned items carefully before deciding whether they were worth purchasing.

Transparency and honesty were also central to effective product communication. Sellers were expected to disclose visible flaws such as stains, holes, discoloration, fading, fabric damage, or other signs of wear. Because secondhand garments were not always in perfect condition, buyers relied on sellers to provide accurate descriptions rather than presenting the items in a misleading way. The study found that honest sellers were more likely to gain buyer trust because customers understood that the clothing had been previously owned. When sellers openly addressed defects and answered questions about the actual condition of an item, they reduced the possibility of buyer disappointment and strengthened the credibility of their online shops [European Proceedings, 2024].

The research further showed that non-verbal elements also influenced the effectiveness of Facebook Live selling. Sellers paid attention to their appearance, posture, presentation, and the background visible during the broadcast. Some sellers attempted to look presentable because they believed that their appearance affected how buyers evaluated the quality and legitimacy of the products being sold. A clean and organized background also helped direct attention toward the clothing

items. Some sellers displayed garments on hangers or mannequins behind them, while others used plain backgrounds so that the products would stand out more clearly on camera. These visual elements contributed to the overall professionalism of the live session and influenced the level of trust and engagement generated among viewers.

The study also identified the use of Facebook's sharing function as an important strategy for expanding the reach of live-selling sessions. Sellers encouraged viewers to share the broadcast with friends, relatives, and members of online groups who might be interested in ukay-ukay products. Sharing allowed sellers to reach potential buyers beyond their immediate followers without requiring substantial advertising expenses. It also helped create a network effect in which existing viewers introduced the live stream to other members of their social networks. As a result, Facebook Live served not only as a selling platform but also as a low-cost promotional tool for small businesses that might not have the resources to use traditional forms of advertising [European Proceedings, 2024].

The study found that effective communication strategies influenced buyer consumption behavior through several related mechanisms. First, real-time interaction reduces purchase uncertainty. Buyers could ask immediate questions about measurements, fabric, condition, color, price, and availability. Sellers could respond directly and provide additional information that might not have been included in a written product listing. This immediate exchange addressed one of the major challenges of online apparel shopping: buyers could not physically touch, inspect, or try on the clothing before making a purchase. Through live demonstrations and direct communication, sellers helped consumers make more informed purchasing decisions.

Second, the interactive nature of live selling created a sense of community among viewers and sellers. Buyers could observe other participants claiming items, asking questions, commenting on products, and reacting to the seller's presentation. This social environment made the shopping experience more engaging and encouraged viewers to remain in the live stream. The presence of other buyers could also create social proof, as customers might interpret active participation and successful purchases as evidence that the seller was trustworthy or that the products were

desirable. In this way, purchasing became a shared social activity rather than an isolated individual decision.

Third, the entertainment value of live selling encouraged prolonged viewer engagement. Games, jokes, giveaways, promotions, and interactive product demonstrations helped sustain viewers' attention during sessions that could last for an extended period. As viewers spent more time watching, they were exposed to a larger number of products and promotions. This increased their opportunities to discover items that matched their interests and could encourage unplanned or impulse purchases. However, the study primarily documented how participants experienced these strategies and did not establish a universal numerical effect on conversion rates or basket size. The findings instead indicated that entertainment and interaction made buyers more likely to stay, participate, and consider purchasing products during the live session [European Proceedings, 2024].

The study also highlighted the importance of seller responsiveness in completing transactions. Buyers expected sellers to acknowledge comments, reserve items accurately, answer questions, and provide clear instructions for payment and delivery. Since several customers might attempt to claim the same item simultaneously, sellers needed to monitor the comment section carefully and respond quickly. Private messaging could also be used to confirm details, clarify concerns, and secure a sale. The researchers noted that some viewers expressed interest in products but did not ultimately complete their purchases. These individuals, sometimes referred to as "joy miners" or irresponsible buyers, created difficulties for sellers by reserving items and later cancelling or failing to pay. Sellers therefore had to develop rules and procedures for managing reservations, confirming payments, and limiting future transactions with unreliable buyers [European Proceedings, 2024].

Despite the advantages of live selling, the research documented several challenges that affected the quality and effectiveness of ukay-ukay broadcasts. Technical difficulties were among the most common problems encountered by sellers. Unstable internet connections could interrupt the broadcast, cause audio or video delays, prevent comments from appearing immediately, or result in a dark or frozen screen. Poor lighting could also make it difficult for buyers to see the actual color, texture, or condition of a garment. Similarly, background noise and audio problems could interfere with communication and prevent viewers from hearing product

descriptions or responding to the seller's instructions. These technical problems could reduce buyer confidence and cause viewers to leave the live session.

Managing large numbers of comments and inquiries also required considerable organizational ability and quick decision-making. Sellers had to respond to questions, identify which buyer claimed an item first, track reserved products, describe new garments, announce promotions, and monitor payment or delivery concerns at the same time. Mistakes in identifying buyers or recording reservations could lead to conflict and dissatisfaction. Sellers also had to manage unpredictable levels of viewer engagement because the number of people attending a live session could vary from one broadcast to another. Some sessions attracted many viewers and generated high levels of interaction, while others received fewer comments and shares. Successful sellers responded to these challenges by refining their strategies through continuous practice, observation, feedback, and experimentation

with different presentation styles and promotional techniques [European Proceedings, 2024].

The research made a significant contribution to local literature by providing empirical evidence on the communication practices involved in digital ukay-ukay commerce in the Philippines. It demonstrated how Facebook Live enabled small businesses to compete through relatively affordable yet highly interactive marketing strategies. Rather than depending only on product prices or static advertisements, sellers could use communication, entertainment, transparency, and personal connection to attract and retain customers. The study also showed that digital selling required more than simply uploading photographs of available products. Sellers needed to perform, communicate, respond, demonstrate, and build relationships in real time.

The findings had practical implications for current and aspiring ukay-ukay entrepreneurs. Sellers could improve their live-selling performance by developing stronger communication skills, presenting products clearly, disclosing defects honestly, using appropriate lighting and backgrounds, responding promptly to inquiries, and creating simple but engaging activities for viewers. They could also establish clear rules for reservations, payments, shipping, and cancellations to reduce misunderstandings and prevent problems with unreliable buyers. Consistent schedules, regular updates, and the effective use of Facebook's sharing features could further help sellers develop a stable audience and increase customer engagement.

For policymakers, educational institutions, and support organizations, the study highlighted the importance of digital literacy and technical training for small businesses operating in the online ukay-ukay sector. Training programs could focus on live-stream presentation, customer communication, digital marketing, online safety, product photography, transaction management, and the use of social media analytics. Improving access to reliable internet infrastructure would also help sellers conduct more stable and professional live broadcasts. Overall, the research suggested that the continued growth of online ukay-ukay would depend not only on the affordability of secondhand clothing but also on sellers' ability to create trustworthy, responsive, entertaining, and socially engaging digital shopping experiences [European Proceedings, 2024].

Level of Satisfaction Among Women on Online Ukay-Ukay

(2023) The study Level of Satisfaction Among Women on

Online Ukay- Ukay (2023), published in the International

The study published in the Journal of Academic Research and Innovative Engineering Education (IJARIE, 2023) investigates female consumer satisfaction with online ukay-ukay purchasing experiences in the Philippines. This research was significant because women comprise a large proportion of ukay-ukay consumers, yet their specific experiences, concerns, expectations, and sources of satisfaction had received limited academic attention. As online shopping continued to influence the way consumers purchased secondhand clothing, it became important to understand the factors that shaped women's perceptions of online ukay-ukay sellers and their overall purchasing experiences. The study employed quantitative research methods to measure customer satisfaction across several dimensions, including product quality, pricing, delivery service, seller communication, and perceived value for money.

The research identified several key factors that influenced women's satisfaction with online ukay-ukay purchases. Product quality emerged as the most significant determinant of satisfaction. Respondents were particularly concerned with the actual condition of garments, the accuracy and completeness of product

descriptions, and the possibility of discrepancies between the items shown in photographs and the products received upon delivery. Since online buyers could not physically inspect or try on the clothing before purchasing, they relied heavily on the seller's descriptions and visual presentation. Any difference between the advertised condition of an item and its actual condition could result in disappointment, distrust, and negative evaluations of the seller.

Price competitiveness also remained an important factor in determining customer satisfaction. Many consumers were attracted to online ukay-ukay because it provided access to affordable clothing, including branded or unique pieces that might otherwise be expensive when purchased new. However, the study found that satisfied customers were willing to pay slightly higher prices when sellers offered guaranteed product quality, accurate descriptions, dependable service, and secure transactions. This finding suggested that low prices alone were not enough to ensure customer satisfaction. Consumers also evaluated whether the price of an item was reasonable in relation to its condition, brand, uniqueness, and the quality of the overall purchasing experience.

Delivery efficiency significantly affected the satisfaction of female online ukay-ukay consumers. Delayed shipments, unclear delivery schedules, high shipping costs, and inadequate packaging contributed to negative purchasing experiences. Because secondhand garments were often sold as individual or limited pieces, customers expected sellers to process and ship their orders within the promised timeframe. Poor packaging could also damage the clothing or create the impression that the seller lacked professionalism and concern for product quality. In contrast, timely delivery, secure packaging, and accurate tracking information helped strengthen customer confidence and contributed to more positive evaluations of the seller.

Customer communication also emerged as a substantial factor affecting overall satisfaction ratings. Respondents valued sellers who responded promptly to inquiries, provided complete product information, and communicated clearly throughout the purchasing process. Effective communication was especially important when customers asked about garment measurements, fabric condition, stains, defects, shipping arrangements, or return procedures. The ability of sellers to address concerns and resolve problems also influenced whether customers viewed the transaction positively. Sellers who ignored messages, provided incomplete

answers, or failed to resolve complaints were more likely to receive unfavorable evaluations, even when the products themselves were reasonably priced [IJARIE, 2023].

The study found that women's satisfaction with online ukay-ukay was generally moderate to high, with the mean satisfaction scores indicating that respondents experienced relatively positive purchasing outcomes overall. Nevertheless, the research identified several specific problems that could undermine customer satisfaction. These included inconsistent or inaccurate sizing information, difficulty determining the true condition of garments from photographs, incomplete product descriptions, and limited options for customers who received items that did not meet their expectations. Because clothing sizes may vary across brands and styles, respondents preferred sellers who provided exact measurements rather than relying only on general labels such as small, medium, or large.

The respondents also expressed a strong preference for sellers who provided detailed descriptions of each garment's condition, multiple photographs taken from different angles, clear sizing charts, and transparent return or exchange policies. These forms of information helped customers make more informed purchasing decisions and reduced the uncertainty associated with buying secondhand clothing online. Clear photographs were particularly important because they allowed customers to inspect the fabric, color, design, visible flaws, and overall appearance of a garment before completing a purchase. Similarly, transparent return policies provided customers with greater confidence because they offered some form of protection in cases where the product differed significantly from its online presentation.

Trust-building mechanisms were also identified as critical in reducing perceived risk and increasing customer satisfaction. Seller ratings, customer reviews, testimonials, and other forms of social proof helped potential buyers evaluate the reliability of online ukay-ukay sellers. Consumers were more likely to trust sellers who had positive feedback, consistent transaction histories, and visible evidence of satisfied customers. Social media engagement also contributed to seller credibility, particularly when sellers regularly posted actual product images, responded to customer comments, and demonstrated transparency in their business practices. These mechanisms were important because online ukay-ukay transactions often

required customers to make purchasing decisions without personally knowing the seller or physically examining the product before payment.

The research also documented demographic patterns in satisfaction levels. Younger women, particularly those between the ages of 18 and 30, reported higher levels of satisfaction with online ukay-ukay compared with older age groups. This generational difference may have reflected younger consumers' greater digital literacy, familiarity with online shopping platforms, and comfort with digital payment and communication systems. Younger women may also have been more culturally accepting of secondhand clothing because of the growing popularity of thrift fashion, online selling, social media styling, and sustainability-related consumer practices.

Their familiarity with digital platforms could have made it easier for them to compare sellers, interpret online reviews, evaluate product photographs, and manage the risks associated with online purchases.

The study further found that repeat customers reported higher satisfaction than first-time buyers. This finding suggested that previous experience with online ukay-ukay helped consumers reduce uncertainty and develop more effective purchasing strategies. Repeat customers may have become more familiar with reliable sellers, common sizing differences, typical product descriptions, shipping procedures, and the quality standards of particular online shops. Over time, this experience allowed them to identify trustworthy sellers and avoid transactions that might result in dissatisfaction. First-time buyers, by contrast, may have experienced greater uncertainty because they had limited knowledge of the seller's reliability, product quality, communication practices, and return policies.

This learning effect had important implications for sellers who wanted to build customer loyalty and encourage repeat business. Providing a positive first transaction could increase the likelihood that customers would return, recommend the seller to others, and develop long-term trust in the online ukay-ukay store. Sellers could strengthen these relationships by maintaining consistent product quality, accurately describing garment conditions, responding promptly to inquiries, packaging orders securely, and addressing complaints fairly. Customer loyalty therefore depended not only on offering affordable products but also on consistently delivering a reliable and transparent shopping experience.

The study contributed important insights to the local literature on female consumer behavior and online secondhand clothing consumption in the Philippines. It provided empirical evidence that women's satisfaction with online ukay-ukay depended on multiple interconnected factors that extended beyond simple price considerations. Product quality, accurate information, delivery reliability, seller communication, customer service, and trust all contributed to the way consumers evaluated their purchasing experiences. The findings demonstrated that women did not assess online ukay-ukay purchases solely according to whether the item was inexpensive. Instead, they considered whether the product met their expectations and whether the seller provided a dependable and satisfactory transaction.

The findings also suggested that sellers who invested in quality control, transparent communication, accurate product presentation, and reliable service delivery could distinguish themselves in an increasingly competitive online market. By providing detailed garment information, clear photographs, accurate measurements, responsive customer service, and reasonable return policies, sellers could reduce perceived risk and improve customer satisfaction. These practices could also help sellers establish sustainable relationships with customers, encourage repeat purchases, and generate positive word-of-mouth recommendations.

For future researchers, the study identified several opportunities for further investigation. These include examining gender-specific consumption patterns, comparing satisfaction drivers across different segments of the ukay-ukay market, and exploring how satisfaction influences customer loyalty over time. Future studies could also investigate differences between social media-based sellers, e-commerce platforms, and live-selling businesses. In addition, researchers could examine how factors such as sustainability awareness, brand preference, income level, body size, and previous online shopping experience influence women's satisfaction with online ukay-ukay purchases. Such research would provide a more comprehensive understanding of the long-term relationship between satisfaction, trust, repeat purchasing, and loyalty in the Philippine online ukay-ukay market.

Filipino Gen Z Consumer Behavior Towards Ukay-Ukay (DLSU,

2024)

The study *Filipino Gen Z Consumer Behavior Towards Ukay-Ukay and Their Sustainable Fashion Practices* (2024), published in the De La Salle University Senior High School Research Convention Proceedings, examines how Filipino Generation Z consumers participate in ukay-ukay and the motivations that influence their engagement in sustainable fashion practices. The study was particularly timely because Generation Z had become one of the fastest-growing consumer groups in the ukay-ukay market. Their participation also contributed to the cultural transformation of secondhand clothing, which gradually shifted from being associated with social stigma or financial necessity to becoming a more accepted and mainstream fashion choice. Using a survey methodology, the researchers investigated the attitudes, purchasing behaviors, and personal values that shaped Filipino Gen Z consumers' relationship with ukay-ukay and sustainable fashion consumption [DLSU, 2024].

The study found that Filipino Gen Z consumers participated in ukay-ukay primarily because of three interconnected motivations: affordability, sustainability consciousness, and unique fashion expression. Affordability remained a fundamental reason for purchasing secondhand clothing, particularly because many Gen Z consumers faced financial limitations such as student loans, entry-level salaries, limited disposable income, and the rising cost of living. Ukay-ukay allowed consumers to purchase clothing at lower prices while still accessing a wide variety of styles, brands, and clothing pieces. However, the research also revealed that affordability was no longer the only factor influencing their purchasing decisions. Sustainability consciousness had become equally important, as many respondents explicitly viewed their ukay-ukay purchases as environmental choices connected to their values concerning climate action, waste reduction, and responsible consumption. A third motivation was unique fashion expression. Gen Z consumers used ukay-ukay to curate distinctive personal styles that allowed them to express their individuality and differentiate themselves from peers who wore mass-produced fast-fashion clothing [DLSU, 2024].

The research also documented that Filipino Gen Z consumers predominantly accessed ukay-ukay products through digital platforms. Facebook Marketplace, Instagram, and TikTok served as important channels for discovering, promoting, and purchasing secondhand clothing. These platforms were consistent with Gen Z's digital-native communication and shopping preferences because they provided quick access to sellers, product information, styling ideas, and curated selections

beyond traditional physical ukay-ukay stores. Through online platforms, consumers could browse clothing items conveniently, compare prices, communicate directly with sellers, and participate in digital communities centered on thrift fashion. The study further found that Gen Z consumers actively researched sellers before completing their purchases. They checked seller ratings, customer reviews, previous transactions, and social media presence to assess credibility and reduce the risks associated with online shopping. In addition, many consumers engaged with ukay-ukay content creators who shared styling tips, thrift hauls, clothing recommendations, and educational content about sustainable fashion. These creators helped establish a digital ecosystem that normalized, promoted, and celebrated secondhand clothing consumption among younger consumers [DLSU, 2024].

The study emphasized that Gen Z's participation in ukay-ukay reflected broader values associated with the circular economy and sustainable consumption. Many respondents expressed concern about the environmental consequences of textile waste, the resource-intensive nature of fast-fashion production, and the ethical issues connected to the clothing industry. By purchasing secondhand clothing, consumers were able to extend the life cycle of garments, reduce the demand for newly produced clothing, and contribute to the diversion of reusable items from landfills. This values-driven form of consumption distinguished Filipino Gen Z consumers from previous generations, who were often perceived as engaging with ukay-ukay primarily because of economic necessity. For Gen Z, ukay-ukay could function not only as an affordable shopping option but also as a form of environmental participation and personal value expression. The study further noted that this alignment between sustainability values and purchasing behavior contributed to stronger consumer loyalty. Gen Z consumers were more likely to support sellers who communicated clear sustainability messages, promoted responsible consumption, and demonstrated concern for environmental protection. This trend suggested that the future growth of the ukay-ukay market would depend partly on its ability to maintain credibility and remain aligned with the sustainability values of Gen Z consumers [DLSU, 2024].

The study made significant contributions to the local literature on sustainable fashion, secondhand consumption, and Filipino consumer behavior. By documenting the changing motivations and purchasing patterns of Gen Z consumers, it provided evidence of a generational shift in the way ukay-ukay was perceived and practiced in the Philippines. The findings showed that Filipino Gen Z

consumers represented a critical demographic in the transition toward more sustainable fashion systems because of their high level of engagement with digital platforms, strong interest in individual self-expression, and growing awareness of environmental concerns. Their participation demonstrated that secondhand clothing could become an important component of sustainable consumption when it was connected to affordability, personal identity, and environmental responsibility.

The findings also had important implications for policymakers, educators, and businesses seeking to encourage circular economy practices. Policymakers could use the findings to support initiatives that reduce textile waste, strengthen secondhand clothing systems, and promote responsible consumer behavior. Educators could incorporate ukay-ukay and sustainable fashion into environmental education programs to demonstrate how everyday purchasing decisions influence waste generation and resource consumption. Businesses and online sellers, meanwhile, could appeal to Gen Z consumers by providing transparent information about their products, adopting responsible selling practices, and communicating genuine sustainability commitments. The study suggested that Gen Z's participation in ukay-ukay could be used as an entry point for broader sustainability education and behavior-change initiatives. For future researchers, the study identified opportunities for longitudinal research examining how Gen Z's sustainable fashion values, shopping habits, and environmental attitudes develop as they grow older, enter the workforce, and gain greater purchasing power [DLSU, 2024].

Experiences of Online Sellers of Pre-Loved Clothes (Zenodo, 2023)

The study *Experiences of Online Sellers of Pre-Loved Clothes (2023)*, archived on Zenodo, explored the lived experiences, challenges, and adaptive strategies of online sellers of pre-loved clothing in the Philippines. This qualitative research provided critical seller-centered perspectives that complemented consumer-focused studies and offered a more comprehensive understanding of the digital ukay ecosystem. Through in-depth interviews and observational methods, the research captured the day-to-day realities of individuals who had transitioned into online ukay entrepreneurship, documenting both the opportunities they encountered and the obstacles they faced within this informal yet highly dynamic market [Zenodo, 2023].

The study documented several critical challenges faced by online sellers of preloved clothing. Inventory sourcing emerged as one of the primary difficulties, with sellers

reporting irregular availability of quality items, competition from other sellers for desirable pieces, and the physical labor involved in sorting, cleaning, and preparing garments for sale. Pricing also presented a significant challenge, as sellers had to balance competitiveness and profitability while considering the varying conditions of individual items and the uncertainty of customer demand.

Platform dependency further created vulnerabilities, as changes in the algorithms of Facebook, Instagram, or Shopee could significantly affect product visibility and sales. Fraud and scam attempts by buyers, including the use of fake payment screenshots and refusal to claim purchased items, also represented ongoing financial risks that required sellers to remain constantly vigilant [Zenodo, 2023].

Despite these challenges, the study identified several adaptive strategies employed by successful online sellers. Multi-channel selling emerged as a critical strategy, with sellers maintaining a presence across multiple platforms, including Facebook Marketplace, Instagram, Shopee, and TikTok. This approach helped reduce their dependence on a single platform while allowing them to maximize their market reach and increase opportunities for sales. Content creation capabilities, including product photography, video showcasing, and the development of engaging captions, also became essential competitive advantages. Relationship-building through personalized communication, follow-up messages, and loyalty programs helped sellers establish and maintain repeat customer bases, which contributed to greater income stability. In addition, financial management skills, such as inventory tracking, cash flow management, and maintaining emergency funds, proved essential in helping sellers manage the income volatility associated with online selling [Zenodo, 2023].

The research further highlighted the entrepreneurial resilience of online ukay sellers, many of whom balanced their selling activities with other forms of employment or education. For some sellers, online ukay selling represented their primary source of livelihood, while for others, it served as a means of generating supplementary income or as a pathway toward full-time entrepreneurship. The study found that sellers developed diverse skill sets that extended across several areas, including product sourcing, digital marketing, customer service, photography, and financial management. This entrepreneurial adaptability reflected broader Filipino cultural values of resourcefulness, flexibility, and improvisation, which enabled sellers to continue operating and developing their businesses despite functioning within an informal and lightly regulated market with limited institutional support [Zenodo, 2023].

This study made important contributions to the local literature by centering the perspectives of sellers in understanding the digital ukay economy. It revealed that online ukay selling required sophisticated capabilities that extended far beyond

simple reselling, as sellers had to demonstrate digital literacy, business management skills, marketing capabilities, and effective interpersonal communication. The findings also had practical implications for organizations seeking to support online ukay entrepreneurs, as they suggested that capacitybuilding programs should address areas such as digital marketing, financial management, fraud prevention, customer relations, and legal compliance. For policymakers, the research highlighted the economic significance of online ukay selling as a source of livelihood and entrepreneurship. Rather than focusing solely on prohibition or restriction, the findings suggested that appropriate formalization and support mechanisms could better serve both sellers and consumers by promoting safer, more sustainable, and more organized online selling practices [Zenodo, 2023].

Consumer Behavior Towards Online Clothes Shopping in San Fernando City, La Union

Torres (2021) conducted a local study entitled *Consumer Behavior Towards Online Clothes Shopping in San Fernando City, La Union*, which described how consumers in the urban barangays of San Fernando City, La Union, behaved when purchasing clothes online. The research primarily aimed to present the demographic profile of online clothing shoppers and identify the factors that influenced their purchasing decisions, including convenience, website features, time considerations, and the security of online transactions. A descriptive research design using a survey questionnaire was employed to gather data from individuals who had previous experience purchasing clothes through online platforms. This approach allowed the researcher to capture the actual experiences and behaviors of online clothing shoppers in the area during the period of the study.

The study involved 382 respondents who were online clothing shoppers residing in selected urban barangays of San Fernando City, La Union. The respondents were selected based on their engagement in online clothing purchases to ensure that the findings reflected actual consumer behavior rather than hypothetical opinions. The questionnaire gathered information regarding their demographic characteristics, access to computers and the internet, and their self-rated internet skills. Through descriptive analysis, the researcher summarized these data to provide a clear

representation of the local online shopping environment and the characteristics of consumers who participated in online clothing purchases.

The results indicated that most of the online clothing shoppers were female employees aged 21 to 25 years old, with a monthly income of at least Php 5,000. Many of the respondents had access to a computer unit and an internet connection either at home or at work, which made it more convenient for them to participate in online shopping activities. They also described their internet skills as good, indicating that they were relatively confident in navigating online shopping websites and completing online transactions. These conditions contributed to their active participation in online clothing purchases and demonstrated how access to digital technology and internet skills could support consumers' engagement with online shopping platforms.

The findings further showed that several factors strongly influenced the respondents' online buying behavior, particularly convenience, website design and features, time-saving advantages, and security concerns. Convenience referred to the ability to shop at any time and from any location without the need to visit physical stores. Attractive, organized, and user-friendly websites also contributed to a more positive online shopping experience and encouraged consumers to engage in online purchasing. These findings indicated that consumers considered not only the products being offered but also the overall quality and accessibility of the online shopping platform when making purchasing decisions.

Time-saving was also identified as an important consideration because online shopping reduced the need for consumers to travel to physical stores and spend extended periods of time in shopping malls. Through online platforms, consumers were able to browse available clothing products and make purchasing decisions without having to leave their homes or workplaces. In addition, the perceived safety and security of personal and financial information played a major role in determining whether customers felt comfortable continuing with their online purchases. Concerns regarding transaction security therefore became an important factor that online clothing businesses needed to address in order to establish consumer trust and encourage continued patronage.

Based on these results, Torres (2021) recommended that entrepreneurs consider engaging in online selling as an alternative approach to conducting business within the clothing industry. The study emphasized that business owners could better respond to the needs and preferences of online consumers by improving website design, providing secure payment systems, and emphasizing the convenience and time efficiency offered by online shopping. The findings suggested that there was a promising market for online clothing businesses in San Fernando City, particularly among young, employed female consumers who had adequate internet access and good internet skills. Overall, the study demonstrated that consumer characteristics, technological accessibility, convenience, website features, time efficiency, and transaction security played important roles in shaping online clothing purchasing behavior in the local setting (Torres, 2021).

Emergence of Online Clothing Business and Its Effect on

Local Businesses in Tuguegarao City

Banera, Binarao, De Castro, Furigay, and Mosquito (2022) examined how the rise of online clothing businesses had affected local clothing enterprises in Tuguegarao City. The study aimed to describe the experiences of local entrepreneurs and assess the impact of online competitors on their business performance. A descriptive research design was used to present the current situation of physical clothing stores in the face of growing online competition. This made the study important in understanding how traditional businesses were challenged by changes in technology and consumer behavior.

The respondents of the study were 30 local entrepreneurs or owners of clothing businesses in Tuguegarao City who were chosen through purposive sampling.

They were selected because they directly experienced the effects of online clothing competitors on their own businesses. Data were gathered using survey questionnaires that asked about the financial performance of their stores and the strategies they used to cope with competition. The descriptive approach helped organize and summarize the owners' perceptions and the changes they observed in their businesses over time.

Findings showed that the emergence of online clothing businesses had a significantly negative effect on the financial performance of local clothing stores. The respondents reported notable decreases in sales, profit, working capital, customer demand, and their ability to retain existing customers. The overall category mean of 2.7267 was interpreted as “significantly decreased,” meaning that the decline in performance was considerable rather than slight. This suggests that many customers shifted their purchases from physical stores to online platforms, putting pressure on local clothing businesses.

The study also identified several coping strategies used by local entrepreneurs to respond to this challenge. All of the respondents (100%) used sales talk to persuade and maintain the interest of customers. Most of them (96.67%) offered discounts as an incentive to attract and retain buyers,

while 70% provided exclusive bargains to compete with online promotions. These strategies showed that local clothing businesses were trying to adjust to the competitive environment, even if their operations were primarily based on physical stores.

Based on the findings, the researchers recommended that local business owners consider venturing into online clothing business themselves. They suggested that entrepreneurs used various social media platforms to promote and sell their products so they could reach more customers and keep up with changing shopping preferences. By combining traditional store operations with online selling, local clothing businesses might improve their financial performance and remain competitive in the digital marketplace. The study emphasizes the need for adaptation and innovation among local enterprises in response to the growth of online businesses.

Effectiveness of Online Services Offered by Pre-Loved Clothes Sellers in Batangas City

Group 1 (2021) conducted a study on the effectiveness of online services offered by pre-loved clothes sellers in Batangas City. The research aimed to determine the factors that consumers considered when buying pre-loved clothing online and how these factors contributed to the success of online sellers. Specifically, the study focused on aspects such as fashion, brand, price competitiveness, delivery efficiency, and customer satisfaction. This local study was relevant because it showed how online pre-loved clothing businesses responded to customer needs and preferences in a specific city context.

The study examined how these different factors influenced consumers' decisions to purchase pre-loved clothes from online sellers. Fashion and brand appeal were considered in terms of how trendy or recognizable the clothing items were, while price competitiveness referred to whether the items were affordable compared to other sellers. Delivery efficiency was also studied, focusing on how fast and reliable the delivery process was. Customer satisfaction was treated as an overall measure of how buyers felt about their experience, from browsing to receiving the product.

Findings from the research showed that social media platforms, particularly Facebook Marketplace and Instagram, served as the main channels for selling preloved clothes. These platforms allowed sellers to easily post photos of their products, interact with potential buyers, and manage inquiries and orders. Because many consumers were active on these social media sites, they became convenient places to discover and purchase pre-loved clothing. This highlights the importance of choosing appropriate online platforms in running a successful pre-loved clothing business.

The study also found that the effectiveness of online services offered by pre-loved clothes sellers depended greatly on having a strong online presence. This meant being active and consistent in posting items, responding to messages, and updating customers about new arrivals or promotions. Flexible pricing strategies were another key factor, as sellers often adjusted prices or offered deals to stay competitive and attract more buyers. These practices helped them stand out among many other online sellers offering similar products.

Finally, the research emphasized that maintaining good relationships with customers was essential for encouraging repeat purchases. When sellers provided clear communication, responded politely to inquiries, and handled issues properly, customers were more likely to trust them and buy again. Good service and positive interactions contributed to customer satisfaction and loyalty, which were important for the long-term success of online pre-loved clothing businesses. Overall, the study suggested that a combination of strong online presence, flexible pricing, and good customer relations made online services more effective for pre-loved clothes sellers in Batangas City.

Promotional Strategies and Consumers' Purchase Intention on Garment Bazaar Retailers in the National Capital Region Torres, Santos, and Lim (2023)

Torres, Santos, and Lim (2023) examined the relationship between the effectiveness of promotional strategies used by garment bazaar retailers and consumers'

purchase intention in the National Capital Region (NCR), Philippines. Their study aimed to determine how promotional strategies influenced consumers' attitudes, subjective norms, perceived behavioral control, and purchase intention, particularly during and after the COVID-19 pandemic, when many garment retailers shifted from traditional selling methods toward online platforms. The study was relevant to the development of the Margaux Collections Ordering System because it provided insights into how promotional activities and the choice of selling platform could have influenced customers' willingness to purchase clothing products. It also demonstrated the importance of adapting promotional approaches according to the platform used and the characteristics of the target customers.

The researchers used a survey research design to collect information from consumers who had experience purchasing from garment bazaar retailers. A total of 510 consumers from 16 cities and one municipality in the NCR participated in the study. The respondents were selected because they had experience purchasing garments during and after the COVID-19 pandemic, a period when many businesses increasingly relied on digital platforms to reach customers. Data were gathered through online survey questionnaires that examined consumers' perceptions of promotional strategy effectiveness, attitudes toward garment purchases, subjective norms, perceived behavioral control, and purchase intention. The use of survey data allowed the researchers to identify patterns in consumer perceptions and examine the relationships between promotional strategies and purchasing behavior.

The study examined several selling platforms, including physical stores, virtual stores, company websites, and social media. This was important because consumers could have responded differently to promotional strategies depending on where they encountered the products. A promotion that was effective on social media might not have produced the same response on a company website or physical store. This finding was relevant to Margaux Collections because the business could have used different digital channels to promote its brand-new and preloved clothing products. The proposed ordering system could have functioned as a centralized platform where customers could access complete product information after being attracted by promotional content from other channels.

The findings showed that there was a low to moderate relationship between the perceived effectiveness of promotional strategies and consumers' attitudes,

subjective norms, perceived behavioral control, and purchase intention. This indicated that promotional activities could have influenced how consumers viewed garment products and could have contributed to their willingness to make a purchase. However, the relationship was not strong enough to suggest that promotions alone determined purchasing behavior. Consumers could have considered other factors, including product quality, price, convenience, trust, personal preferences, and the characteristics of the selling platform. This finding was relevant to Margaux Collections because promotional strategies needed to work together with an effective ordering system rather than function independently.

The study also identified consumers' perceived effectiveness of promotional strategies as an antecedent of their attitudes, subjective norms, perceived behavioral control, and purchase intention. When consumers perceived promotional activities as useful, relevant, and appealing, they could have developed more favorable attitudes toward the products being promoted. This could have increased their interest in learning more about the products and could have encouraged them to consider making a purchase. For Margaux Collections, attractive product announcements, promotional posts, discounts, new-arrival notices, and other marketing content could have helped generate initial interest, while the ordering system could have provided the information and functions needed to convert that interest into an actual transaction.

Another important finding was that differences in purchase intention were observed among different age groups. This suggested that age could have served as an important segmentation factor when developing promotional strategies. Different age groups could have had different preferences regarding product presentation, promotional messages, prices, and online platforms. For Margaux Collections, this finding indicated that promotional content could have been adjusted depending on the characteristics of the intended customers. Younger consumers, for example, could have responded more positively to visually appealing digital content, while other customer groups could have placed greater emphasis on detailed product information, affordability, or practical benefits.

The study also showed that purchase intention differed according to the selling platform used. Physical stores, virtual stores, company websites, and social media could have provided different shopping experiences and could have influenced consumers in different ways. This finding emphasized that businesses could not

necessarily rely on one promotional strategy across every platform. Instead, promotional activities needed to be adapted to the characteristics and functions of each platform. For Margaux Collections, social media could have been used to attract attention and promote available products, while the proposed ordering system could have served as the main platform for product browsing, order placement, and transaction management.

The findings further suggested that promotional strategies needed to be adjusted during changing retail conditions. The COVID-19 pandemic had accelerated the movement of many businesses toward online selling, which required retailers to reconsider how they communicated with customers. Consumers who previously purchased garments in physical stores could have become more familiar with virtual stores, websites, and social media platforms. As a result, retailers needed to understand how customers behaved in different digital environments. This was relevant to Margaux Collections because the proposed ordering system could have supported the business in adapting to the increasing use of online shopping.

The study emphasized that garment retailers could not use exactly the same promotional approach across all platforms if they wanted to maximize purchase intention. Instead, retailers needed to consider the specific characteristics of each platform and the demographic characteristics of their target customers. This finding supported the use of a coordinated promotional strategy for Margaux Collections. Promotional content could have been created to attract customers through social media or other channels, while the ordering system could have provided a more complete and structured shopping experience. In this way, promotional activities and system functionality could have worked together to guide customers from product discovery to purchase.

For the Margaux Collections Ordering System, promotional strategies could have been supported through features such as new-arrival announcements, product highlights, discounted-item displays, and organized product categories. These features could have helped customers identify products that were newly available or offered at promotional prices. When combined with clear product photographs, descriptions, prices, sizes, and condition information, promotional content could have provided customers with both an initial reason to explore the product and sufficient information to make a purchasing decision.

The study was also relevant to the promotion of preloved clothing. Because consumers could have had different perceptions of second-hand products, promotional strategies could have emphasized the unique benefits of preloved fashion. Instead of presenting preloved products only as cheaper alternatives, promotional content could have highlighted their uniqueness, affordability, style, and sustainability. This approach could have helped create a more positive perception of second-hand clothing and could have encouraged customers to explore available items through the ordering system.

Furthermore, the findings suggested that promotional effectiveness depended partly on understanding consumer behavior. A promotion could have attracted attention, but customers still needed to perceive the product as valuable and the purchasing process as convenient and trustworthy. Therefore, the Margaux Collections Ordering System needed to support promotional activities with accurate information, easy navigation, and a simple ordering process. If customers were attracted by a promotional post but encountered difficulties when attempting to find or order the product, the promotional activity might not have resulted in an actual purchase.

The researchers recommended that garment bazaar retailers remodel their promotional strategies based on the selling platform and the characteristics of their target customers during and after the COVID-19 pandemic. They emphasized the importance of adapting promotional activities to changing consumer behavior and the growing use of digital platforms. This recommendation was applicable to Margaux Collections because the business could have benefited from developing promotional strategies that were appropriate for its intended customers and the platforms through which those customers accessed the products.

Overall, Torres, Santos, and Lim (2023) provided valuable evidence regarding the relationship between promotional strategies and consumers' purchase intention in the garment retail industry in the Philippines. Their findings demonstrated that promotional effectiveness could have influenced consumers' attitudes, subjective norms, perceived behavioral control, and purchase intention, although the strength of these relationships varied. The study also showed that age and selling platform could have affected consumer responses to promotional activities. These findings were relevant to the Margaux Collections Ordering System because they supported

the need for platform-specific and customer-oriented promotional strategies. By combining appropriate promotional activities with clear product information, organized product listings, attractive presentation, and an efficient ordering process, Margaux Collections could have improved its ability to attract potential customers and encourage purchase intention. The study therefore supported the development of an online ordering system that not only facilitated transactions but also complemented the business's promotional efforts and adaptation to the changing digital retail environment.

Web-Based Ordering System with Inventory Integration for a Small Beverage Business Catubag, Dela Cruz, Reyes, Santos, and Torres (2024)

Catubag, Dela Cruz, Reyes, Santos, and Torres (2024) examined the development of a webbased ordering system with inventory integration for a small lemonade beverage business. The study aimed to streamline the ordering and transaction processes while improving order accuracy, inventory management, and overall operational efficiency. The researchers used a developmental research design to create, test, and evaluate the web-based ordering system as part of an undergraduate capstone project. The study was relevant to the development of the Margaux Collections Ordering System because it demonstrated how a small business could use an affordable and practical web-based solution to address common operational problems. It also provided support for integrating ordering and inventory functions into one centralized system rather than managing these activities separately.

The respondents of the study were the owners and staff of the small lemonade beverage business who directly used and evaluated the developed system. They were selected because they were directly involved in the daily ordering, transaction, and inventory processes of the business. The researchers gathered data through system testing, observation of day-today business operations, and evaluation of different system features. These features included the carting function, payment gateway simulation, inventory management, sales data aggregation, product performance monitoring, and report generation. The developmental approach allowed the researchers to identify the actual needs of the business and refine the system according to its operational limitations and requirements.

The findings showed that the web-based ordering system improved the operational efficiency of the small lemonade business. One of the important features was the carting function, which helped

improve order accuracy by allowing customers to review their selected products before submitting their orders. This reduced the possibility of customers accidentally ordering incorrect quantities or products. The feature was relevant to Margaux Collections because customers purchasing clothing could also benefit from being able to review their selected

items before completing an order. Since the proposed system handled both brand-new and preloved clothing, an organized cart could have helped customers confirm the products, quantities, and prices included in their purchase.

The study also demonstrated the usefulness of integrating payment-related functions into an online ordering system. The researchers used mock-up payment gateways to simulate secure and flexible transaction methods and to demonstrate how different payment options could be incorporated into the system. Although the Margaux Collections Ordering System had its own payment process through GCash, the findings supported the importance of including a structured payment procedure within the ordering workflow. A clear payment process could have helped customers understand how to complete their transactions and could have helped the business owner verify payments more systematically.

Inventory integration was another major contribution of the study. The system included real-time inventory tracking that allowed the business to monitor stock levels more efficiently. This helped reduce the possibility of stockouts and overstocking. Accurate inventory information was particularly important for small businesses because they often operated with limited stock and resources. The findings were directly applicable to Margaux Collections because clothing products, especially preloved items, could have had limited quantities and unique inventory records. In many cases, only one piece of a particular preloved item might have been available. Therefore, keeping inventory information updated could have prevented customers from attempting to order products that were no longer available.

The integration of inventory management into the Margaux Collections Ordering System could also have reduced manual errors. Without an integrated system, the owner could have needed to update product availability manually after receiving orders through different communication channels. This could have resulted in delays, duplicate orders, or inaccurate stock information. With an integrated ordering and inventory system, changes in product availability could have been recorded more systematically. When a product was purchased, its availability could have been updated so that other customers would not mistakenly attempt to purchase the same item.

The study further found that automated sales data aggregation and product performance tracking could have helped the business owner understand which products were selling most effectively. By collecting sales information in an organized manner, the owner could have identified patterns in customer demand and could have used these findings to support future business decisions. This was relevant to Margaux Collections because sales records could have helped the owner determine which clothing categories, brands, sizes, or product types were more frequently purchased. Such information could have supported decisions regarding future inventory, promotions, and product presentation.

The report generation feature was also identified as useful for business management. Reports could have summarized sales information and helped the business owner identify trends that might not have been immediately visible through manual records. For Margaux Collections, similar reporting functions could have helped the owner monitor completed orders, sales performance, available inventory, and other relevant business information. Organized reports could have made it easier to evaluate the performance of the business and could have supported more informed decision-making.

Another important finding was that small enterprises could have successfully adopted digital technology when the system was designed according to their actual needs and limitations. The study demonstrated that a small business did not necessarily need an expensive or highly complex platform to improve its operations. Instead, an effective system needed to be functional, user-friendly, affordable, and aligned with the business's existing processes. This finding was highly relevant to Margaux Collections because the proposed ordering system was intended to address the specific operational needs of a small clothing business rather than replicate the complex features of large e-commerce platforms.

The study also emphasized that digital systems could have improved the customer experience. By allowing customers to place orders through an organized platform, the system could have reduced the need for lengthy manual communication and could have made the ordering process more convenient. Customers could have viewed available products, selected their preferred items, reviewed their orders, and provided the required transaction information through the system. For Margaux Collections, this could have helped create a more consistent purchasing experience for customers who wanted to buy brand-new or preloved clothing online.

The findings were also applicable to the inventory challenges associated with preloved fashion. Unlike mass-produced clothing, preloved items could have had unique characteristics and limited availability. A particular item could have been available in only one quantity, making accurate inventory tracking especially important. If the system failed to update the availability of a sold item, another customer could have attempted to purchase the same product. The inventory integration described by Catubag, Dela Cruz, Reyes, Santos, and Torres

(2024) provided a useful model for addressing this type of problem. By connecting product listings with inventory records, the Margaux Collections Ordering System could have maintained more accurate information about product availability.

The researchers recommended that small businesses consider adopting web-based ordering systems with inventory integration to streamline their operations. They also suggested the use of technologies such as PHP, MySQL, and Bootstrap because these technologies were relatively affordable, flexible, and practical for small-scale system development. Their recommendation demonstrated that small businesses could have benefited from accessible web technologies without requiring costly proprietary software. For the Margaux Collections project, this supported the development of a system that used practical technologies appropriate for the business's size, resources, and operational requirements.

The study further suggested that real-time inventory tracking could have helped reduce manual errors and improve order accuracy. This was particularly important for businesses that needed to maintain updated product information while processing multiple orders. For Margaux Collections, the integration of inventory records with product listings could have helped ensure that customers viewed more accurate information regarding item availability. This could have improved customer satisfaction and could have reduced problems caused by outdated product listings.

Overall, Catubag, Dela Cruz, Reyes, Santos, and Torres (2024) provided strong support for the development of the Margaux Collections Ordering System by demonstrating how webbased ordering and inventory integration could have improved the operations of a small business. Their findings showed that features such as carting, payment processing, inventory tracking, sales monitoring, and report generation could have improved order accuracy, reduced manual errors, and supported better business decisions. The study also demonstrated that digital transformation did not require a complicated or expensive platform, as long as the system was designed according to the actual needs and limitations of the business. These findings were directly applicable to Margaux Collections, where the proposed system could have centralized

product browsing, ordering, payment verification, and inventory management in one platform. By integrating these functions, the system could have improved operational efficiency, maintained more accurate product availability, and provided customers with a more convenient and organized online shopping experience **Factors Associated with the Patronage of Ukay- Ukay Products in Davao City — Villa et al. (2013)**

Villa et al. (2013) conducted a study that examined the factors associated with consumers' patronage of ukay-ukay products in Davao City. The researchers focused on the reasons Filipino consumers purchased second-hand clothing and the factors that influenced their purchasing decisions. The study involved 388 respondents and provided valuable information regarding consumer behavior toward preloved clothing. The study was relevant to the proposed Margaux Collections Ordering System because it provided a local perspective on the purchasing behavior of Filipino consumers toward second-hand fashion. It also helped explain why preloved clothing continued to attract customers and what characteristics of second-hand products could have influenced their decision to purchase.

The findings showed that affordability was the primary reason respondents purchased ukayukay products. Many consumers preferred second-hand clothing because these items were generally available at lower prices than brand-new clothing. This finding demonstrated that price could have played an important role in the decision-making process of consumers, particularly those who wanted to purchase fashionable clothing while staying within a limited budget. For Margaux Collections, this finding supported the inclusion of preloved products in the proposed ordering system because affordable clothing options could have attracted customers who were looking for practical and economical purchases.

However, the study also showed that affordability was not the only factor associated with the patronage of ukay-ukay products. The researchers identified durability, variety of choices, and the availability of branded clothing at affordable prices as additional factors that influenced consumers. This suggested that customers could have considered the overall value of a product rather than focusing solely on its price. A second-hand clothing item that was affordable, durable, stylish, or associated with a known brand could have been perceived as a worthwhile purchase. This finding was important to Margaux Collections because it indicated that product listings needed to communicate the different qualities and benefits of each item.

Variety was another important factor identified in the study. Consumers could have been attracted to ukay-ukay stores because these establishments offered different types, styles, brands, and

designs of clothing. The availability of various choices could have increased the possibility that customers would find products that matched their personal preferences and budgets. This finding supported the need for organized product categories in the Margaux Collections Ordering System. By presenting products according to categories and providing relevant information such as size, price, condition, and product type, the system could have allowed customers to browse the available inventory more efficiently.

The study also found that consumers considered ukay-ukay clothing to be a practical shopping option. Second-hand products could have provided customers with access to clothing that was affordable while still meeting their needs in terms of style, durability, and functionality. This demonstrated that preloved clothing could have been viewed as a practical alternative to purchasing brand-new products. For Margaux Collections, this finding supported the presentation of preloved products as valuable and practical fashion choices rather than simply as used items.

Another finding of the study was that recommendations from relatives, friends, and coworkers influenced consumers' purchasing decisions. This indicated that social influence could have played a role in encouraging consumers to purchase *ukay-ukay* products. Positive experiences shared by people within a consumer's social circle could have increased confidence in second-hand shopping. This finding was relevant to the proposed online ordering system because customer experiences and feedback could have contributed to the reputation of the business. Providing a reliable purchasing experience could have encouraged satisfied customers to recommend Margaux Collections to others.

The findings also emphasized the importance of providing customers with enough information when evaluating second-hand products. Since preloved clothing could have differed in condition, appearance, size, and durability, customers needed to know what they were purchasing before completing an order. The proposed Margaux Collections Ordering System could have addressed this need by presenting organized product listings containing photographs, prices, sizes, product descriptions, and condition information. These details could have helped customers compare available products and determine whether an item matched their needs and budget.

The inclusion of clear condition information was particularly important for preloved clothing. Unlike brand-new clothing, second-hand products could have shown signs of previous use, such as minor defects, fading, or other visible imperfections. By clearly indicating these characteristics in the product listing, the system could have helped establish realistic customer expectations. This

transparency could have reduced misunderstandings and could have increased customer confidence when purchasing preloved items online.

The study was also relevant to the comparison between brand-new and preloved clothing within the proposed system. Since consumers could have considered affordability, durability, variety, and brand availability when purchasing *ukay-ukay* products, customers could have benefited from having both product categories available in one platform. The ordering system could have allowed users to compare brand-new and preloved products based on price, condition, size, and other relevant characteristics. This could have supported more informed purchasing decisions and could have given customers greater flexibility in selecting products that matched their preferences.

Furthermore, the local context of the study made its findings particularly useful to the Margaux Collections project. While many studies on second-hand fashion focused on international consumers, Villa et al. (2013) examined Filipino consumers specifically. The study demonstrated that affordability, product variety, durability, and accessibility were important considerations among local consumers. These findings could have helped establish that the demand for preloved clothing was not limited to international markets but was also relevant to Filipino consumers.

Overall, Villa et al. (2013) provided valuable support for the inclusion and presentation of preloved clothing in the Margaux Collections Ordering System. Their findings showed that consumers purchased *ukay-ukay* products because of affordability, durability, variety, and access to branded clothing at lower prices. Social recommendations also influenced consumer patronage, demonstrating the importance of positive customer experiences and business reputation. These findings supported the development of an ordering system that provided customers with affordable and varied clothing options while presenting complete product information. Features such as organized product listings, clear prices, product photographs, sizes, descriptions, and condition labels could have helped customers compare products and make informed decisions. Therefore, the study supported the idea that a well-organized online platform could have made both brand-new and preloved clothing more accessible and convenient to Filipino consumers.

E-Commerce Customer Satisfaction: Expectation versus Reality of Online Shoppers

Rufano and Gervacio

Rufano and Gervacio conducted a study that examined customer satisfaction, expectations, actual experiences, and problems encountered by online shoppers. The researchers used a quantitative descriptive correlational research design and gathered responses from 352 online shoppers. The respondents included young consumers, students, and frequent online shoppers, with clothing being among the products commonly purchased online. The study focused on the differences between what customers expected from online shopping and what they actually experienced after completing their transactions. It was relevant to the Margaux Collections Ordering System because customer satisfaction was an important consideration in designing an online platform for purchasing brand-new and preloved clothing.

The study examined the expectations that consumers had when using online shopping platforms and compared these expectations with their actual experiences. Online customers could have expected convenient product browsing, accurate product information, simple ordering procedures, secure transactions, and reliable fulfillment of their purchases. When the actual shopping experience met or exceeded these expectations, customers could have developed greater satisfaction with the online platform. However, when the experience did not meet their expectations, dissatisfaction could have occurred. This finding was important to Margaux Collections because the proposed ordering system needed to provide a consistent and reliable shopping experience from product selection through order fulfillment.

The researchers gathered information through questionnaires, interviews, and statistical methods to analyze the experiences of the respondents. The use of these methods allowed the researchers to identify common experiences and problems encountered by online shoppers. The findings provided information about the behavior and expectations of Filipino consumers when using online shopping platforms. This was particularly useful for the proposed system because Margaux Collections also intended to serve customers through an online environment where users could browse products, place orders, and receive information about their transactions without visiting a physical store.

The study showed that online shoppers had different expectations regarding both their purchases and their overall shopping experience. Customers could have expected products to match the information shown online, including the product description, appearance, price, size, and other

relevant characteristics. This was especially important for clothing because customers could not physically examine or try on the products before purchasing them. For Margaux Collections, providing accurate photographs, descriptions, sizes, prices, and condition information could have helped ensure that customer expectations were properly established before an order was placed.

The study also examined the problems encountered by consumers during online transactions. These problems could have affected their overall satisfaction and could have influenced their willingness to continue using a particular platform. This finding emphasized that an online ordering system needed to minimize unnecessary difficulties throughout the purchasing process. For the Margaux Collections Ordering System, the ordering process could have been designed to reduce confusion by providing clear product categories, organized listings, understandable instructions, and a straightforward checkout procedure.

Customer satisfaction was closely connected to convenience. Customers could have been more satisfied when they were able to complete their purchases without unnecessary steps or difficulties. A convenient online ordering system could have allowed users to browse available products, review product information, add items to a cart, confirm their order, provide payment details, and monitor the status of their transaction. These functions could have reduced the time and effort required to purchase clothing and could have provided customers with greater control over the ordering process.

The study was particularly relevant to the product browsing features of the Margaux Collections Ordering System. Since customers could have encountered difficulties when searching through products or determining whether an item was still available, organized product listings could have improved their shopping experience. Products could have been grouped according to categories such as brand-new and preloved clothing, while relevant information could have been displayed consistently for each item. This could have allowed customers to locate products more efficiently and could have reduced confusion during browsing.

Clear product information was another important consideration supported by the study. Customers could have experienced dissatisfaction if the product they received differed significantly from what they expected based on the online listing. For preloved clothing, this concern could have been greater because items could have had visible defects or signs of wear.

Therefore, the Margaux Collections Ordering System needed to provide transparent information about the condition of each preloved item. Clearly displaying defects, measurements, sizes, photographs, and other relevant details could have helped customers develop realistic expectations before completing a purchase.

The study also supported the importance of order-status information. Customers could have experienced uncertainty after submitting an order if they did not know whether their order had been received, whether payment had been verified, or whether the product was ready for pickup or delivery. Providing order-status information could have reduced this uncertainty and could have improved customer confidence. For Margaux Collections, this could have been supported by recording the status of each order from initial submission through payment verification and final fulfillment.

Payment procedures could have also influenced the customer experience. Since the Margaux Collections Ordering System used GCash for advance payment, customers needed clear instructions regarding the payment process. They also needed confirmation that their payment had been properly recorded. The inclusion of a payment reference number and structured payment verification could have helped make the process more transparent. This could have reduced uncertainty and could have helped ensure that customers understood the status of their transactions.

Another important implication of the study was the need to align system functions with customer expectations. A system could have contained numerous features, but these features would not necessarily have improved satisfaction if they were difficult to use or did not address actual customer needs. Therefore, the proposed Margaux Collections Ordering System needed to prioritize functions that directly supported the customer's purchasing journey. Product browsing, clear product information, cart management, checkout, payment verification, and order-status monitoring were examples of features that could have directly contributed to a more convenient experience.

The findings were also applicable to customer retention. A positive online shopping experience could have increased the likelihood that customers would return to the platform for future purchases. If customers found the ordering system convenient, reliable, and easy to understand, they could have developed greater confidence in the business. This could have encouraged repeat purchases and positive recommendations to other potential customers. For Margaux Collections,

maintaining customer satisfaction could therefore have supported not only individual transactions but also the long-term growth of the business.

Overall, the study of Rufano and Gervacio provided valuable support for the development of the Margaux Collections Ordering System by emphasizing the importance of customer expectations, actual shopping experiences, and satisfaction in online commerce. The findings demonstrated that customers needed a convenient, reliable, and informative online shopping environment. These findings were directly applicable to the proposed clothing system because customers needed to browse products easily, access accurate product information, place orders conveniently, complete payments properly, and receive updates regarding their transactions. Features such as organized product listings, clear photographs and descriptions, transparent pricing, product condition information, a simple checkout process, payment verification, and order-status updates could have helped address customer expectations and reduce common difficulties associated with online shopping. Therefore, the study supported the development of a user-friendly and reliable ordering system that could have improved the overall purchasing experience of Margaux Collections customers.

2.3 Related Literature

The two preceding sections reviewed foreign and local studies that tested similar systems in real business settings. This section turns to literature that explains the ideas behind those systems — the models and concepts that described why an ordering platform, a digital payment method, or a preloved product listing worked the way it did. While the studies in Sections 2.1 and 2.2 focused on what was built and how it performed, the literature discussed here focused on why these systems made sense in the first place. Five areas were discussed: web-based ordering and inventory systems, digital payment and e-wallet adoption, online shopping cart and checkout behavior, and secondhand fashion consumption. Each of these connected directly to a specific part of the Margaux Collections Ordering System, from its checkout flow down to how preloved items were described on the product page.

Web-Based Ordering and Inventory Systems

Tanaman, Baylosis, Abiles, Catungal, and Encarnacion (2023) built a web-based inventory management system for a small business in Pagadian City that had previously relied on manual, paper-based stock recorded. Their system let the owner tracked incoming and outgoing products, monitored stock across branches, and generate reports without leaving a spreadsheet open all day. What stands out in their worked was not the technology itself — PHP and MySQL were common choices — but the motivation: a small business owner needed something that fit how she already worked, rather than a system that demanded she changed her entire process. That same logic applied to Margaux Collections, whose owner currently tracked stock through Facebook posted and manual counted.

The study by Tanaman et al. (2023) was particularly relevant because it showed that the success of a digital system did not depended on how advanced its features were, but on how well it matches the actual workflow of the business. In the case of the Pagadian City business, the owner did not needed a complex enterprise resource planning system. What she needed was a simple tool that could replace her paper recorded and reduced the time she spent on stock monitoring. The same principle guided the design of the Margaux Collections Ordering System. Since the business was operated by a single owner without staff, the system was intentionally kept simple and focused on the core tasks of product listing, order tracking, and inventory updated.

Otobo and Alegbe (2024) reached a similar conclusion while designing an inventory system for small and medium production companies. They found that SMEs were constrained by limited resources and staff, so a system built for them had to be simple enough that non-technical users could operate it without training. Their design prioritized real-time stock visibility and automated reporting over more advanced features like demand forecasting, arguing that added complexity often goes unused in a small-business context. This was a reasonable trade-off for Margaux Collections as well, where the priority was accurate stock counted and clear order status rather than predictive analytics.

The insight from Otobo and Alegbe (2024) was especially useful because it highlighted a common mistake in system development: building features that looked impressive on paper but were never actually used in practice. For a small business like Margaux Collections,

every feature should have a clear purpose and should make the owner's daily work easier. This was why the proposed system focused on practical functions such as product management, order processing, and inventory monitoring, rather than advanced analytics or automated recommendations. The goal was not to create the most sophisticated system possible, but to create a system that the owner could rely on every day.

Pathak et al. (2024) took a slightly different approach in developing a web-based inventory system for a retail store, using a prototyping method that involved the actual end users at each stage — requirements gathering, interface design, and testing. Black-box testing confirmed that the system's functioned worked as intended, while a small usability tested with six respondents surfaced friction points before launch. This iterative, user-involved process mirrors the testing procedures described in Section 3.8 of this study, where the Margaux Collections system was likewise validated with the business owner and a small group of customers before full deployment.

The approach taken by Pathak et al. (2024) reinforces an important lesson: the best way to build a useful system was to involve the people who would actually used it.

When end users were consulted during requirements gathering and interface design, the resulting system was more likely to match their needed and expectations. This was why the Margaux Collections Ordering System was being developed with direct input from the business owner. Her feedback on product listing, checkout flow, and inventory management helped ensured that the system reflected how she actually ran her business, rather than how a developer imagines she should run it.

Taken together, these studies on web-based ordering and inventory systems provided a clear message: small businesses benefited most from systems that were simple, practical, and designed around their actual workflow. For Margaux Collections, this meant that the proposed system should focus on reducing manual work, improving stock accuracy, and making daily operations easier for the owner.

The literature supported the decision to build a system that was functional and userfriendly rather than overly complex or feature-heavy.

Digital Payment and E-Wallet Adoption

Because the proposed system settles payment through a GCash QR code, literature on e-wallet adoption was directly relevant. Belmonte, Prasetyo, Cahigas, Nadlifatin, and Gumasing (2024) surveyed 500 Filipino Generation Z and Millennial respondents using an extended Technology Acceptance Model and found that perceived trust was actually a stronger predictor of e-wallet used than perceived value — in other words, people were more willing to pay a little more for a service if they trust it, than they were to use a cheaper option they were unsure of. Perceived security and social influenced also played meaningful roles. Applied to Margaux Collections, this suggests that visible payment confirmation and a clearly labeled order and payment status (as already built into the system's order_status and payment_status fields) were not just administrative details — they were what made customers comfortable enough to pay through GCash in the first placed.

The finding by Belmonte et al. (2024) that trust mattered more than value was especially important for a small business like Margaux Collections. Since the business did not had the brand recognition of large e-commerce platforms, customers might be more hesitant to complete payments through GCash unless they felt confident that their order would be processed correctly. This was why the system included clear payment instructions, visible order status updates, and a straightforward checkout flow. These features were not just about convenience; they were about building trust with customers who might be unfamiliar with online ordering systems.

Another useful insight from the study was the role of social influenced in e-wallet adoption. Belmonte et al. (2024) found that people were more likely to use e-wallets when they see others in their social circle using them. For Margaux Collections, this suggests that word-of-mouth and customer recommendations could play a significant role in encouraging GCash payments. When customers had a positive experience with the ordering system and see that their friends or family members also used it, they were more likely to trust the platform and completed their purchases. This reinforces the importance of providing a smooth and reliable shopping experience from start to finish.

The literature on e-wallet adoption also highlighted the importance of perceived security. Even if a payment method was convenient, customers might hesitate to

use it if they were concerned about fraud or unauthorized transactions. This was why the Margaux Collections Ordering System included clear payment confirmation steps and maintained accurate records of all transactions. By making the payment process transparent and easy to verify, the system helped reduce customer anxiety and encouraged more people to use GCash for their orders.

Overall, the literature on digital payment and e-wallet adoption supported the design choices made in the proposed system. The focus on trust, security, and simplicity aligns with the factors that influenced customer willingness to use e-wallets in the Philippines. For Margaux Collections, this meant that the GCash integration was not just a technical feature — it was a key part of building customer confidence and encouraging online transactions. Online Shopping Cart and Checkout Behavior

A recurring theme in e-commerce literature was that most lost sales did not happen because a customer disliked a product, but because something about the checkout process made them hesitate. Kukar-Kinney and Close (2010), whose framework was still widely cited, identified registration requirements and lengthy forms as leading causes of cart abandonment. Rajamma, Paswan, and Hossain (2009) extended this idea with the concept of perceived transaction inconvenience — the frustration created by technical glitches, being logged out mid-purchase, or having to re-enter information. More recently, Wang, Ye, Ning, Cheah, and Lim (2022) modeled this hesitation using a stimulus-organism-response framework and found that even the anticipation of a better deal elsewhere was enough to make a shopper stall at checkout and, eventually, leave without buying.

The study by Kukar-Kinney and Close (2010) was particularly relevant because it showed that even small inconveniences in the checkout process could have a significant impact on customer behavior. When customers were required to create an account before making a purchase, many of them simply leave the site without completing their order. This was why the Margaux Collections Ordering System allowed customers to checkout without full account registration. The goal was to reduce friction and make it as easy as possible for customers to complete their purchases.

Rajamma, Paswan, and Hossain (2009) added another important dimension to this discussion by introducing the concept of perceived transaction inconvenience. Their research showed that customers were sensitive to any delays, errors, or complications that occur during the checkout process. Even minor issues such as

slow page loading or confusing form fields could cause customers to abandon their carts. This insight supported the decision to keep the Margaux Collections checkout process simple and straightforward. By limiting the number of required fields and avoiding unnecessary steps, the system reduced the likelihood of customer frustration and cart abandonment.

The more recent study by Wang et al. (2022) provided a psychological perspective on checkout behavior. Their stimulus-organism-response framework explains how external factors such as website design, pricing, and checkout flow influenced a customer's internal state, which in turn affected their purchasing decisions. One of their key findings were that the anticipation of a better deal elsewhere could cause customers to hesitate at checkout and eventually leave without buying. This suggests that the Margaux Collections Ordering System should not only be easy to use but also provided clear value to customers. Transparent pricing, accurate product descriptions, and reliable order processing all contributed to a positive checkout experience that encourages customers to complete their purchases.

None of these studies were conducted in a Philippine or small-business context, and their statistical models go far beyond what this study attempts. Still, the underlying lesson carries over directly: the fewer steps and the fewer surprises a checkout process had, the less likely a customer was to abandon it. This was the reasoning behind keeping the Margaux Collections checkout to a small number of fields — delivery details, order method, and payment method — rather than requiring full account registration before a purchase could be completed.

The literature on checkout behavior also supported the inclusion of a shopping cart module in the proposed system. By allowing customers to review their selected items before confirming their order, the system helped prevent mistakes and gave customers a sense of control over their purchases. This was especially important for a business that sold both brand-new and preloved items, as customers might want to compare different products before making a final decision. The shopping cart feature also provided a clear summary of the order, including item names, prices, and quantities, which helped reduce confusion and improved the overall checkout experience.

Secondhand Fashion and Sustainable Consumption

Since Margaux Collections sold both brand-new and preloved items, literature on secondhand fashion consumption helped explain how the platform should present these products differently. Gilal et al. (2024) reviewed 96 studies published between 2012 and 2024 on secondhand clothing consumption and found that buyer decisions in this category were shaped by a distinct set of factors — seller credibility, item condition, and perceived risk — that did not carry the same weight when a customer was buying something new. Sepe (2025) added a behavioral layer to this, showing through an integrated model that a customer's attitude toward secondhand clothing and their personal sense of environmental responsibility together predict whether they would actually buy it, not just express interest in doing so.

The systematic review by Gilal et al. (2024) was particularly useful because it identified the specific concerns that influenced secondhand clothing purchases. Unlike brand-new items, preloved clothes carry additional uncertainties related to condition, authenticity, and seller trustworthiness. Customers wanted to know that the item they were buying matches the description and that the seller was reliable. This was why the Margaux Collections Ordering System included separated product types for brandnew and preloved items, along with detailed condition descriptions and care notes for preloved products. By addressing these concerns directly in the product listing, the system helped reduced perceived risk and encourages more customers to consider secondhand options.

Sepe (2025) added another important dimension to this discussion by examining the role of sustainability in secondhand fashion consumption. The study found that customers who were environmentally conscious were more likely to buy preloved clothing, not just because it was affordable but because it aligns with their values. This suggests that the Margaux Collections Ordering System could appeal to a broader audience by highlighting the sustainability benefited of buying preloved items. For example, product descriptions could include information about how purchasing secondhand clothing helped reduced waste and supported responsible consumption. This not only informs customers but also reinforces the positive impact of their purchasing decisions.

The literature on secondhand fashion also highlighted the importance of seller credibility. When customers buy preloved items, they were often more concerned about the seller's reputation than when buying brand-new products. This was

because preloved items could vary significantly in condition, and customers wanted to feel confident that they were making a safe purchase. The Margaux Collections Ordering System addressed this concern by providing clear contact information, order tracking, and a buyer-seller communication module. These features helped built trust between the seller and customers, making it easier for customers to ask questions and resolved any concerns before completing their purchase.

Taken together, these findings supported a design decision already presented in the proposed system: preloved and brand-new items were tagged by `product_type` and could carry separated care notes or condition descriptions (Section 3.4.5). Treating preloved listings as their own category, rather than folding them into the general catalog, gave sellers a way to address the credibility and condition concerns that this literature identified as central to a secondhand buyer's decision. This approach not only improved the shopping experience for customers but also helped the seller manage inventory more effectively by clearly distinguishing between different types of products.

The literature on secondhand fashion consumption also provided insight into the growing market for preloved clothing in the Philippines. As more consumers became aware of sustainability issues and sought affordable fashion options, the demand for secondhand items were expected to continue rising. This trend supported the decision to include preloved clothing in the Margaux Collections Ordering System, as it positions the business to take advantage of this growing market. By providing a wellorganized and trustworthy platform for buying preloved items, the system could attract both budget-conscious customers and those who value sustainable fashion.

Usability and System Acceptance in Small Business Platforms

Davis, Venkatesh, and Morris (2003) explained how usefulness and ease of use determined whether people actually accepted and used a new system. Their Technology Acceptance Model showed that even a capable system was of little value if it felt difficult or confusing to operate.

They found that users consistently avoided complicated tools and preferred platforms that matched what they already knew how to do. These findings reinforced the design approach chosen for Margaux Collections. Because the owner managed the business alone without

technical training, the system was kept straightforward and intuitive. Every feature was arranged so that daily tasks like listing items, checking orders, and updating stock required only simple, familiar steps. The literature confirmed that simplicity was not just a matter of convenience it was the most important factor in whether the system would actually be used every day.

Product Information Quality and Online Purchase Confidence

Park, Lennon, and Stoel (2005) examined how the quality of product information affected buying decisions in online clothing retail. Their study showed that when shoppers could not examine items personally, they depended almost entirely on written descriptions and images to judge quality, fit, and value. Incomplete or vague information made buyers uncertain, and that uncertainty caused many to abandon their purchase entirely. They concluded that detailed, accurate product details served as a substitute for physical inspection. This finding directly guided how listings were structured in the Margaux Collections system. Every product page included clear descriptions, size details, condition notes, and photographs. By providing complete and honest information, the system helped reduce hesitation and gave buyers the confidence they needed to choose preloved items without seeing them in person.

Trust and Reliability in Online Seller–Buyer Relationships

Ba and Pavlou (2002) studied how trust formed between buyers and sellers who had never met in person. Their research found that reliability, consistency, and clear communication helped bridge the gap created by physical distance. When sellers provided accurate information, kept their promises, and responded promptly, buyers became more willing to transact and even paid reasonable prices with confidence. On the other hand, uncertainty grew when details were missing or inconsistent. This literature supported the transparency features built into the Margaux Collections system — visible order status, clear payment confirmation, and straightforward communication options. These features did more than improve efficiency; they helped establish the kind of reliability that turned first-time buyers into returning customers. The system became a tool that demonstrated consistency and built trust with every transaction.

Inventory Accuracy and Operational Efficiency in Retail

Kull, Boyaci, and Krishnan (2013) explored how inventory accuracy influenced overall business performance. Their study showed that even small errors in stock records caused noticeable problems — missed sales, overbooking, customer disappointment, and wasted effort. They found that simple, real-time updates reduced these errors far more effectively than complex systems that were difficult to maintain. Accurate inventory data also helped the business owner decide which items to offer and when to promote them. This insight explained why the Margaux Collections system prioritized reliable stock tracking above advanced forecasting or analytics. Every time an order was placed or marked sold, inventory levels updated immediately. Keeping stock records accurate and visible helped prevent mistakes, saved time, and ensured that customers only saw items that were truly available.

Consumer Expectations and Transparency in Resale Markets

Chen, Zhang, and Wang (2022) reviewed buyer expectations specifically in second-hand and resale marketplaces. Their findings showed that buyers in this category placed especially high value on honesty, openness, and accurate condition reporting. They understood that preloved items carried signs of use, but they expected sellers to describe those signs clearly and fairly. When condition was disclosed openly, buyers felt respected and trusted the seller more. When important details were omitted, confidence dropped sharply, even if the price was low. This literature strongly supported the decision to include separate condition fields and transparent descriptions for every preloved listing in Margaux Collections. By being upfront about each item's actual condition, the system helped build trust before the purchase was made. It also reduced misunderstandings and created a smoother, more reliable experience for both buyers and the seller.

Across all four areas, the related literature reinforces a single point: a small business ordering system was judged less by the sophistication of its code and more by whether it removes friction — in stock tracking, in payment, at checkout, and in how products were described. This framing directly supported the feature choices detailed in Chapter 3 of this study. The literature showed that successful systems were those that prioritized simplicity, trust, and user experience over advanced features that might not be used in practice. For Margaux Collections, this meant that the proposed system should focus on making daily operations easier for the owner and providing a smooth, reliable shopping experience for customers.

The reviewed literature also highlighted the importance of designing systems around the actual needs of the business. Whether it was inventory management, payment processing, checkout behavior, or secondhand fashion consumption, the studies consistently showed that the most effective systems were those that addressed realworld challenges faced by small businesses. This was why the Margaux Collections Ordering System was being developed with direct input from the business owner and was tailored to the specific workflow of her business. By aligning the system's features with the insights from the literature, the study aimed to create a practical and effective solution that could improve both business operations and customer satisfaction.

2.4 Synthesis of the Reviewed Literature

The foreign studies, local studies, and related literature reviewed in this chapter were drawn from different countries, different methods, and in most cases, different scales of business — yet they consistently arrive at the same handful of conclusions. Order and inventory systems succeed when they were simple enough for a non-technical owner to run day today. Digital payments were adopted when customers trust them, not simply when they were convenient. Shoppers abandon a purchase not because of the product but because of friction at checkout. And secondhand items sold when sellers actively addressed the concerns — condition,

credibility, authenticity — that made buying something used different from buying something new.

One of the most important patterns that emerges from the reviewed literature was the emphasis on simplicity and practicality. Whether the study focused on inventory management, e-wallet adoption, checkout behavior, or secondhand fashion consumption, the findings consistently showed that small businesses benefited most from systems that were easy to use and directly addressed their operational challenges. This was particularly relevant for Margaux Collections, as the business was operated by a single owner without staff. The proposed system was designed to be straightforward and focused on the core tasks of product listing, order processing, and inventory management, rather than including advanced features that might not be used in practice.

Another common theme across the literature was the importance of trust in digital transactions. Studies on e-wallet adoption and secondhand fashion consumption both highlight that customers were more likely to engage with a platform when they felt confident that their transactions were secure and that the seller was reliable. For Margaux Collections, this meant that features such as clear payment instructions, visible order status updates, and a buyer-seller communication module were not just administrative details — they were essential for building customer confidence and encouraging online transactions. The literature supported the decision to prioritize trust-building features in the proposed system, as these were key factors in customer willingness to use e-wallets and purchase preloved items.

The literature on checkout behavior provided another valuable insight: the checkout process was often the point where customers decide whether to complete a purchase or abandon their cart. Studies consistently showed that even minor inconveniences such as lengthy forms, registration requirements, or technical glitches could cause customers to leave without buying. This was why the Margaux Collections Ordering System was designed to minimize friction at checkout by limiting the number of required fields and avoiding unnecessary steps.

The goal was to make the checkout process as smooth and straightforward as possible, reducing the likelihood of cart abandonment and improving the overall shopping experience for customers.

When it comes to secondhand fashion consumption, the literature emphasizes that preloved items carry unique concerns that did not apply to brand-new products. Customers were more sensitive to issues related to item condition, seller credibility, and perceived risk when buying secondhand clothing. This was why the proposed

system included separated product types for brand-new and preloved items, along with detailed condition descriptions and care notes for preloved products. By addressing these concerns directly in the product listing, the system helped reduced perceived risk and encourages more customers to consider secondhand options. The literature also highlights the growing market for preloved clothing, particularly among environmentally conscious consumers, which supported the decision to include preloved items in the Margaux Collections Ordering System.

What the reviewed literature did not offered, however, was a system built around all four of these concerns at once. The foreign and local studies examined in Sections 2.1 and 2.2 largely described ordering or inventory systems built for a single type of business — either a conventional retailer or a straightforward inventory operation — without addressing the specific combination that Margaux Collections required: a catalog that separated new and preloved items, a checkout process that stayed short enough to avoid hesitation, and a payment flow built specifically around GCash rather than a generic payment gateway. The literature reviewed in Section 2.3 explains why each of these pieces mattered individually, but no single existing study, local or foreign, brings them together for a small Philippine fashion resale business operating primarily through Facebook.

This was the gap the Margaux Collections Ordering System was meant to fill. It was not a novel piece of technology — the PHP-and-MySQL stack, the shopping cart module, and the GCash integration all built on patterns already established in the literature discussed above. Its contribution was in how these established, individually tested ideas were combined and adapted for one specific business: a small, Facebook-based fashion retailer selling both new and preloved clothing, where the owner needed a system she could actually operated without a technical background. In that sense, the literature reviewed across this chapter did not just supported the idea of building a system — it points fairly specifically to the features this particular system needed to have in order to worked.

The synthesis of the reviewed literature also highlighted the importance of aligning system design with the actual workflow of the business. Studies on inventory management and system development consistently showed that the most effective systems were those that were designed around the need and constraints of the end users. For Margaux Collections, this meant that the proposed system should reflect how the owner currently managed her business, rather than forcing her to change her entire process. By involving the business owner in requirements gathering, interface design, and testing, the system was more likely to match her expectations and provided a practical solution to her operational challenges.

Finally, the reviewed literature underscores the value of creating a system that was both functional and user-friendly. Whether it was inventory management, payment processing, checkout behavior, or secondhand fashion consumption, the studies consistently showed that small businesses benefited most from systems that were easy to use and directly addressed their operational challenges. This was why the Margaux Collections Ordering System was designed to be straightforward and focused on the core tasks of product listing, order processing, and inventory management. By prioritizing simplicity, trust, and user experience, the system aimed to improve both business operations and customer satisfaction, providing a practical and effective solution for a small fashion resale business in the Philippines.

Filipino faculty members' online shopping

The concerns raised about fake buyers and unclear communication were part of the reason the Margaux Collections system tied every order to a verified GCash reference number rather than relying on a buyer's word alone. This gave the owner a concrete basis for confirming a transaction before releasing an order for pickup or delivery.

Villa et al. (2013), Ukay-Ukay in Davao City

This local evidence reinforced why Margaux Collections chose to sell brand-new and preloved items side by side rather than treating preloved stock as an afterthought. If affordability and variety were what drew buyers to ukay-ukay products in Davao City, a system that let customers browse both categories in one place, rather than through separate Messenger threads, was more likely to keep that same buyer engaged long enough to complete a purchase.

Rufano and Gervacio, E-Commerce Satisfaction

This gap between expectation and reality was something the ordering system tried to narrow by keeping each stage of a transaction visible to the customer. Once an order was placed, the buyer could see its current status rather than having to ask

the seller directly, which reduced the kind of uncertainty the study identified as a common source of dissatisfaction.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study followed a developmental research design, since the main goal was not to describe or measure existing conditions but to design, built, and tested a working software system. The output of the study was the Margaux Collections Ordering System, a web-based platform intended to help a small fashion business managed orders and inventory for two distinct product lines: brand-new items such as bikinis, coordinated sets, and accessories, and preloved items such as pants, skirts, and shorts. Because the project centers on solving an actual operational problem rather than simply observing or explaining one, a developmental approach fit the nature of the worked.

A developmental design was chosen for the following reasons:

It was appropriate when the primary goal was to build and refine a working system rather than to describe or correlate variables.

It allowed the design process to be iterative, with continuous testing and adjustment based on feedback rather than a single fixed plan.

It supported a structured evaluation of the finished system's performance, usability, reliability, and security once it was used in an actual business setting.

Guided by this design, the study evaluated the Margaux Collections Ordering System against four practical questions: whether it accurately recorded and processes orders, whether it managed both new and preloved inventory without confusion, whether the interface was usable for both the shop owner and her customers, and whether it fit naturally into the Facebook-based marketing and selling process the business already relied on. Answering these questions through actual development and testing, rather than through description alone, was what made the developmental design the appropriate choice for this study.

3.2 System Development Methodology

The Margaux Collections Ordering System was developed using the Agile methodology, applied through short, iterative development cycles rather than a single long build-and-release phase. Agile was selected over a strictly sequential model such as Waterfall because the requirements of a small business system tend to shift as the owner sees working features and reacts to them. Specifically, Agile was chosen because it:

allowed work to be broken into short, focused iterations instead of one long development phase; kept the business owner involved throughout development rather than only at the start and the end; made it easier to adjust features as requirements become clearer; and built in regular testing and review, so problems were caught while they were still small.

The methodology was carried out through five phases, described below.

3.2.1 REQUIREMENTS ANALYSIS

The first phase focused on understanding how the business actually operated before any code was written. This involved:

conducting interviews with the business owner to learn about her current process for receiving orders, tracking stock, and dealing with the mix of new and preloved items; observing the existing manual workflow — mostly conducted through Facebook Messenger and handwritten or spreadsheetbased recorded — to identify recurring pain points such as missed orders or unclear stock counted; reviewing existing product lists, price sheets, and order records to understand how items and transactions were already being organized; and translating these observations into user stories (for example, "As a customer, I wanted to view preloved items separately from new items so that I know what I am buying") and, from there, into concrete technical requirements, such as supporting two distinct product categories, allowing full product and order management, and providing a dashboard for daily sales and stock status.

This phase produced a working requirements specification that guided the design and development work that followed.

3.2.2 SYSTEM DESIGN

With the requirements defined, the next phase focused on planning the system's structure. This included laying out the overall architecture across the front end, back end, and database; sketching wireframes for both the customer-facing shop and the admin panel; designing the database schema and the relationships between tables; and preparing a Data Flow Diagram and Entity Relationship Diagram to map out how information would move through the system before any of it was built.

3.2.3 DEVELOPMENT

Development work followed the plans set in the design phase. The front end was built using HTML, CSS, and JavaScript to handle product browsing, the shopping cart, checkout, and order history. The back end was implemented in PHP, handling authentication, order processing, inventory updated, and the analytics features. The MySQL database was built and connected securely to the PHP scripts, and periodic self-review of the code was carried out to catch errors early rather than after the fact.

3.2.4 TESTING

Testing was carried out at multiple levels throughout development rather than only at the end. Individual PHP functions and database queries were checked in isolation through unit testing; the front end, back end, and database were then tested together through integration testing to confirm that the layers worked correctly as a whole; the system was checked for common security weaknesses such as injection attacks and unauthorized access; and, once a working version was available, the business owner and a few customers tried the system directly to evaluate whether it was accurate and easy to use. A fuller account of the testing process was given in Section 3.6.

3.2.5 DEPLOYMENT (PROTOTYPE)

Once testing showed the system was stable enough for real use, a working prototype was deployed to a local test environment and presented to the business owner for hands-on evaluation. She was able to place tested orders, update inventory, and review the analytics dashboard herself. Her feedback from this stage informed a round of minor refinements before the system was finalized, which kept the final product closely aligned with how the business actually operated rather than with assumptions made early in the project.

3.3 System Architecture

The Margaux Collections Ordering System followed a three-layer architecture — a presentation layer, an application layer, and a data layer — so that the interface, the business logic, and the stored data remained separated from one another. Keeping these concerns apart made the system easier to build, test, and maintain, since a change in one layer, such as a redesign of the product page, did not require touching how orders were processed or how data was stored.

3.3.1 PRESENTATION LAYER

The presentation layer was what customers and the shop owner actually see and interact with. It was built using HTML5 for page structure, CSS3 for styling and

layout, and JavaScript for dynamic behavior such as live cart updated and form validation. On the customer side, this layer provided a product browsing page that separated new items (bikinis, coordinated sets, accessories) from preloved items (pants, skirts, shorts), a shopping cart, a checkout page for entering delivery details and choosing a payment method, and an order history page. On the admin side, it provided a dashboard summarizing sales and inventory, along with pages for managing products, orders, and customer accounts. The interface used responsive CSS so it displayed properly on desktops, tablets, and phones, which mattered given that most customers reach the shop through links shared on Facebook and browse primarily on mobile.

3.3.2 APPLICATION LAYER

The application layer was the back end, built in PHP, and sat between the interface and the database. It handled user authentication, order processing (including order numbering, total computation, and inventory updated), status changed such as moving an order from "pending" to "for delivery," and a set of security measures: prepared statements to prevent SQL injection, secure session handling with timeouts, and validation of all user-submitted input. This layer also enforced role-based access — customers could browse, added items to their cart, checked out, and view their own order history, while the owner, as administrator, could manage products, orders, users, and the analytics dashboard — so that sensitive operations remained restricted to the account that should be able to perform them.

3.3.3 DATA LAYER

The data layer stored user accounts, the product catalog, order transactions, and the individual line items within each order. The database followed a normalized structure to avoid duplicating information, such as repeating product details inside every order records, and indexes were applied to fields that were queried often, such as order and product IDs, to keep lookups fast as the number of orders grew.

3.3.4 CORE SYSTEM MODULES

Within these three layers, the system's functionality was organized into modules, each responsible for one part of the ordering workflow:

Authentication Module — handled account registration, login, and session management, and, at minimum, secures admin access to the owner's dashboard even if customer accounts were kept simple.

Product Management Module — let the owner added, edit, and deactivated products, tagged items by category and condition (for example, "preloved, good condition"), and kept stock counted current as items sold.

Shopping Cart Module — managed a per-session cart, checking stock before items were added, letting quantities be adjusted, and calculating running totals.

Order Processing Module — assigned order numbers, recorded customer and product details, computed totals, supported cash and GCash payment, and updated stock once an order was confirmed.

Preloved-Specific Management Module — since preloved fashion was central to the business, this module let the owner tagged condition and added notes specific to secondhand items, distinguishing them from new stock.

User Management Module — allowed the owner to view, search, and managed customer accounts.

Analytics and Sales Reporting Module — gave the owner a dashboard of revenue, best-selling items, order volume, and low-stock alerts to support restocking and pricing decisions.

Together, these modules covered the full cycle of listing products, taking orders, and monitoring how the shop was performing.

3.3.5 DATABASE DESIGN

The database was built in MySQL and organized around five core tables:

Users — user_id (primary key), name, email, hashed password, contact number, address, role, and date created.

Products — product_id (primary key), product name, description, price, product type (new or preloved), condition, image path, stock quantity, and date added.

Orders — order_id (primary key), user_id (foreign key), customer details, fulfillment method (meet-up, drop-off, shipping, or pick-up via rider), payment method and status, order status, total amount, pinned map location where relevant, and order date.

Order_Items — item_id (primary key), order_id and product_id (foreign keys), quantity, and unit price at the time of purchase.

User_Payment_Accounts — account_id (primary key), user_id (foreign key), account type (GCash), account name and number, and addressed.

Primary and foreign keys enforced the relationships between these tables, the schema was normalized to avoid redundant data, and indexes were placed on frequently queried fields such as order and product IDs, which together kept the data consistent and queries efficient as the shop's order volume grew.

3.4 Tools and Technologies Used

The development of the Margaux Collections Ordering System utilized various programming languages, frameworks, development tools, and technologies that supported the implementation, testing, and maintenance of the system. These tools were selected based on their suitability for developing a web-based ordering system, managing customer and product information, supporting interactive features, and maintaining an organized development workflow. The following tools and technologies were used throughout the development process:

PHP — was used as the primary server-side programming language for implementing the system's business logic and backend functions. It handled processes such as user authentication, product management, order processing,

inventory updates, payment verification, and communication between the application and database.

MySQL — was used as the relational database management system for storing and retrieving system information. It managed records related to users, products, categories, inventory, orders, order items, payments, messages, notifications, and other system data. The relational structure helped maintain consistency and establish connections among related records.

HTML5 and CSS3— were used to structure and design the system's web pages and user interfaces. HTML5 provided the structure for the storefront and administrative pages, while CSS3 was used to implement the dark noir visual theme, layouts, typography, colors, spacing, and responsive design. These technologies allowed the system to provide a consistent interface across desktop and mobile screen sizes.

JavaScript — was used to provide interactive and dynamic behavior within the system. It supported functions such as cart updates, form validation, interface interactions, and other client-side processes. Its use helped make the ordering process more responsive and convenient for users.

Chart.js — was used to create graphical representations of sales and performance data within the administrative analytics dashboard. The charts helped present business information in a more understandable visual format, allowing the administrator to review sales-related data and identify relevant trends.

XAMPP — was used as the local development and testing environment. It provided the Apache web server, PHP runtime, and MySQL database services required to run and test the web application locally. It also allowed the developers to verify system functions before deployment.

Visual Studio Code — served as the primary source-code editor throughout the development process. It was used to create, edit, organize, and review the PHP, HTML, CSS, JavaScript, and other project files.

Git and GitHub — were used for version control and collaborative development. Git allowed the developers to track changes, manage different versions of the project

files, and revert changes when necessary, while GitHub provided a centralized repository that helped keep the codebase synchronized across the development machines.

3.5 Data Gathering Techniques

During the requirements analysis phase, several data gathering techniques were used to obtain relevant information about the operations, existing processes, and system needs of Margaux Collections before the development of the proposed ordering system. Interviews were conducted with the business owner through structured and semi-structured questions to understand how orders were received and processed, particularly through Facebook Messenger and in-person transactions. The interviews also covered existing inventory management practices, common challenges encountered during order processing, payment procedures, and the features that the business owner considered important for the proposed system. Observation was also conducted by examining the actual ordering, product preparation, and packing processes. This allowed the researchers to identify operational difficulties such as disorganized product listings, manual recording errors, difficulties in monitoring inventory, and challenges in distinguishing preloved products from brand-new stock.

Document analysis was conducted by reviewing existing product lists, price sheets, and previous order records to understand how products were categorized, how prices were determined, and how customer orders and transactions had been recorded. These documents provided additional information that helped the researchers identify the data that needed to be included in the proposed system. ****Surveys**** were also used, where feasible, to gather feedback from existing customers regarding their interest in online ordering and their preferred system features. The survey responses helped identify customer expectations, including the need for clearer product photographs, organized product information, and a separate section for preloved items. The information gathered through these techniques served as a basis for defining the system requirements, designing the database structure, and developing features that addressed the actual operational needs of Margaux Collections and its customers.

3.6 Testing Procedures

The Margaux Collections Ordering System underwent four types of testing to determine whether the developed system functioned correctly, performed reliably under normal conditions, and adequately protected customer and business information. The testing procedures were conducted throughout the development process to identify functional errors, usability issues, integration problems, and potential security concerns. Each testing type focused on specific aspects of the system, and the results were used to identify necessary corrections and refinements before the system was considered ready for deployment and evaluation.

3.6.1 UNIT TESTING

Individual PHP functions, such as order creation and stock updates, were tested with a range of inputs to confirm they returned the expected results.

Database queries were checked to make sure records were inserted, updated, and retrieved correctly, and basic front-end validation, such as required-field checks, was verified as well. This confirmed that each component worked correctly on its own before being combined with the rest of the system.

3.6.2 INTEGRATION TESTING

Once individual components passed unit testing, they were tested together to confirm that the front end, back end, and database worked correctly as a whole. This included confirming that adding an item to the cart correctly stored it, that placing an order created the right order and order-item recorded while updating stock accordingly, and that changed made in the

admin panel, such as price edits, were reflected properly on the customerfacing side.

3.6.3 SECURITY TESTING

Because the system stored personal and transactional data, it was checked against common vulnerabilities, including SQL injection (mitigated through prepared statements), cross-site scripting (mitigated by escaping user input), cross-site request forgery (mitigated with tokens where applicable), and session hijacking (mitigated through proper session handling). Access controlled were also tested to confirm that customers cannot reach the admin panel and that authentication cannot be bypassed.

3.6.4 USER ACCEPTANCE TESTING (UAT)

After the prototype was deployed, the business owner and a small group of customers used the system through real or simulated tasks — placing orders, updating products, and reviewing the dashboard — and their feedback on ease of used, accuracy, and overall appearance was collected. This step confirmed that the system holds up not just in testing conditions, but in the kind of everyday used it was actually built for.

3.7 Evaluation Criteria

The finished system was assessed against five criteria, described below.

3.7.1 FUNCTIONALITY

The system had to correctly perform its required functions, including accurately recording orders, properly updating inventory, correctly computing totals, and clearly displaying order history and status.

3.7.2 USABILITY

The interface had to be intuitive for both customers and the owner, keeping the learning curve low, providing clear navigation and feedback, and remaining fully usable on mobile devices.

3.7.3 SECURITY

The system had to protect user data and restrict access appropriately, with information properly hashed, transmitted securely, and accessible only to authorized users.

3.7.4 PERFORMANCE

The system had to respond quickly under normal use, loading pages within a reasonable time and handling multiple concurrent users without noticeable slowdown.

3.7.5 RELIABILITY

The system had to process orders correctly and maintain data integrity, with orders and inventory changes recorded consistently and reversibly only through a proper process, such as a formal cancellation.

Supporting Diagrams

The following diagrams illustrated the structure and expected behavior of the system in more detail.

Data Flow Diagram (DFD — Level 0)

The Level 0 Data Flow Diagram showed how customers and the administrator interacted with the system and how their data ultimately reached the MySQL database. Customers submitted registration details, login credentials, order information, and checkout confirmations; the system validated product availability, created order records, and updated inventory before storing everything in the database and returning confirmations, product listings, or order status updates. The administrator, on the other hand, submitted product updates, inventory adjustments, and order status changes, which the system processed and reflected in the database in real time. Read together, the diagram laid out the system's overall workflow in a way that was easier to follow than a written description alone.

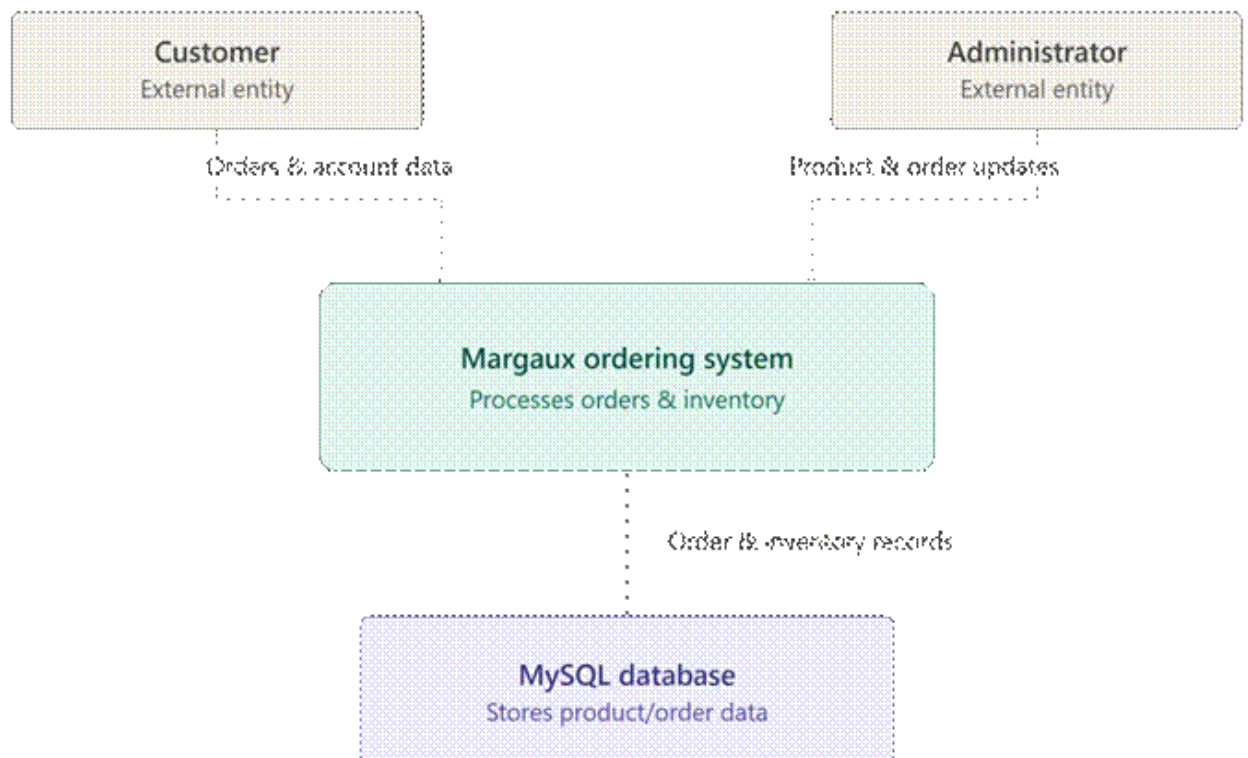


Figure 3.1. Level 0 Data Flow Diagram of the Margaux Collections Ordering System

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram defined the core entities in the database User, Product, Order, OrderItem, and UserPaymentAccount — along with how they related to one another. A user could place multiple orders, an order could contain multiple order items, a product could appear across multiple order items, and a user could save multiple payment accounts. Laying this out clearly helped confirm that the tables connected correctly and that data could be retrieved accurately, which in turn supported a database design that was easier to maintain as the system grew.

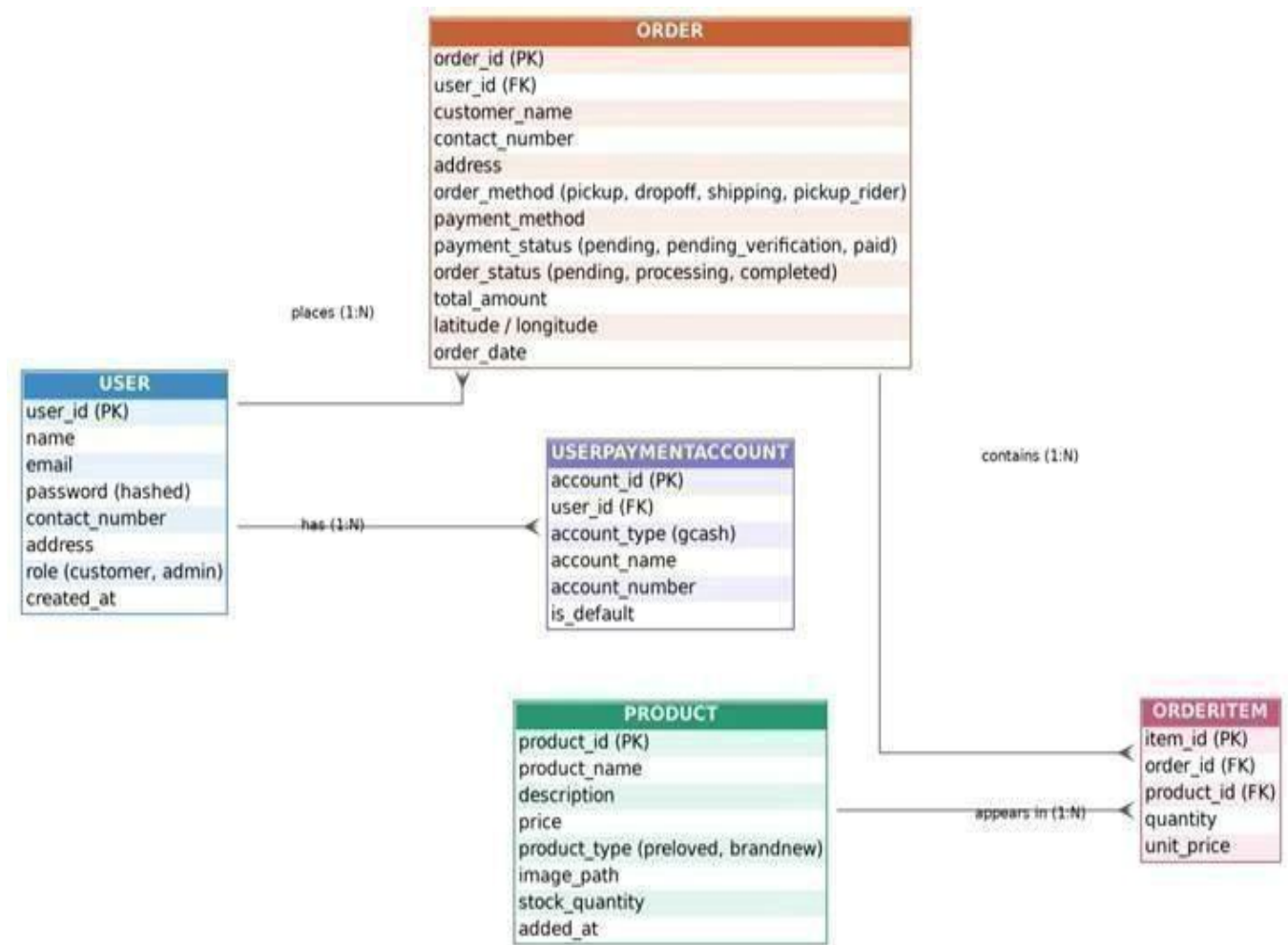


Figure 3.2. Entity Relationship Diagram of the Margaux Collections Ordering System

Use Case Diagram

The Use Case Diagram presented the system's two main actors, the customer and the administrator, and the actions available to each. Customers could register, log in, browse products, managed their cart, checked out, placed orders, viewed

order history, updated their profile, and save payment methods. Administrators could log in to the admin panel, manage products, manage orders, manage users, and view analytics. Together these use cases outlined the functional scope of the system and made clear what each type of user was able to do within it.

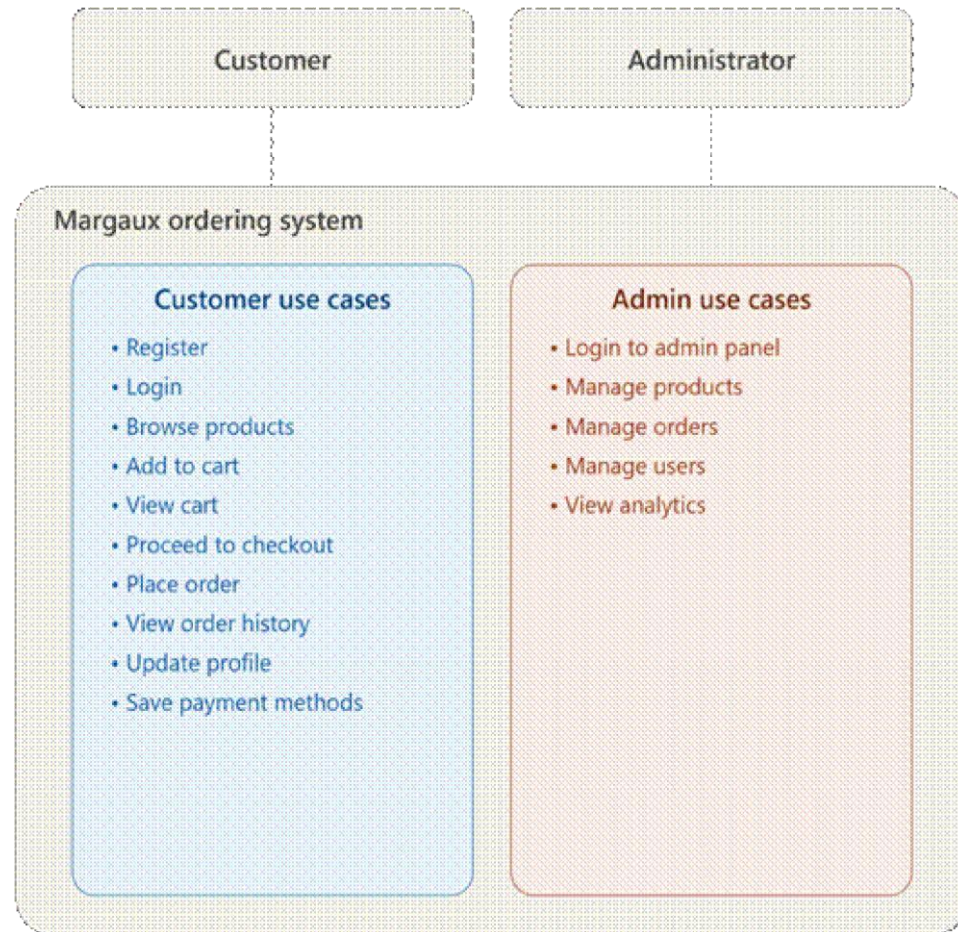


Figure 3.3. Use Case Diagram of the Margaux Collections Ordering System

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 OVERVIEW OF THE DEVELOPED SYSTEM

Margaux Collections was a web-based e-commerce and boutique management system developed for a fashion retail business to modernize its

product presentation, sales transactions, and order handling. The system replaced manual and social-media-based selling practices with a structured online platform that allowed customers to browse the boutique's product catalog, place orders, and complete payment through a GCash QR code checkout process.

Unlike generic e-commerce templates, Margaux Collections was designed with a distinct visual identity tailored to the boutique's brand. A dark noir theme was implemented throughout the storefront and administrative interfaces using custom CSS variables for color theming and the Playfair Display and Jost typefaces for headings and body text, respectively, giving the platform a fashion-forward and elegant presentation consistent with the boutique's market positioning.

The system was built entirely from scratch, encompassing the database schema, authentication pages, customer-facing storefront, and a full administrative panel. The application processed product listings, customer orders, and homepage promotional content, and presented this information through role-based interfaces for customers and administrators. The major objectives of the developed system were as followed:

Provided an online storefront where customers could browse and purchase boutique products.

Automate the recording, storage, and tracking of customer orders.

Support a secure, QR-code-based GCash payment process, alongside Cash on Pickup and Cash on Delivery options depending on the customer's chosen order fulfillment method.

Enabled administrators to manage products, orders, customer accounts, and site content through a dedicated panel.

Allowed administrators to control homepage promotional slots to feature selected products or campaigns.

Provide a messaging feature for direct communication between customers and administrators.

Maintain a consistent, brand-aligned visual design across all customer and administrative pages.

The development of Margaux Collections followed modern web development practices, incorporating responsive interface design, secure database management, user authentication, and version-controlled collaborative development. The completed system consisted of several interconnected modules that worked together to manage products, process orders, and maintained the boutique's online presence.

Figure 4.1 illustrated the overall workflow of the developed system.

Figure 4.1 System Workflow of Marguax Collections

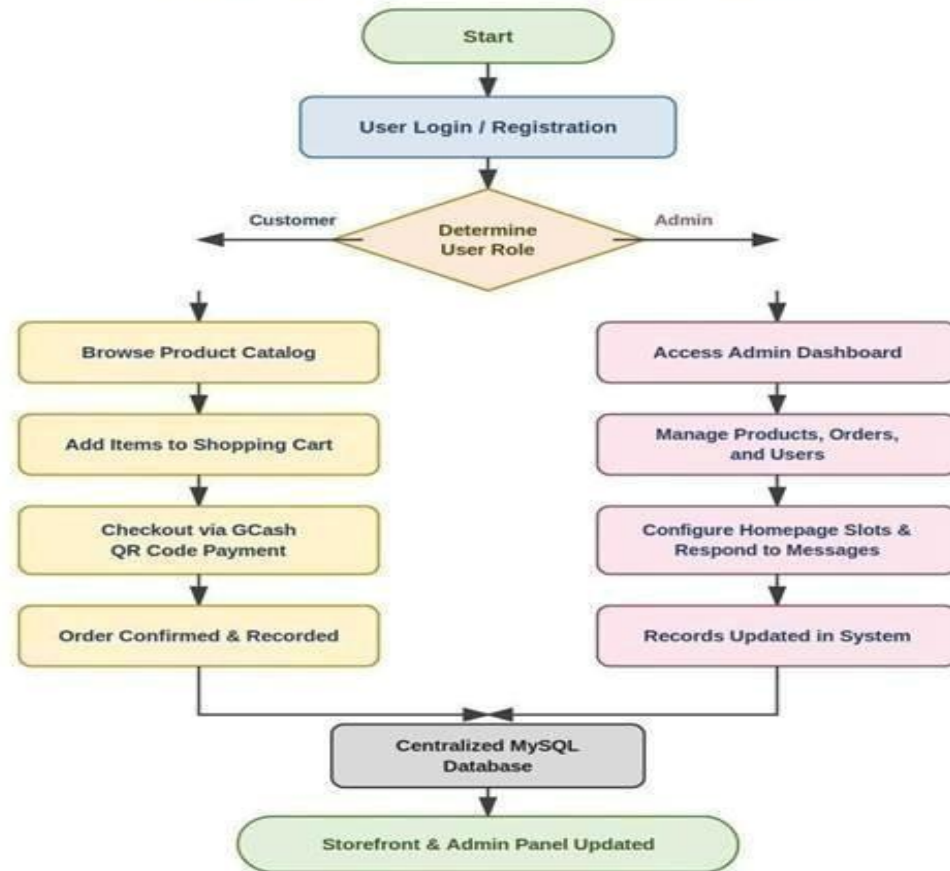


Figure 4.1. System Workflow of Margaux Collections

The workflow began with user authentication, after which customers could browse products, add items to their cart, and proceed to checkout using the GCash QR payment method. Administrators, upon logging in, could manage products, monitor and update order status, configured homepage promotional slots, and respond to customer messages through the administrative dashboard.

4.3 SYSTEM DEVELOPMENT METHODOLOGY

4.1 DEVELOPMENT METHODOLOGY

The development of Margaux Collections adopted the Agile Software Development Methodology, an iterative approach that emphasized incremental development, continuous testing, evaluation, and refinement throughout the software development life cycle. Agile was selected because the system requirements, particularly its functional features, user interface, and visual design, evolved as the development progressed. This approach provided the developers with the flexibility to add, modify, improve, or remove features based on the identified needs of the business without significantly disrupting the overall development process. It also allowed the team to respond to issues discovered during implementation and testing, ensuring that the final system remained aligned with the operational requirements of Margaux Collections.

The Agile methodology allowed the development process to be organized into a series of iterative cycles in which individual modules were planned, developed, tested, evaluated, and refined before being integrated into the completed system. Each iteration provided an opportunity to identify technical issues, usability concerns, and unnecessary features at an early stage. This iterative process proved particularly useful when previously implemented features required significant modification. One example was the eventual removal of the tiered membership access feature, which had initially been included in the system design but was later determined to be unnecessary for the actual business requirements. Removing the feature simplified the customer experience, database structure, and administrative controls while allowing development efforts to focus on the system's core functions.

Requirements Gathering

The requirements gathering phase involved identifying and documenting the operational needs of the Margaux Collections boutique. The development team determined the required functions by considering the business's existing selling and ordering processes, the needs of customers, and the administrative tasks performed by the business owner. These requirements included the desired look and feel of the online storefront, product catalog management, inventory monitoring, customer account management, order processing, payment verification, and administrative controls. The preferred GCash payment method was also considered in designing the checkout and transaction verification process. The information gathered during this phase served as the basis for defining the system's core modules, database structure, user roles, and overall functionality. In addition, the requirements guided the development of the system's dark noir visual theme, which was intended to maintain a consistent and recognizable brand identity for Margaux Collections.

System Planning

Following the requirements gathering phase, the project scope, feature set, development priorities, and schedule were established. The planning phase identified the major components that needed to be developed, including the customer-facing storefront, product catalog, shopping cart, checkout and payment verification, user accounts, inventory management, order management, messaging, and administrative dashboard. The technical resources required for development were also identified, including the local XAMPP development environment, MySQL database platform, PHP programming language, and Git and GitHub for version control and collaborative development. This planning process provided a clear direction for the development team and helped ensure that the system was developed according to the identified business requirements.

System Design

During the system design phase, the database schema, system structure, page layouts, navigation flow, and administrative panel were designed based on the requirements identified during the previous phase. The database structure was planned to organize information related to users, products, inventory, orders, payments, messages, notifications, and other supporting functions while maintaining appropriate relationships among records. The customer-facing pages and administrative interfaces were also designed to provide clear navigation and organized access to system functions. In addition, the color palette, typography, spacing, and other visual elements of the dark noir theme were defined through reusable CSS variables and design rules. This approach helped maintain a consistent visual identity across the storefront and administrative panel while allowing the interface to be modified more efficiently during later development and refinement.

Software Development

The software development phase involved implementing the planned system components using PHP and MySQL within a XAMPP local server environment. Development proceeded incrementally, beginning with the creation of the database schema and authentication functions, followed by the development of the customer storefront, product catalog, shopping cart, and checkout process with GCash QR-based payment. The administrative

panel was then developed to include the dashboard, analytics, order management, product management, user management, inventory-related functions, promotional content, and messaging modules. Each module was developed according to the established requirements and was gradually integrated with the other components of the system. During development, several features and interface elements were refined based on testing results and the actual needs of the boutique. This included modifying existing functions and removing the tiered membership feature when it was determined to be unnecessary for the intended operation of the system.

Testing

The testing phase involved evaluating the functionality, usability, integration, and consistency of the developed system. Each module was tested individually after it was completed to identify errors and confirm that its intended functions operated correctly. Additional testing was conducted after the modules were integrated to determine whether the different components worked properly together. Particular attention was given to changes that affected multiple parts of the system. For example, when the tiered membership feature was removed, the developers checked the database structure, authentication processes, administrative panel, and customer-facing pages to ensure that the removal did not create broken links, unnecessary records, or inconsistencies in system behavior. Interface and stylesheet issues were also identified and corrected during testing to maintain the consistency of the dark noir theme. The testing process allowed the developers to refine the system and address issues before deployment.

Deployment

Following the testing and refinement phases, the Margaux Collections Ordering System was configured and deployed within a local XAMPP server environment. The deployment process involved setting up the application files, MySQL database, server configuration, and required system components to ensure that the application could operate properly in its intended environment. The completed system was made accessible to its intended users for demonstration, evaluation, and operational testing. The deployment phase also included verifying that the database connection,

authentication functions, product listings, ordering process, GCash payment verification, and administrative features continued to operate correctly in the configured environment. This ensured that the final system was properly prepared for use by the Margaux Collections business and its intended customers.

Deployment Process

The deployment process began with the configuration of the XAMPP environment, where the Apache web server and MySQL database service were prepared to host the Margaux Collections Ordering System locally. The completed database schema, which included the Users, Products, Categories, Orders, Order Items, Homepage Slots, and Messages tables, was exported and imported through phpMyAdmin to establish a consistent database state across the development machines used by the project. This process helped ensure that the application operated using the same database structure and records during development, testing, and deployment.

The application files, including the customer-facing storefront, administrative panel, shared stylesheet files, configuration files, and other required resources, were placed inside the XAMPP `htdocs` directory so that they could be accessed through the local Apache server. The database connection settings were configured to connect the application to the local MySQL service. The folder structures and file permissions for product images and other uploaded content were also verified to ensure that both the storefront and administrative panel could properly retrieve and display stored media. These configurations helped ensure that the system's product listings, images, administrative functions, and other resources operated correctly within the deployed environment.

Since development was conducted across two separate laptops, Git and GitHub were used to synchronize the project's source code and maintain consistency between the development environments. During this process, several practical technical issues were encountered and resolved. One issue involved multiple `mysqld.exe` processes running simultaneously on one machine, which prevented the XAMPP MySQL module from starting properly. The issue was resolved by terminating the duplicate MySQL process before restarting the MySQL service through XAMPP. Occasional merge conflicts were also encountered in the stylesheet files when changes from different development environments were combined. These conflicts were resolved through manual review and reconciliation of the affected CSS rules to ensure that necessary changes were preserved while maintaining the intended dark noir visual theme of the system.

After the application files and database were successfully configured and deployed on the local server, end-to-end testing was conducted to verify the functionality of the completed system. Both customer and administrator workflows were simulated to ensure that the different components operated correctly after deployment. Customer-side functions, including product browsing, product viewing, shopping cart management, and GCash QR-based checkout, were tested, while administrative functions such as order management, product management, and homepage slot configuration were also verified. The testing process confirmed that the deployed system maintained the expected behavior observed during development and that the database, application files, user interfaces, and system functions were properly integrated within the local XAMPP environment.

Maintenance

The maintenance phase involved monitoring the performance, functionality, and consistency of the Margaux Collections Ordering System after its initial development and deployment. This phase focused on identifying and resolving issues that were discovered through continued testing and actual system use. Maintenance activities included correcting styling inconsistencies, addressing data-related issues, refining existing features, and ensuring that the customer-facing storefront and administrative panel continued to operate as intended. These activities helped maintain the system's reliability and usability while ensuring that its functions remained aligned with the operational requirements of the boutique.

Particular attention was given to resolving stylesheet conflicts between the customer-facing storefront and administrative interfaces to preserve the consistency of the dark noir visual theme. Database synchronization issues encountered during collaborative development were also addressed to ensure that information remained consistent across the development environments. In addition, existing system features were reviewed and refined whenever testing identified areas that could be improved. Through continuous maintenance and refinement, the system remained functional, organized, and suitable for supporting the online ordering and management needs of Margaux Collections.

Figure 4.2 presented the Agile Software Development Life Cycle adopted in the study.

Figure 4.2 Agile Software Development Life Cycle

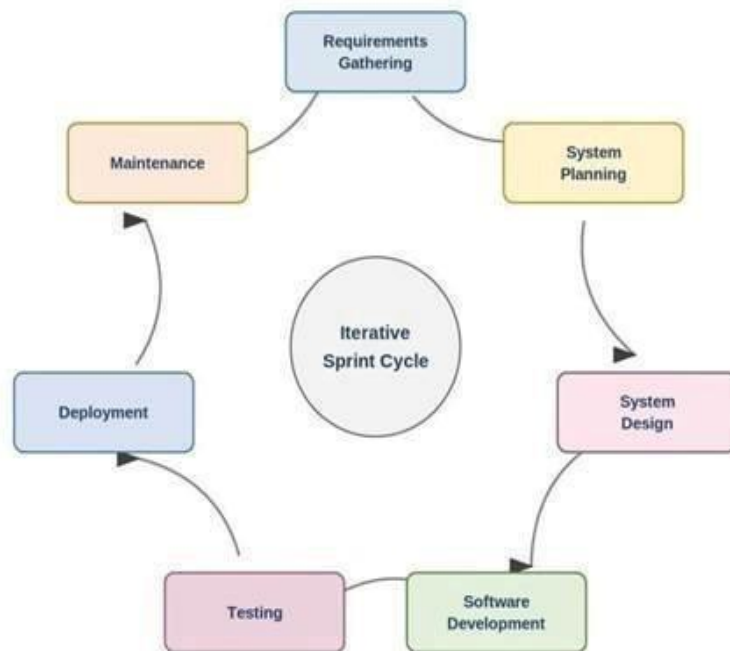


Figure 4.2. Agile Software Development Life Cycle

4.4 SYSTEM ARCHITECTURE

Margaux Collections utilized a Three-Tier Client-Server Architecture, which divided the system into three interconnected layers: the presentation layer, application layer, and data layer. The presentation layer handled the user interface and interactions of customers and administrators with the system, including product browsing, ordering, checkout, and administrative functions. The application layer manages the system's business logic and core processes, such as authentication, product and order management, payment verification, inventory management, messaging, and other application functions. Meanwhile, the data layer was responsible for storing, retrieving, and managing system information through the MySQL database. This architectural approach improved the organization, maintainability, security, and scalability of the system by assigning specific responsibilities to each layer. It also allowed individual components to be modified, tested, or improved with minimal impact on the other layers, making the system easier to manage and maintain throughout its development and future enhancement.

Figure 4.3 illustrated the architecture of the developed system.

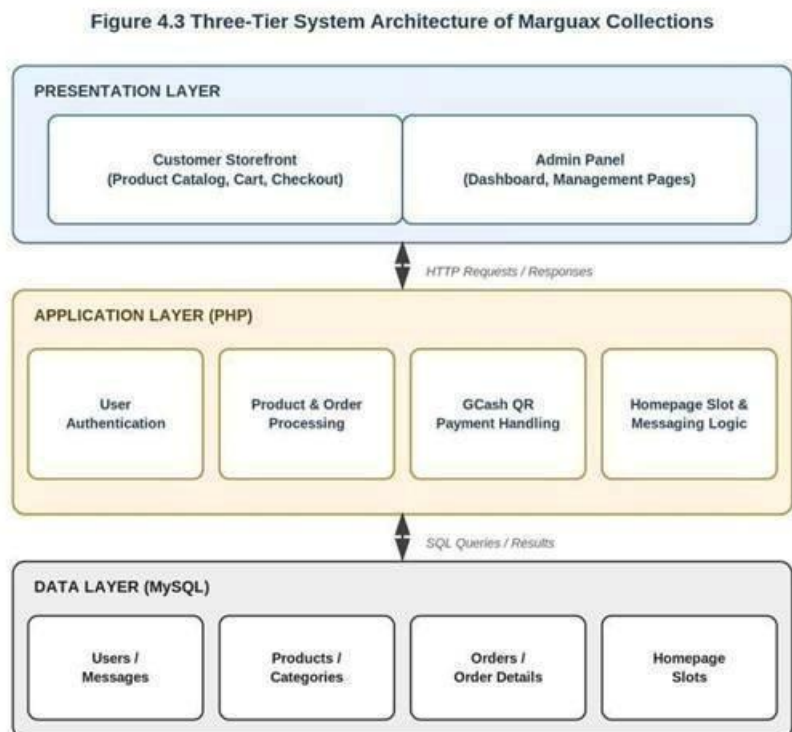


Figure 4.3. Three-Tier System Architecture of Margaux Collections Layer

Presentation

The presentation layer served as the primary interface between users and the Margaux Collections Ordering System. It was responsible for presenting the product catalog, product details, shopping cart, checkout process, order information, and administrative dashboards in an organized and userfriendly manner. It also handled user interactions, allowing customers to browse available brand-new and preloved clothing, review product information, place orders, and access relevant transaction details. For administrators, the layer provided interfaces for managing products, users, orders, inventory, and promotional content. The presentation layer also implemented the system's dark noir visual theme consistently throughout the storefront and administrative pages while using a responsive layout to provide a suitable viewing experience across different screen sizes and devices.

Major interface components included: Login and
Registration Pages
Customer Storefront and Product Catalog

Shopping Cart and Checkout (GCash QR Payment)

Admin Dashboard

Product Management

Order Management

User Management

Analytics

Messages

Conversations

Notifications

User Payment Accounts

OTP Tokens Password Resets

Application Layer

The application layer contained the business logic responsible for processing user requests, validating input data, and managing the interactions between the storefront, the administrative panel, and the database. Its primary responsibilities included:

User Authentication

Product and Category Management • Order Processing

Payment Verification (GCash QR)

Homepage Promotional Slot Management

Customer–Admin Messaging

Analytics and Reporting

The application server, built using PHP, acted as the intermediary between the presentation layer and the MySQL database server. *Data Layer* The data layer stored all business information in a centralized relational database managed through MySQL. It was responsible for maintaining data integrity and supporting efficient retrieval of the information required by the storefront and administrative panel. Stored information included:

User Accounts

Products and Categories

Orders and Order Items

Homepage Promotional Slots

Customer–Admin Messages

The architecture ensured that communication between the application server and the database remained organized and consistent, restricting unauthorized access to administrative functions and customer data.

4.5 HARDWARE AND SOFTWARE REQUIREMENTS

The implementation of Margaux Collections required appropriate hardware and software resources to ensure efficient performance during product browsing, order processing, and administrative operations.

Component	Minimum Specification	Recommended Specification

Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	128 GB SSD	256 GB SSD or higher

Monitor	1366 x 768 Resolution	Full HD Display
Internet Connection	10 Mbps	25 Mbps or higher
Mobile Device	Android 8	Android 11 or higher

Software	Purpose
Windows 10 / 11	Operating System

Visual Studio Code

Source Code Editor

MySQL (via XAMPP)

Database Management
System

PHP	Backend Development
HTML, CSS, JavaScript	Frontend Development
Google Chrome	Web Browser

Git and
GitHub

Version Control and
Collaboration

XAMPP

Local Development
Server

4.5.1 HARDWARE REQUIREMENTS

Table 4.1 presented the recommended hardware specifications.

4.5.2 SOFTWARE REQUIREMENTS

Table 4.2 presented the software requirements.

The selection of these software tools was based on their stability, accessibility, and suitability for developing and locally hosting a PHP and MySQL web application.

4.6 DATABASE DESIGN

The database served as the foundation of the Margaux Collections Ordering System by organizing, storing, and managing essential information required for the operation of the online platform. It contained records related to product listings, customer accounts, orders, order details, payments, inventory, promotional content, messages, notifications, and other supporting system functions. A relational database model was implemented using MySQL, allowing related information to be stored in separate but interconnected tables. This structure helped reduce data redundancy, improve data organization, and maintain consistency among related records.

The database design was developed based on the functional requirements of the system and the actual processes involved in the operations of Margaux Collections. Primary entities were established for important business information, while supporting tables were created for authentication, communication, payment, and security-related functions. Relationships between tables were established using primary and foreign keys to ensure that records remained properly connected. For example, customer accounts could be associated with their orders, while order details could be linked to the corresponding products. This relational structure allowed the system to retrieve and update information efficiently while minimizing the possibility of duplicate or inconsistent data.

The use of MySQL also provided a structured environment for managing the increasing amount of information generated by the system. Product records could be updated as items became available or were sold, while order and payment records could be maintained for transaction monitoring and

administrative reference. Customer and administrative information could also be managed through appropriate database relationships and security measures.

Overall, the database design provided a reliable and organized foundation for the Margaux Collections Ordering System and supported its major functions, including product management, customer management, inventory monitoring, order processing, payment verification, communication, and administrative operations.

Figure 4.4 presented the Entity Relationship Diagram (ERD) of the system.

Figure 4.4. Entity Relationship Diagram of Margaux Collections The primary entities of the database included:

Users

Products

Categories

Orders

Order Items

Homepage Slots

Messages

Conversations

Notifications

User Payment Accounts

OTP Tokens Password

Resets

4.6.1 Database Tables *Users Table*

Field	Description
user_id	Primary Key

name	Complete Name
email	Login Email Address
password	Encrypted Password

role	User Access Level (Customer / Admin)
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This table stored account credentials and role information for both customers and administrators.

Products Table

This table stored the boutique's product listings.

Filed	Description
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product_id	Primary Key
product_name	Product Name

category_id	Foreign Key — Product Category
price	Selling Price

Field	Description
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stock	Available Inventory
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image	Product Image
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description	Product Description
product_type	Product Type (Pre-loved / Brandnew)
sold_out_at	Timestamp When Product Went Out of Stock

Categories Table

This table stored the product classifications used to organize the boutique's catalog.

Field	Description

category_id	Primary Key
name	Category Name

Orders Table

This table recorded customer purchase transactions.

Field	Description
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order_id	Primary Key

Field	Description
user_id	Foreign Key

order_date	Date of Purchase
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order_method	Order Fulfillment Method (Meet up / Drop off / Shipping Delivery / Pick-up Via Rider)
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payment_method	Selected Payment Method (Cash on Pickup / Cash on Delivery / GCash)
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total_amount	Total Order Amount
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latitude	Pinned Delivery Latitude (Shipping / Pick-up Via Rider only)
longitude	Pinned Delivery Longitude (Shipping / Pick-up Via Rider only)

payment_status	Payment Status (Pending / Pending Verification / Paid)
order_status	Order Fulfillment Status

Order Items Table

This table recorded the individual items included in each order.

Field	Description
item_id	Primary Key

order_id	Foreign Key
product_id	Foreign Key
quantity	Quantity Ordered

price	Unit Price at Time of Purchase
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Homepage Slots Table

This table stored administrator-configured promotional content displayed on the storefront's homepage.

Field	Description
slot_id	Primary Key

label	Slot Label (e.g. Featured, Dresses, Tops)
image_path	Uploaded Promotional Image Path

price_label	Optional Display Price Text (e.g. "Starting at ₱299")
updated_at	Date Last Updated

Messages Table

This table stored messages exchanged between customers and administrators.

Field	Description
message_id	Primary Key

conversation_id	Foreign Key — Linked Conversation
sender_type	Sender Role (Customer / Admin)
sender_id	Foreign Key — Sending User

message	Message Text
created_at	Date and Time Sent

In addition to these primary entities, the database included several supporting tables that handled authentication, communication, notifications, and account security. These included the Conversations table, which grouped customer–admin messages into threads referenced by the Messages table; the Notifications table, which stored order-related alerts and updates sent to customers; the OTP Tokens and Password Resets tables, which supported one-time password verification for login, registration, and password recovery; and the User Payment Accounts table, which stored customers’ saved GCash payment details for checkout. These supporting tables were excluded from Figure 4.4 for clarity and to maintain a more readable representation of the primary database structure. However, they remained connected to the appropriate user, order, and transaction records through established database relationships.

The relationships among the database tables allowed the application to associate customers with their orders, connect order details with specific products, maintain transaction histories, and

manage the promotional content displayed on the storefront. Foreign key constraints were implemented to maintain referential integrity among related records and to ensure that data remained properly associated with its corresponding users, products, orders, and messages. These relationships also helped prevent inconsistent or incomplete records while supporting the accurate retrieval and management of information throughout the system. As a result, the database structure provided a reliable foundation for the major functions of the Margaux Collections Ordering System, including product management, customer communication, ordering, payment processing, notifications, and account security.

CHAPTER 5 CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the overall conclusion of the development of the Margaux Collections Ordering System. It synthesized the results obtained from the development, testing, and evaluation of the system and assessed the extent to which the identified research objectives were achieved. The chapter also summarized how the developed system addressed the operational needs of the boutique, particularly in terms of online product presentation, ordering, inventory management, payment verification, and administrative monitoring. The findings provided an overall assessment of the system's functionality, usability, and relevance to the business operations of Margaux Collections.

Furthermore, this chapter discussed the contributions and impact of the developed system on the boutique's transition from traditional and manual selling processes to a more organized web-based ordering environment. It highlighted how the system could have improved the accessibility of products, streamlined transaction procedures, supported inventory monitoring, and provided customers with a more convenient and reliable shopping experience. The chapter also presented recommendations for further system improvements, future research directions, and implementation strategies that could have strengthened the system's effectiveness, security, reliability, and long-term sustainability for Margaux Collections.

5.2 Summary of Findings

The development and evaluation of the Margaux Collections Ordering System yielded findings that were generally aligned with the objectives of the project. The developed system successfully provided an online storefront where customers could browse the boutique's available product catalog, view relevant product information, and place orders through a GCash QR-based checkout process. The system provided a more organized alternative to manual ordering and allowed customers to access product information and complete their purchases through a centralized online platform. This demonstrated that the system was capable of supporting the basic online selling requirements of Margaux Collections.

The administrative panel also effectively supported the management of important business operations, including product, order, and user management. The administrator was able to manage product information, monitor customer orders, and maintain user-related records through a centralized interface. The system also included configurable promotional content on the homepage, which allowed the boutique to highlight selected products, announcements, or promotional materials. These functions provided the business owner with greater control over the content and operations of the online store and helped reduce the dependence on separate manual processes for managing products and orders.

The system's dark noir visual theme was also implemented consistently across both the customer-facing storefront and administrative interfaces. During development, stylesheet conflicts and inconsistencies affected certain aspects of the interface, but these issues were identified and resolved through testing and refinement. The correction of these styling problems helped maintain a more consistent visual identity throughout the system and ensured that the design remained aligned with the branding of Margaux Collections. This demonstrated the importance of continuous testing and interface refinement in achieving both functional and visual consistency in a web-based system.

The removal of the tiered membership feature also contributed to simplifying the overall customer purchasing experience and reducing unnecessary system complexity. The feature was removed after considering its relevance to the actual operations and requirements of the boutique. Its removal reduced the number of membership-related processes that needed to be managed by customers and administrators and also simplified the associated database structures and administrative controls. As a result, the system could have focused more directly

on its primary functions, such as product browsing, ordering, payment verification, and inventory and user management.

Lastly, collaborative development through Git and GitHub supported the maintenance and organization of the project across multiple development machines. The use of version control allowed the developers to manage changes, maintain project files, and coordinate development activities more systematically. During the development process, synchronization issues were encountered, particularly when changes were made across different machines, but these issues were identified and resolved through proper version-control procedures. This experience demonstrated the importance of collaborative development tools in maintaining consistency, tracking changes, and supporting efficient teamwork throughout the development of the Margaux Collections Ordering System.

5.3 Conclusion

Margaux Collections proved to be a functional, user-friendly, and brand-consistent solution for transitioning the boutique's traditional sales process into an organized online platform. The developed system provided a structured digital storefront that allowed customers to browse available brand-new and preloved clothing products in a more convenient and organized manner. It also incorporated a secure GCash-based checkout and payment verification process, which helped provide greater transparency and reliability during online transactions. In addition, the comprehensive administrative panel enabled the business owner to manage product listings, monitor inventory, process orders, verify payments, and maintain important transaction records more efficiently.

The system successfully addressed several operational challenges experienced in the boutique's previous selling process. By centralizing product information and ordering activities, it reduced the need to rely solely on manual records and scattered conversations through messaging platforms. Customers were able to access important product information, including photographs, prices, sizes, descriptions, and condition details, before placing their orders. This was particularly beneficial for preloved clothing because customers needed accurate information regarding the actual condition of each item before making a

purchase. The system therefore provided not only a platform for completing transactions but also a structured way of presenting products and communicating essential information to customers.

The GCash-based checkout process further strengthened the system's ability to support reliable online transactions. Customers were able to submit their payment information through the established ordering procedure, while the business owner could verify transactions using the provided payment reference number. This process helped reduce uncertainty regarding advance payments and provided a more organized method of matching payments with customer orders. As a result, the system could have contributed to greater accountability and reduced the possibility of errors associated with manually confirming payments through screenshots and separate messaging conversations.

The administrative panel also played an important role in improving the business's internal operations. Through centralized management of products, inventory, orders, and payment information, the owner could have monitored business activities more efficiently. The system helped ensure that product information remained organized and that available items could be managed more systematically. This was especially important for preloved clothing because many items were unique or available only in limited quantities. Proper inventory management could have reduced the possibility of accepting orders for products that were already sold or unavailable.

Furthermore, the developed system maintained the visual identity of Margaux Collections while incorporating the necessary functions of an online ordering platform. The design allowed the boutique to maintain a professional and consistent presentation of its products and brand. This was important because the online platform served not only as a transaction tool but also as a digital representation of the business. By combining organized product listings, attractive product presentation, and functional ordering features, the system could have helped present preloved clothing as valuable and desirable fashion products rather than simply as used items.

The project also demonstrated that a small-scale fashion retail business could have benefited from a purpose-built web application designed around its actual operational needs. Instead of relying entirely on generic or template-based e-commerce solutions, Margaux Collections was able to use a system that was specifically developed to address its product categories, ordering procedures, payment method, inventory requirements, and administrative

processes. This demonstrated the potential of customized information systems to support small businesses that required functions tailored to their particular operations.

Overall, the Margaux Collections Ordering System successfully demonstrated how a purposebuilt web application could have supported the digital transformation of a small fashion retail business. By integrating an organized storefront, product and inventory management, GCashbased payment verification, order processing, and administrative functions, the system provided a more structured and convenient approach to online selling. It addressed both customer-facing and business-management needs while maintaining the boutique's brand identity. The project therefore demonstrated that an appropriately designed and affordable webbased ordering system could have improved the efficiency, organization, accessibility, and reliability of Margaux Collections' sales operations without requiring dependence on generic ecommerce platforms.

5.4 Recommendations System

The Margaux Collections Ordering System could have been further improved by implementing additional features that strengthened inventory management, account security, and reporting capabilities. Automated low-stock notifications could have been integrated to alert the administrator when product quantities reached a specified level, allowing timely inventory updates and reducing the possibility of stock shortages. Account security could also have been enhanced through additional authentication measures, such as one-time password (OTP) verification, to provide better protection for administrator and customer accounts. In addition, reporting functions could have been improved by allowing order and sales information to be exported in formats such as PDF or CSV, which could have supported recordkeeping, monitoring, and business analysis.

For future research and system development, automated payment verification could have been explored to reduce the need for manual confirmation of GCash transactions. This feature could have allowed the system to verify payment information more efficiently and

could have reduced delays in order processing. The feasibility of developing a mobile-native application could also have been investigated to make the system more accessible to customers who had limited access to desktop computers. Furthermore, a follow-up study could have been conducted to evaluate the long-term impact of the system on the boutique's sales performance, customer satisfaction, operational efficiency, and customer retention. The results of such a study could have provided additional information for identifying areas that required further improvement.

For successful implementation and continued operation, boutique staff could have been provided with proper training on the consistent use of the administrative panel. Training could have helped ensure that product information, inventory records, orders, and payment details were entered and updated accurately. Regular database backups could also have been performed to protect important business information from accidental loss, system errors, or other technical problems. Maintaining updated backups and conducting periodic system maintenance could have supported the reliability, security, and long-term sustainability of the Margaux Collections Ordering System.

5.5 Impact of the Study

The Margaux Collections Ordering System contributed to the practical application of web development principles in supporting the operations of a small-scale fashion retail business. The study demonstrated how a purpose-built e-commerce platform could have addressed the specific needs of the boutique, particularly in product presentation, online ordering, inventory management, payment verification, and administrative monitoring. By providing a brand-consistent and organized digital storefront, the system could have helped improve the accessibility of the boutique's products while providing customers with a more convenient way to browse and purchase both brand-new and preloved clothing. The project also demonstrated that small businesses could have benefited from customized web-based solutions that were designed according to their actual operational requirements rather than relying entirely on generic ecommerce platforms.

The study also contributed to improving the efficiency and organization of the boutique's business processes. The integration of product listings, inventory records, ordering functions, payment verification, and administrative tools could have reduced dependence on manual processes and separate communication channels. This could have helped minimize errors, improve the accuracy of product and order information, and provide the business owner with a more systematic method of managing daily transactions. For customers, the system could have provided clearer product information, organized categories, convenient ordering procedures, and more transparent transaction updates. These improvements could have contributed to a more reliable and satisfactory online shopping experience.

Furthermore, the development of the system demonstrated the importance of iterative refinement in software development. Throughout the development process, adjustments such as removing unnecessary features, correcting stylesheet and interface issues, improving system functions, and refining the overall user experience could have been necessary to ensure that the final system remained functional and aligned with the needs of Margaux Collections. These refinements demonstrated that developing an effective information system required continuous evaluation, testing, and improvement rather than relying solely on the initial design. Overall, the study showed that a customized and continuously refined web-based ordering system could have supported the digital transformation of a small fashion business while maintaining its unique brand identity and addressing its specific operational challenges.